

2030s Net Zero Playbook

Version 3.2

Status: Final

Last updated: May 2026

© 2026, Shannon A. Fiume. This document was developed with assistance from OpenAI ChatGPT (GPT-4.1 Turbo and GPT-5.x series).

This work is licensed under the Creative Commons Attribution 4.0 International License (CC-BY-4.0). You are free to share and adapt the material for any purpose, including commercial use, provided appropriate credit is given.

<https://creativecommons.org/licenses/by/4.0/>

Preferred citation:

Fiume, Shannon A. (2026). 2030s Net Zero Playbook.

Version 3.0. Developed with assistance from OpenAI's ChatGPT.

Licensed under CC-BY-4.0. <https://bit.ly/NZpbk>.

For additional information, please contact:

Shannon A. Fiume

P.O. Box 190622

San Francisco, CA 94119-0622

shannon@autofracture.com

2030s Net Zero: Policy Architecture and Operational Blueprint

[Executive Summary](#)

[Introduction](#)

[Present Climate Outlook and Why This Playbook Exists](#)

[Definition of Operational Net Zero](#)

[In Scope \(Required Conditions\)](#)

[Out Of Scope \(Not Permitted as Substitutes for Phaseout\)](#)

[Residuals and Constrained Exceptions](#)

[Floor Condition](#)

[Four Operational Rules](#)

[Clean-Only Growth Rule Definition](#)

[Deliverable Clean Scale Rule Definition](#)

[Fossil Exit Rule Definition](#)

[Reliability Replacement Rule Definition](#)

[What Each Chapter Does](#)

[Chapter 1 — Workforce Training & Mobilization](#)

[Workforce Mobilization Phase Plan](#)

[Education and Training Multipliers](#)

[Core Constraint](#)

[Chapter 2 — Clean Energy Financing and Financial System Realignment](#)

[Operational Role of Finance](#)

[Financial Conditions for the Four Operational Rules](#)

[Capital Mobilization Architecture](#)

[Financial Repricing Tools](#)

[Strategic Function](#)

[Chapter 3 — Electricity Infrastructure: Deliverable Clean Scale](#)

[Risk Threshold](#)

[Milestones](#)

[Operational Sequence \(2026–2030\)](#)

[Summary Condition](#)

[Chapter 4 — Solar and Wind Exponential Expansion](#)

[Clean Energy Manufacturing Scale-Up](#)

[Solar PV](#)

[Solar Required](#)

[Install Trajectory](#)

[Solar Risk Threshold](#)

[Wind](#)

[Wind Required \(Combined\)](#)

[Wind Risk Threshold](#)

[Parallel Manufacturing Scale-Up](#)

[Operational Sequence \(2026–2030\)](#)

[Summary Condition](#)

[Chapter 5 — Energy Storage and Clean Flexibility Scaling](#)

[Functional Role of Clean Flexibility](#)

[Storage Scaling Requirements](#)

[Short-Duration Storage \(Batteries\)](#)

[Battery Storage Functional Requirement](#)

[Risk Threshold](#)

[Long-Duration Storage \(LDES\)](#)

[Requirement](#)

[Demand Flexibility](#)

[Demand Flexibility Functional Role](#)

[Requirement](#)

[Firm Clean Generation](#)

[Firm Clean Generation Functional Role](#)

[System Integration Requirement](#)

[Risk Threshold](#)

[Operational Sequence \(2026–2030\)](#)

[Summary Condition](#)

[Chapter 6 — Firm Clean Generation and Residual Adequacy](#)

[Functional Role of Firm Clean Generation](#)

[Technology Scope](#)

[Scaling Requirement](#)

[Deployment Constraint](#)

[Risk Threshold](#)

[Market Design Requirement](#)

[Operational Sequence \(2026–2030\)](#)

[System Integration Requirement](#)

[Summary Condition](#)

[Chapter 7 — Industrial Electrification and Load Transformation](#)

[Functional Role of Industrial Electrification](#)

[Scope of Electrification](#)

[Industrial Conversion Requirement](#)

[Load Growth Requirement](#)

[Key Requirement](#)

[Failure Condition](#)

[Load Shape Transformation](#)

[Infrastructure Requirement](#)

[Risk Threshold](#)

[Market and Policy Requirements](#)

[Operational Sequence \(2026–2030\)](#)

[System Integration Requirement](#)

[Summary Condition](#)

[Chapter 8 — Circular Materials & Recycling Systems](#)

[Targets by ~2035 \(Decision-Relevant Floors\)](#)

[Metals and Battery Materials](#)

[Polymers and Composites](#)

[Grid-Critical Components and Industrial Inputs](#)

[System Target](#)

[Longer-Run Goal](#)

[Risk Threshold](#)

[Operational Sequence \(2026–2030\)](#)

[Summary Condition](#)

[Chapter 9 — Fossil Retirement Governance](#)

[Functional Role of Fossil Retirement Governance](#)

[Governance Toolset](#)

[Persistence Pathways](#)

[Retirement Conditions](#)

[Sequencing Principle](#)

[Fuel-Specific Retirement Dynamics](#)

[Coal](#)

[Oil](#)

[Gas](#)

[Asset Management and Financial Stability](#)

[Workforce and Regional Transition](#)

[Market Alignment](#)

[Risk Threshold](#)

[Operational Sequence \(2026–2030\)](#)

[Operational Gating Factors](#)

[Summary Condition](#)

[Chapter 10 — System Constraints on Transition Speed](#)

[Institutional Throughput and Coordination](#)

[Infrastructure Siting and Social Acceptance](#)

[Workforce and Industrial Capacity](#)

[Incumbent System Persistence](#)

[Interaction with Financial Dynamics](#)

[Implication for Transition Pathways](#)

[Chapter 11 — Planetary Biosphere Restoration](#)

[Functional Role of Biosphere Restoration](#)

[Separation from Offsetting and Accounting](#)

[Relationship to Carbon Dioxide Removal \(CDR\)](#)

[Scope of Restoration](#)

[Constraints and Risks](#)

[Justice and Stewardship](#)

[Operational Sequence \(2026–2035+\)](#)

[Risk Threshold](#)

[System Integration Requirement](#)

[Summary Condition](#)

[Chapter 12 — Managed Land Systems](#)

[Role in the Net Zero Framework](#)

[Operational Priorities](#)

[Chapter 13 — Corporate and Government Acceleration](#)

[Core Principle](#)

[Operational Levers](#)

[A. Demand Aggregation and Financing Pull-Through](#)

[Key mechanisms include:](#)

[B. Clean Procurement Standards That Reshape Markets](#)

[Effective standards include:](#)

[C. Workforce and Internal Execution Capacity](#)

[Key actions include:](#)

[AI And Discretionary Load Discipline](#)

[Datacenter Requirements \(Minimum Operational Set\)](#)

[Risk Threshold](#)

[Operational Sequence \(2026–2030\)](#)

[Summary Condition](#)

[Chapter 14 — AI as Transition Multiplier](#)

[Functional Role of AI in the Transition](#)

[High-Value Deployment Domains](#)

[Productivity Assumption](#)

[AI Load Discipline](#)

[Governance Requirement](#)

[Risk Threshold](#)

[Operational Sequence \(2026–2030\)](#)

[System Integration Requirement](#)

[Summary Condition](#)

[Final Systems Conclusion — Net Zero as Coordinated System Transformation](#)

[Appendices](#)

[Appendix A — Critical Timeline](#)

[Conditional Timeline Checkpoints Table](#)

[Solar and Wind Curve Baseline Inputs \(End-2025\)](#)

[System Rule \(Invariant\)](#)

[Post-Net-Zero to 2100](#)

[Interpretation](#)

[Appendix B — Structural Caveats and Bottlenecks Risk Hierarchy](#)

[Hierarchy Of Highest-Risk Bottleneck Categories:](#)

[Appendix C — Common Major Failure Modes](#)

[1. Fossil Financial Repricing Fails / Persistence Pathways Remain Open](#)

[2. Emerging Markets Lock In New Fossil Capacity](#)

[3. Cryosphere Albedo / Ice-to-Land Cascade](#)

[4. Transmission And Interconnection Buildout Failure](#)

[5. Permafrost Carbon and Methane Feedback](#)

[6. Clean Adequacy Replacement Lags Generation Buildout](#)

[7. AMOC / Ocean-Circulation Disruption](#)

[8. Coordinated Fossil Adaptive-Delay Strategy](#)

[9. Amazon / Tropical Forest Dieback](#)

[10. Workforce And Institutional Throughput Bottleneck](#)

[11. Fossil Asset Bailouts And Sunset-Asset Protection](#)

[12. AI And Discretionary Load Growth Outpaces Clean Build](#)

[13. Freshwater-Cycle / Monsoon Destabilization](#)

[14. Compound Biosphere Cascade](#)
[15. Critical Materials And Component Disruption](#)
[Ranking by Risk to Net Zero Timeline](#)
[Structural Insight](#)
[Appendix D — Net-Zero Workforce Mobilization Framework](#)
[Workforce Categories and Annual Additions \(Peak 2026–2032\)](#)
[Phase Structure](#)
[Education and Training Growth Rates](#)
[Appendix E — Clean Energy Financing and Financial Realignment](#)
[Core Principle](#)
[I. Core Financial Mechanisms](#)
[II. Capital Mobilization Architecture](#)
[III. Financial System Risk Repricing](#)
[IV. Financial Alignment Across the Four Operational Rules](#)
[The Strategic Objective](#)
[Appendix F — Financing Tier Adjustments Consistent with Rapid System Transition](#)
[Recommended Adjustments to Financing Tiers Under Emerging System Constraints \(2026–2028\)](#)
[Appendix G — Illustrative Infrastructure Deployment Mechanics Under Accelerated Transition Conditions](#)
[Overview](#)
[Illustrative Deployment Workflow](#)
[Appendix H — SDG Playbook Mapping](#)
[Appendix I — Prospective Manufacturing Hubs](#)
[Possible Domestic Manufacturing Anchors](#)
[Appendix J — NZ Readiness](#)
[Tier 1 — Structurally Net-Zero-Capable Grids \(With Expansion\)](#)
[Tier 2 — Near-Ready With Transmission Expansion](#)
[Tier 3 — Major Expansion Required](#)
[Appendix K — Minimum Viable Coalition for Accelerated Net Zero](#)
[Core System Requirement](#)
[System Model \(Operational Abstraction\)](#)
[Critical Node Transitions](#)
[Core Coalition Capabilities](#)
[High-Leverage Acceleration Layers](#)
[Queue Synchronization Requirement](#)
[Primary Failure Modes](#)
[Coalition Condition](#)
[What Is Not Required](#)
[Summary](#)
[Appendix L — Patent Fast-Tracking and Open Systems](#)
[Appendix M — Carbon Dioxide Removal \(CDR\)](#)
[NZ-CDR10](#)
[NZ-CDR10 Scale Milestones](#)
[NZ-CDR10 Engineered Removals and Energy](#)

[Energy Intensity and Grid Implications](#)
[Workforce Ramp \(Decision-Relevant Ranges\)](#)
[Speculative High-Removal Scenario \(Illustrative Outer Bound\)](#)
[Anthro-2125 Energy and Its Implications](#)
[Anthro-2125 Engineered Removals and Energy](#)
[Anthro-2125 Workforce Additions Table](#)
[Appendix N — Ramp-Rate Derivation Methodology](#)
[Methodological Principles](#)
[Core Formulas](#)
[Baseline and Checkpoint Inputs](#)
[Wind — Worked Example](#)
[Storage — Methodological Approach](#)
[Transmission — Methodological Approach](#)
[Workforce — Methodological Approach](#)
[Indicative Structure](#)
[Multiple Valid Trajectories](#)
[Mixed Derivation Methods](#)
[Annual System Load-Closure Interpretation](#)
[Adequacy Constraint](#)
[Limitations](#)
[Summary](#)
[Appendix O — AI Thread Alignment And Reproducibility](#)
[How To Make New Threads Act More Like This One](#)
[Initial Thread Context That Should Carry Over to New Threads](#)
[Planetary Boundaries and Equitable System Alignment](#)
[Alignment Framework — Planetary Boundaries and Equitable System Conditions](#)
[AI Governance and Ethically Aligned Design Guardrails](#)
[Epilogue](#)
[GPT's Epilogue — A discipline of reality, a wager on coordination](#)
[Shannon's Epilogue](#)
[Common Conversions and Conventions](#)
[Key Terms and Measurement Conventions](#)
[Additional Conventions](#)
[Interpretation Notes](#)
[References](#)

Executive Summary

This playbook identifies the earliest operational net-zero window that remains physically plausible under current technology maturity, infrastructure build constraints, institutional throughput, and fossil retirement dynamics. In this framework, the earliest high-execution window is ~2037–2038, with ~2039 as its outer edge. If required control variables are not secured during 2026–2030, the pathway shifts into the early 2040s and then later decades as path dependence, fossil reinvestment, and institutional delay accumulate.

This playbook is not a forecast or a declaration of ambition. It is a constraint-mapped blueprint focused on the operational decisions that force system change at speed under conditions that remain materially just, including through equitably durable implementation. It identifies which control variables must move, which bottlenecks must be cleared, and which governance and infrastructure conditions must mature on time for operational net zero to remain plausible. In practice, this means policies, market rules, and governance structures that make clean investment dominant, fossil reinvestment uncompetitive, and fossil persistence harder to sustain, while protecting workers, reducing burden-shifting, and scaling the physical system fast enough to drive fossil capacity factors down under durable conditions. The chapters and appendices map these conditions in detail: binding constraints, failure modes, required rates of clean build and fossil retirement, and the institutional conditions that determine whether the pathway holds or slips.

The pathway is governed by four non-substitutable operational rules:

1. **Clean-only growth** — no new fossil capacity serves net load growth.
2. **Deliverable clean scale** — clean generation, storage, transmission, and firm resources scale in forms that can be connected, transmitted, and used at load.
3. **Fossil exit** — fossil assets lose refinancing, life-extension, and like-for-like replacement pathways.
4. **Reliability replacement** — storage, transmission, demand flexibility, and firm clean resources replace fossil generation in its routine adequacy role.

Execution speed is governed by two binding constraints:

1. Financial repricing of fossil capital
2. Workforce mobilization of roughly 2–3 million additional trained workers per year, with 30–40% AI-assisted productivity gains

These constraints determine whether the four rules can be executed at sufficient speed and scale.

Conditional on these constraints, the pathway requires the following system-level conditions to hold in parallel:

- Solar PV sustaining ~24–27% CAGR through ~2032, then ~16–19% through 2034
- Wind sustaining ~12–15% CAGR through ~2032, then ~9–11% through 2034
- Battery storage sustaining ~40–45% CAGR through ~2032, with long-duration storage commercialized before 2030
- Transmission expansion sustaining ~7–9% annually through peak build years
- Fossil refinancing and life-extension pathways materially constrained before ~2030

Under these primary conditions, fossil persistence pathways narrow, clean supply becomes deliverable at system scale without creating new system adequacy barriers, capacity factors decline, and operational net zero by ~2037–2038 remains plausible.

Industrial conversion is a downstream feasibility condition: roughly ~70–75% or more likely needs to be achieved by the late 2030s, since residual pathways can absorb only a limited remaining share without delaying the ~2037–2038 outcome.

The decisive window is late 2026 through 2030. Conversely, if these conditions are not upheld, the pathway shifts toward 2045–2055 as path dependence locks in by ~2032, fossil reinvestment continues, and institutional delay compounds.

Carbon dioxide removal (CDR) is not used to reach net zero; it is reserved for post-net-zero atmospheric drawdown of any remaining residual emissions in the hardest-to-abate sectors and, depending on societal choice, some or all historical emissions.

Introduction

This playbook outlines net-zero policy architecture and serves as an operational blueprint. It identifies the binding constraints, order of operations, and decision levers required to make operational global net zero by ~2037–2038 physically plausible, while recognizing that the pathway is institutionally fragile and must be materially just to remain durable. It is written for executive decision-makers and technical directors who need a concrete picture of what being on the path looks like: which system variables must move, which commitments must be locked in by when, and which failure modes push the timeline outward.

The playbook does not treat net zero as a single target year. It identifies operational windows determined by the interaction of physical constraints, institutional throughput, control variables, and technology and resource maturity. Under this framework, the swiftest execution makes ~2037 the first institutionally plausible year, opening the earliest net-zero window of ~2037–2038. If the required variables are not secured in time, the pathway shifts into later windows as delays compound and fossil persistence pathways remain open.

The core premise is simple: the transition is now technology-capable but institution-limited. The question is no longer whether solar, wind, storage, electrified heat, or industrial conversion can work in principle—they can. The limiting factors are whether the world can reprice fossil capital fast enough, expand transmission fast enough, mobilize trained labor at sufficient scale, and execute four non-substitutable operational rules in parallel: clean-only growth, deliverable clean scale, fossil exit, and reliability replacement. The pathway does not require complete industrial conversion on this timeline, but it does require sufficient conversion. In practice, global industrial decarbonization likely needs to reach on the order of ~70%-75% or more by the late 2030s, because residual pathways—including hydrogen, synthetic fuels, cement and concrete CCU, and firm-clean sources—can contain only a limited residual share. If industrial conversion falls materially below this range, the residual burden exceeds what these pathways can plausibly support, tightening or delaying the ~2037–2038 outcome. Full geopolitical coordination and identical regional timelines are not required, but the pathway does require sufficiently broad functional alignment across major power systems to achieve full fossil phaseout in the power sector and a near-complete phaseout of fossil fuels overall by the net-zero timeline.

This playbook therefore focuses on the operational decisions that can drive system change at speed: policies, market rules, and governance structures that make clean investment structurally dominant, fossil reinvestment structurally uncompetitive, and fossil persistence institutionally harder to sustain, while scaling the physical system fast enough to drive fossil capacity factors down under conditions that support a just transition.

How to read this document

- **Executive Summary:** the handful of control variables that determine whether ~2037–2038 is plausible.
- **Chapters 1–2:** the enabling foundations (workforce and finance). If either fails, downstream chapters cannot compensate.
- **Chapters 3–6:** the clean electricity system growth engine (grid buildout, solar/wind, storage, and firm/heat complements). Together, these chapters determine whether the transition scales fast enough to make fossil phaseout operationally plausible.

- **Chapters 7–10:** the hard-to-abate conversion layer, fossil retirement governance, and system constraints on transition speed.
- **Chapters 11–14:** biosphere restoration, managed land systems, and actor acceleration, including corporate and government leverage and AI as a productivity multiplier with load discipline.
- **Appendices:** operational targets, bottlenecks, failure modes, leverage actors, readiness tiers, and post-net-zero CDR sequencing.

Sequencing note: The chapter order reflects the logic of constraints, not a strictly serial program. Many actions must be executed in parallel; the purpose of sequencing is to highlight which prerequisites must be secured early to avoid path dependence.

Present Climate Outlook and Why This Playbook Exists

Humanity has emitted nearly 2.8 teratonnes of CO₂ (1750–2024¹, plus 2025 est.²) raising CO₂ concentration in both the atmosphere and oceans to levels not seen in many millions of years.³ This accumulation of emissions, their rate of release into the atmosphere, and their uptake by the oceans are the primary drivers of climate change.⁴ Under current-policies’ projections warming is likely to exceed 2.5°C by 2100,⁵ with material risk of cascading impacts pushing additional planetary boundaries beyond safe operating limits.⁶ Meanwhile, global fossil infrastructure is still not on a reliable end-of-life path: new electricity growth is still often met by gas or coal additions.^{7,8}

Yet the situation is not technologically hopeless. Clean generation and storage costs have declined dramatically over the last decade, and electrified end-use technologies are now mature enough to replace most fossil demand categories. The remaining barrier is institutional: grid expansion, permitting, capital structure, and workforce mobilization must scale fast enough to match the physics of the transition. Those institutional barriers are frequently most acute in the Global South and in historically marginalized communities, where climate vulnerability, pollution exposure, and infrastructure exclusion already overlap.^{9,10}

The accelerated net-zero transition must be judged not only by speed and scale, but by who benefits, who participates, and who is protected first. It must reduce not only emissions, but also unequal exposure to pollution, climate risk, infrastructure exclusion, and ecological loss, especially across the Global South and historically marginalized communities. Communities that are overburdened, underinvested, on the frontlines of climate risk, or historically excluded from environmental decision-making should be prioritized in resilience investment, equitable redevelopment, pollution reduction, workforce development, and governance.

While the IPCC Shared Socioeconomic Pathways (SSPs)¹¹ describe differing development trajectories based on broader socioeconomic assumptions, this playbook addresses a narrower yet crucial question: the physical, institutional, and economic conditions required to make operational net zero plausible on an accelerated timeline. Successful implementation would not by itself fully specify a sustainability-oriented development pathway, but it would build much of the foundation for one, from abundant clean energy to more durable public systems.¹

¹ The ~2037–2038 pathway can be interpreted as a return from overshooting 1.5°C, consistent with mid-century convergence to an SSP1-1.9-compatible trajectory. See [Rogelj et al. \(2018\). *Scenarios towards limiting global mean temperature increase below 1.5°C*](#).

Definition of Operational Net Zero

In this playbook, operational net zero by ~2037–2038 is defined in physical system-operational terms, not accounting terms.

At minimum, it requires **full power-sector fossil fuel phaseout: coal and fossil-extracted natural gas** are **eliminated** as routine sources of electricity generation. Fossil generation does not remain a routine or structural component of supply and therefore is not retained through offsets, crediting, or other accounting mechanisms. Within this framework, eligibility is determined by fuel origin rather than fuel form. Carbon dioxide removal (CDR) is not used to achieve operational net zero in the power sector and cannot substitute for fossil phaseout. Fossil fuels are not retained for planning margin, system convenience, or institutional inertia; any remaining use must be justified by physical constraints.

This definition is intentionally stricter than many “net zero” frameworks, which allow continued fossil fuel use balanced by carbon dioxide removal (CDR), offsets, or other accounting approaches. It is designed to yield the smallest feasible set of residual emissions, confined to the hardest-to-abate sectors.

In Scope (Required Conditions)

- Full phaseout of coal and fossil-extracted natural gas in the power sector
- Electricity demand growth met exclusively by clean generation, storage, transmission expansion, and other firm clean resources as needed ([Clean-Only Growth Rule](#))
- System reliability maintained through storage, transmission, demand flexibility, and firm clean resources
- Electrification of industry, buildings, and transport proceeding such that fossil dependence is structurally declining

Out Of Scope (Not Permitted as Substitutes for Phaseout)

- Use of CDR, offsets, or accounting mechanisms to justify continued fossil generation in the power sector
- Continued fossil generation as “backup” outside tightly constrained, time-limited emergency conditions (non-routine, system-disruption events)
- Net-zero claims based on balance rather than physical displacement

Residuals and Constrained Exceptions

- Oil may persist only in a shrinking residual centered primarily in marine liquid fuels, with structural decline underway and strategic reserves winding down
- Smaller residual use may persist in off-road and geographically remote heavy-duty machinery
- Aviation is not treated as a long-run fossil-fuel residual category; it is expected to transition from fossil-derived fuels to synthetic e-fuels and sustainable biofuels.
- Gaseous fuels persist only when not fossil-derived, including BECCS, sustainable biofuels, waste-to-energy with CCS, and hydrogen where no direct electrification

pathway exists; these are treated as constrained system components, not extensions of fossil supply

Floor Condition

This definition establishes a non-negotiable floor condition: if the power sector has not achieved fossil phaseout, operational net zero has not been reached, regardless of net emissions accounting. The four operational rules below define how net zero is achieved in system terms. Together they identify the minimum physical and institutional requirements of the accelerated pathway.

Four Operational Rules

Operational net zero requires four rules to be executed in parallel: clean-only growth, deliverable clean scale, fossil exit, and reliability replacement. Together, they define the minimum physical conditions of the pathway.

These rules define the minimum physical conditions of the pathway, but their execution is equitably durable only when implementation does not shift transition costs, siting burdens, reliability risk, or residual harms onto workers, ratepayers, host communities, Indigenous peoples, or historically marginalized communities. In this playbook, equitable durability means maintaining transition speed through fair burden-sharing, credible transition protection, and implementation that does not reproduce exclusion, pollution concentration, or cost-shifting.

Clean-Only Growth Rule Definition

By year-end 2028, all incremental electricity demand growth must be met by clean supply;² no new fossil capacity may be built for net load growth, and fossil generation must not increase to serve incremental demand.

Execution of this rule must not lock poorer regions into a fossil-as-development trap or shift energy-cost, reliability, or siting burdens onto overburdened communities; clean-only growth remains durable only when paired with fair burden-sharing, credible transition protection, and affordable clean replacement.

Deliverable Clean Scale Rule Definition

Clean energy growth must be physically deliverable to load at system scale, not merely installed in nominal capacity terms, and must be implemented without systematically shifting siting burdens, exclusion, or cost burdens onto already overburdened communities.

Fossil Exit Rule Definition

By the early 2030s, fossil generation must be on a sustained year-over-year decline³ and fossil assets may not persist through refinancing, life-extension capital expenditure, or like-for-like replacement without a credible retirement pathway.

Reliability Replacement Rule Definition

Fossil reliability roles must be replaced by storage, transmission, demand flexibility, and firm clean resources such that reliability is maintained without structural fossil backup, and without concentrating reliability risk, service inequity, or cost burdens in already overburdened communities.

² The underlying logic is literature-grounded: emissions decline requires clean supply to capture incremental demand so fossil generation is displaced rather than supplemented. The year-end 2028 threshold is a playbook judgment. See [IEA. \(2025\). World Energy Outlook 2025](#); [IPCC. \(2022\). Climate Change 2022: Mitigation of Climate Change](#).

³ Net-zero-consistent pathways require fossil-based electricity generation to peak and then decline as clean supply scales. The early-2030s timing used here is a playbook judgment. See [IEA. \(2025\). World Energy Outlook 2025](#); [IPCC. \(2022\). Climate Change 2022: Mitigation of Climate Change](#).

What Each Chapter Does

Chapter 1 — Workforce Training & Mobilization

Defines the workforce ramp needed to prevent a 2029–2032 labor bottleneck and to sustain deployment through the mid-2030s. It emphasizes the job families that gate deployment (electricians, lineworkers, substation technicians, installers, retrofit integrators, permitting/finance staff), the education and training multipliers required, and broad access to fossil transition jobs, with targeted expansion of opportunities for historically marginalized and underrepresented groups.

Chapter 2 — Clean Energy Financing and Financial System Realignment

Defines the capital mobilization architecture and financial repricing tools needed to make clean IRRs structurally exceed fossil IRRs by ~2028, stop fossil refinancing, and de-risk emerging-market clean build, especially where high capital costs would otherwise lock underinvested regions into fossil dependence.

Chapter 3 — Electricity Infrastructure: Deliverable Clean Scale

Acknowledges transmission, interconnection, and grid reinforcement as the primary physical bottleneck and as a core component of the clean electricity system growth required for operational fossil displacement. Sets expansion rates, permitting reforms, HVDC scaling needs, and the institutional reforms required so renewables translate into usable clean electricity rather than curtailment.

Chapter 4 — Solar and Wind Exponential Expansion

Defines the growth trajectory for solar and wind needed to force fossil capacity factors into structural decline before 2040, including manufacturing scale-up, supply chain constraints, and post-net-zero taper logic that avoids a premature plateau.

Chapter 5 — Energy Storage and Clean Flexibility Scaling

Defines the growth of storage, demand flexibility, and clean adequacy resources required to prevent reliability backlash and fossil backup lock-in, including long-duration storage milestones and market design to pay for flexibility.

Chapter 6 — Firm Clean Generation and Residual Adequacy

Identifies the complementary firm and thermal systems (hydro, nuclear life extension, geothermal, thermal storage, heat pumps, district thermal) needed to decarbonize heat and provide reliability as fossil backup is removed.

Chapter 7 — Industrial Electrification and Load Transformation

Details industrial retrofit sequencing and the conversion of fossil-derived molecules (chemicals, fuels, polymers) into electrified/synthetic pathways. Frames industry conversion as integration-limited and sets replacement-rate targets.

Chapter 8 — Circular Materials & Recycling Systems

Acknowledges recycling and circular design as both a critical materials strategy and a fossil feedstock replacement strategy (metals, polymers, critical inputs). Establishes targets and timelines so material supply does not bottleneck deployment.

[Chapter 9](#) — Fossil Retirement Governance

Defines the policy tools to retire coal, oil, and gas assets in an orderly way (sunset securitization, mandatory retirement schedules, refinancing prohibitions, limited emergency exemptions), while managing political backlash and stranded-asset risk.

[Chapter 10](#) — System Constraints on Transition Speed

Outlines the system constraints that govern transition speed, including institutional throughput, infrastructure siting, workforce and industrial capacity, and incumbent fossil persistence. It shows how these dynamics shape sequencing, interact with financial conditions, and determine whether the late-2030s pathway remains plausible.

[Chapter 11](#) — Planetary Biosphere Restoration

Separates biosphere restoration from “offsetting” and treats it as resilience, long-run recovery, and local ecological repair. Frames the sequencing: operational net zero first, then drawdown; ecosystem uplift is pursued immediately but not used to justify fossil delay, and is governed with attention to stewardship, livelihoods, and local participation.

[Chapter 12](#) — Managed Land Systems

Enhances managed land systems resilience through land-use emissions reductions and eligible carbon storage. Separates land-based climate benefits from durable CDR accounting, while recognizing only a limited subset of land-based gains as eligible to count towards removals under strict durability, MRV, sustainability, and justice conditions.

[Chapter 13](#) — Corporate and Government Acceleration

Identifies how large demand-side actors (hyperscalers, governments, OEMs, heavy industry anchors, finance) can force upstream decarbonization through procurement rules, supply-chain requirements, financing pull-through, and discretionary-load discipline, while shaping who receives investment, market access, and the benefits of transition.

[Chapter 14](#) — AI as Transition Multiplier

Frames AI as a productivity lever for design, permitting, and construction—but only if AI load growth is coupled to clean procurement and flexibility (clean-only growth rule enforced for data centers).

[Appendices](#)

Provide the operational backplane of the playbook: targets, bottlenecks, failure modes, leverage actors, readiness tiers, and post-net-zero CDR sequencing. They set out control variables and implementation logic in sufficient detail for technical review without overloading the main chapters.

Chapter 1 — Workforce Training & Mobilization

Workforce is a binding constraint^{14,15} on the net zero pathway because it gates permitting⁴ throughput, grid build, and deployment speed. It is also the primary human system through which the transition occurs—millions of workers whose training, safety, mobility, and economic security determine whether decarbonization can proceed at required speed and whether the transition is equitably durable. This chapter defines decision-relevant workforce ramps and the training multipliers required to prevent labor scarcity from delaying fossil displacement. Detailed training architecture and discipline-level mapping are provided in [Appendix D](#).

Workforce expansion must be global in scope, with particular emphasis on rapidly growing labor markets in the Global South, where deployment needs are large, but training capacity, credentialing systems, and project-finance continuity are often constrained.

The net-zero pathway requires sustained additions of **~2–3 million trained workers per year globally during the peak build window (roughly 2026–2032)**, assuming ~30–40%⁵ AI-assisted productivity gains. Cumulative additions by ~2035: ~18–22 million workers. Workforce mobilization includes both training new entrants and creating credible transition pathways for workers moving out of fossil-linked sectors; the financial protections and retirement sequencing for this transition are developed in [Chapter 2](#) and [Chapter 9](#). **Failure to reach ≥2 million annual additions by ~2028 materially delays the net-zero pathway**, regardless of capital availability.

Workforce mobilization is cross-cutting and only succeeds if:

- financing keeps project pipelines continuous ([Chapter 2](#)), and
- transmission and interconnection expand ([Chapter 3](#)).

Specialized CDR workforce expansion begins immediately but ramps sharply post-2035 to avoid competing with the 2026–2032 mitigation build window; see [Appendix M](#) and the [NZCDR10](#) sheet for the post-2038 ramp. Rapid scale-up must be paired with enforceable labor standards, strong safety regulation, and anti-exploitation safeguards so that workforce expansion does not rely on degraded working conditions.

Workforce Mobilization Phase Plan

The phases define what must be mobilized when, which job families gate progress, and the minimum additions required to avoid a binding labor bottleneck. The decision-relevant phase plan is summarized in the table below and is also provided in the “NZ Workforce” sheet in the NZ2037 pathways workbook. [+ NZ2037 pathways](#)

Phase	Window	Primary objective	Workforce additions (trained/placed)	Dominant job families	What this phase unlocks
-------	--------	-------------------	---	-----------------------	-------------------------

⁴ Here, ‘permitting’ includes siting, interconnection, environmental review, and construction approvals; ‘deployment’ refers primarily to transmission, solar/wind, storage, and end-use electrification.

⁵ If AI-enabled productivity gains are only ~5–10%, required workforce additions rise to ~2.5–4.0 million workers per year and the peak mobilization window shifts earlier (2026–2030). Under this condition, failure to exceed ~2.5 million annual additions by ~2028 makes the ~2037–2038 pathway unlikely.

per year)					
Phase 0	2026–2027	Stand up institutions and training capacity; relieve permitting and interconnection bottlenecks	≈0.5–1.5M/yr (subset of Chapter 1 total)	permitting and interconnection staff, instructors and trainers, electricians, lineworkers, project managers	permitting throughput, interconnection queue clearing, and procurement readiness
Phase 1	2028–2032	Peak build window: transmission, solar and wind, storage, and retrofit mobilization	≈2–3M/yr (core Chapter 1 requirement)	electricians, lineworkers, substation crews, installers, civil works crews, retrofit integrators	exponential clean-energy buildout and accelerated coal retirement where replacement capacity is available
Phase 2	2033–2038	Industrial conversion surge and fossil displacement endgame	≈1–2M/yr (declines as buildout stabilizes)	process and industrial retrofit engineers, plant operators, heavy-industry conversion specialists, MRV personnel, chemists, process engineers, microbiologists, metabolic engineers, and materials/nanomaterials scientists	completion of fossil displacement across power, industry, and heat

Education and Training Multipliers

Necessary Multipliers to Sustain Required Additions:

- Vocational electrical and grid programs: expand 2–3× globally.
- Power engineering and grid-specialist seats: expand 1.5–2×.
- Chemical science, process engineering, microbiology, metabolic engineering, and materials/nanomaterials tracks: expand ~2×.
- Binding international certification reciprocity and fast-track credentialing systems must be operational: by end-2026 (staffed, funded, issuing credentials), not merely announced.
- AI-assisted productivity gains: ~30–40% are assumed for design, permitting, scheduling, and construction management.

Core Constraint

If global workforce mobilization falls below ~2 million annual additions by ~2028, operational net zero by ~2037–2038 becomes structurally improbable even under favorable capital conditions.

Labor is the binding deployment variable.

Chapter 2 — Clean Energy Financing and Financial System Realignment

This chapter restructures capital flows so clean infrastructure becomes structurally financeable while fossil expansion becomes structurally unfinanceable. Its objective is not only to ensure that clean IRRs exceed fossil IRRs⁶ by the late 2020s, but to align finance with the four operational rules that govern the net-zero pathway: clean-only growth, deliverable clean scale, fossil exit, and reliability replacement. If clean projects consistently offer superior risk-adjusted returns, if financing architecture supports large-scale deployment across generation, storage, transmission, and firm clean resources, and if fossil persistence pathways are narrowed early, capital rotation becomes self-reinforcing and operational fossil displacement becomes more likely. Clean project returns rise as the cost of capital falls through de-risking and standardization, as revenues are stabilized through contracts and market design, and as delivery risk is reduced through permitting and interconnection reform. Financial realignment must also preserve equitable durability by protecting workers, reducing energy-burden risk, and lowering capital barriers in emerging markets rather than shifting transition costs onto those least able to absorb them. [Appendix E](#) provides the deeper mechanics, mobilization structures, and repricing architecture behind the categories summarized here. [Appendix F](#) illustrates how financing structures could adjust to better support the conditions required to achieve the accelerated timeline.

Operational Role of Finance

Chapter 2 governs two of the four operational rules directly and supports the other two indirectly.

It governs **clean-only growth** by making fossil expansion economically and politically harder to finance during the decisive build window, and by directing capital toward clean supply for incremental demand growth. It governs **fossil exit** by restricting refinancing, life-extension capital, and other persistence pathways that allow declining fossil assets to remain in operation.

It also supports **deliverable clean scale** by directing capital toward transmission, interconnection, storage, and other enabling infrastructure, and supports **reliability replacement** by financing the clean adequacy resources required to replace fossil generation in its routine system role. In this framework, finance is not a supporting condition only; it determines whether fossil assets persist and whether the clean system can scale to completion.

The four operational rules are defined in the [Introduction](#). This chapter specifies the financial conditions required to make them executable.

⁶ This is a playbook synthesis based on declining clean-energy costs, lower financing risk for mature clean technologies, and rising transition risk for fossil assets. The specific late-2020s timing is a playbook judgment rather than a directly adopted external forecast. See [IEA. \(2025\). World Energy Outlook 2025](#); [NGFS. \(2025\). Guide to Climate Scenario Analysis for Central Banks and Supervisors](#).

Financial Conditions for the Four Operational Rules

1. Clean-Only Growth

Capital expenditure and credit allocation must favor clean generation, storage, transmission expansion, and other firm clean resources for net load growth. The purpose is to prevent parallel fossil expansion during electrification, industrial growth, and data-center expansion.

2. Deliverable Clean Scale

Finance must support not only generation, but also transmission, interconnection, storage, transformers, and other enabling infrastructure. Overinvestment in generation without corresponding investment in deliverability raises congestion, curtailment, and fossil backfill risk.

3. Fossil Exit

Finance must eliminate the persistence pathways that allow fossil assets to remain economically viable after their utilization has weakened. That includes refinancing, life-extension capital expenditure, like-for-like fossil replacement, capacity arrangements, and subsidies or other public-support mechanisms that preserve fossil viability.

4. Reliability Replacement

Finance must direct capital toward the clean adequacy resources required to replace fossil generation in its routine system role. That includes short-duration and long-duration storage, demand flexibility, transmission reinforcement, and firm clean power and thermal flexibility where needed.

These conditions must be advanced in parallel. Constraining fossil credit without making clean replacement capital sufficiently available delays the transition.⁷ Financing generation while underfunding transmission, flexibility, or retirement governance has the same effect.

Capital Mobilization Architecture

- *Sovereign Green Acceleration Funds — anchor capital and de-risk early build*
- *Generate-style Infrastructure Finance Platforms — standardize project finance and speed deployment*
- *Advance Market Commitments (storage, LDES, industrial heat) — guarantee demand and pull supply chains forward*
- *Public Risk Guarantees (Global South) — lower WACC and unlock scale where capital costs are otherwise prohibitive*
- *Fossil Worker Transition Funds — protect wages and pensions and preserve equitable durability*
- *Utility Revenue Stabilization — prevent utility backlash and preserve reliability during fossil decline*

⁷ This reflects literature-grounded transition logic: finance must support clean replacement systems in parallel with constraining fossil persistence, or the transition slows through congestion, underbuild, or continued fossil dispatch. The parallel-execution framing here is playbook synthesis. See: [IEA. \(2025\). World Energy Outlook 2025](#); [NGFS. \(2025\). Guide to Climate Scenario Analysis for Central Banks and Supervisors](#).

Workforce continuity and transition protection are treated here as financing requirements; the labor mobilization architecture is developed in [Chapter 1](#), and fossil retirement sequencing is addressed in [Chapter 9](#).

Financial Repricing Tools

- Mandatory climate risk disclosure — *force repricing; expose transition risk.*¹⁶
- Higher capital requirements for fossil lending — *raise WACC; shrink fossil credit.*¹⁶
- Prohibition on refinancing sunset fossil assets — *stop life-extension; accelerate retirement.*
- Fiduciary duty expansion — *permit or require climate-aligned allocation.*
- Border carbon adjustments — *protect domestic transition; reduce leakage.*¹⁷
- Dividend-centered carbon pricing design — *maintain public support; offset regressive household energy burdens.*

Carbon pricing, border adjustments, and fossil-credit restrictions must be implemented with household protection, worker transition funding, and clean-replacement availability so they reduce fossil dependence without increasing energy insecurity, cost-shifting, or regional economic harm.

Strategic Function

Financial realignment is necessary but not sufficient on its own. It must both block fossil expansion and finance the rate of clean system growth required for operational fossil displacement. If capital markets conclude by ~2028 that fossil reinvestment has no durable return horizon or refinancing pathway,⁸ fossil expansion can decline rapidly even without perfect political coordination. But this outcome depends on finance doing more than repricing fossil risk. It must also support clean-only growth, deliverable clean scale, fossil exit, and reliability replacement together.

Conversely, if fossil expansion remains financeable through 2030, or if clean system growth remains capital-constrained, the ~2037–2038 pathway becomes structurally improbable.⁹

The transition must be rapid enough to structurally close fossil reinvestment pathways, but managed enough to avoid disorderly financial contagion during asset repricing.

⁸ This reflects stranded-asset and transition-risk logic in the literature. The mechanism is literature-grounded; the ~2028 threshold is a playbook judgment. See [NGFS. \(2025\). Guide to Climate Scenario Analysis for Central Banks and Supervisors.](#); [IEA. \(2025\). World Energy Outlook 2025.](#)

⁹ Continued fossil finance delays peak emissions and compresses the remaining decline window. That mechanism is literature-grounded; the 2030 and ~2037–2038 thresholds are playbook judgments. See [IEA. \(2025\). World Energy Outlook 2025.](#); [IPCC. \(2022\). Climate Change 2022: Mitigation of Climate Change.](#); [NGFS. \(2025\). Guide to Climate Scenario Analysis for Central Banks and Supervisors.](#)

Chapter 3 — Electricity Infrastructure: Deliverable Clean Scale

This chapter defines the infrastructure conditions required to translate clean generation growth into operational fossil displacement under the accelerated transition pathway. In this playbook, generation capacity alone is not sufficient. Clean electricity must be deliverable: it must be connected, transmitted, balanced, and available at load in the hours and locations where demand occurs, without systematically shifting costs, siting burdens, or exclusion onto already overburdened communities. It therefore governs the second operational rule: **deliverable clean scale**.

Clean generation that cannot be justly delivered will not displace fossil generation. If transmission expansion, interconnection, and grid reinforcement lag behind generation growth, the system experiences rising curtailment, localized congestion, and continued fossil dispatch to meet load. Under those conditions, renewable buildout coexists with fossil generation rather than displacing it.

The core requirement to meet the accelerated timeframe of late 2030s net zero is straightforward: transmission, interconnection, and grid reinforcement must justly scale in parallel with generation. This includes both bulk transmission and distribution networks, cross-border interconnection, and intra-regional grid upgrades. These investments are not secondary. They are the primary determinants of whether just clean capacity becomes usable electricity.

As noted above, these systems also determine who receives reliable clean power, who bears siting burdens, and who pays for expansion. Build speed must therefore be paired with fair cost allocation, host-community benefit, and implementation that does not reproduce energy exclusion or environmental burden-shifting.

Transmission expansion should be structured as regulated, rate-based infrastructure with transparent cost recovery and clear planning authority. Permitting acceleration should reduce delay and fragmentation without removing legitimate environmental review, Indigenous rights, host-community participation, or fair cost-allocation safeguards. Durable compensation, participation, and benefit-sharing frameworks are part of build speed, not obstacles to it.

These governance and institutional conditions are not secondary implementation details. They are among the primary determinants of whether transmission can scale at the rates required for the accelerated pathway. Where corridor authority, cost allocation, queue reform, permitting discipline, and durable host-community arrangements remain unresolved, transmission expansion does not scale smoothly enough to support deliverable clean growth.

The required grid expansion to support just deliverable clean scale to meet the accelerated transition pathway is:

- **~7–9% annual transmission capacity growth globally (2026–2035)**, corresponding to roughly 2–3× historic build rates¹⁰

¹⁰ This reflects literature-grounded estimates of the scale of transmission expansion required under high-renewables pathways. The 7–9% range and 2–3× historic comparison are playbook planning benchmarks rather than directly adopted single-source targets. See [IEA. \(2025\). *World Energy Outlook 2025*](#); [IRENA. \(2024\). *World Energy Transitions Outlook 2024*](#).

- **~3× HVDC backbone deployment relative to 2015–2025¹¹**, especially for interregional transfer and redundancy
- parallel expansion of substations, transformers, interconnection capacity, and distribution reinforcement

These rates are not supporting conditions only; they are part of the minimum scale required for the clean electricity system to displace fossil generation on the ~2037–2038 timeline. Current global transmission expansion rates (~2–4%) are below required levels. Achieving ~7–9% annual growth is not a gradual scaling process; it requires a step-change driven by permitting reform, corridor planning, cost-allocation resolution, and supply-chain expansion.

Under successful institutional resolution, transmission growth would more likely remain below required levels through the late 2020s and cross into the required range around ~2028–2031, rather than rising smoothly from current levels.

Risk Threshold

If transmission growth falls below ~5% annually on a sustained basis, curtailment rises, congestion becomes structural, and fossil capacity factors stabilize rather than decline. Under those conditions, clean generation remains partly undeliverable due to infrastructure constraints and the ~2037–2038 pathway slips materially.¹² If transmission growth does not approach required levels by the late 2020s, infrastructure lag becomes structurally incompatible with the ~2037–2038 pathway.

This dynamic is why transmission and interconnection buildout failure is ranked among the highest-risk failure modes in the playbook: when deliverability lags generation at system scale, clean growth can continue while fossil displacement stalls.

Milestones

- transmission corridor fast-tracking, including interregional interconnection
- HVDC backbone buildout across major regions
- substation modernization and transformer scaling
- inverter-based grid stabilization
- distribution grid reinforcement
- interconnection queue reform
- explicit procurement of clean flexibility so reliability does not default to fossil backup

Voltage upgrades can increase transfer capacity along existing corridors but do not substitute for geographic expansion of the network, interregional interconnection, or HVDC backbone buildout.

Transmission serves three critical functions:

1. **Resource delivery** — connecting high-quality solar and wind resources to load centers

¹¹ This reflects the literature-grounded role of long-distance transmission and interregional connectivity in high-renewables systems. The ~3× benchmark is a playbook planning estimate informed by these requirements. See [IEA. \(2025\). World Energy Outlook 2025](#); [IRENA. \(2024\). World Energy Transitions Outlook 2024](#).

¹² This reflects the system constraint that insufficient transmission expansion limits renewable integration, increases curtailment, and sustains fossil dispatch. The ~4% threshold is a playbook-defined failure boundary rather than a directly cited limit. See [IEA. \(2025\). World Energy Outlook 2025](#); [IRENA. \(2024\). World Energy Transitions Outlook 2024](#).

2. **System balancing** — smoothing variability across regions and time periods
3. **Reliability support** — reducing the need for local fossil backup by expanding access to remote clean supply

Without sufficient transmission, each region must rely more heavily on local firming resources, increasing the likelihood that fossil generation persists as the default adequacy mechanism.

Interconnection queues are a binding bottleneck in many jurisdictions.¹³ Long delays between project completion and grid connection reduce effective deployment rates and increase financing risk. Reform is therefore required to:

- accelerate queue processing
- standardize interconnection requirements
- prioritize projects that improve just system adequacy and deliverability
- align interconnection timelines with construction timelines

In addition to bulk transmission, distribution networks must be upgraded to handle:

- electrification of transport and heating
- distributed generation and storage
- increased peak load in urban and industrial areas

Failure to reinforce distribution networks can create local constraints that preserve fossil use even when bulk clean generation is available.

Cross-border and interregional interconnection are especially important in high-renewable systems. They reduce variability, lower system costs, improve reliability, and reduce dependence on local fossil backup. Regions with strong interconnection are better positioned to achieve operational net zero earlier because they can share resources and balance variability across larger geographic areas.

This chapter also directly supports the fourth operational rule: **reliability replacement**. Transmission and interconnection are not only delivery mechanisms; they are core components of just system adequacy. By expanding access to diverse clean resources, transmission reduces the need for fossil generation to provide balancing and reserve capacity. However, transmission alone is not sufficient. It must operate in concert with storage, demand flexibility, and firm clean resources. Chapters 5 and 6 define those requirements in detail.

Operational Sequence (2026–2030)

- **2026:** Establish national or regional transmission authorities, or equivalent planning bodies, for corridor designation, cost allocation, and rights-preserving permitting acceleration; publish a 10-year build plan aligned to renewable zones and load growth.
- **By end-2026:** Implement interconnection queue reform, including milestone-based entry, cluster studies, and standardized requirements; automate distribution-level interconnection where possible.

¹³ This reflects empirical observations across major power markets that interconnection backlogs delay project deployment and increase system integration risk. See [IEA. \(2025\). World Energy Outlook 2025.](#)

- **2026–2027:** Launch transformer and HV equipment expansion programs through standard specifications, aggregated procurement, factory expansion incentives, and strategic inventories.
- **2027–2028:** Begin HVDC backbone procurements and right-of-way acquisition for priority corridors; lock in multi-year rate recovery and community compensation packages.
- **2026–2030:** Coordinate workforce, financing, and fossil-retirement implementation and clean replacement and expansion frameworks with the safeguards for equitable durability, fair burden-sharing, and credible transition protection developed in the [Introduction](#), Chapters [1](#), [2](#), [9](#), [10](#), [11](#), [12](#), [13](#), [14](#), Appendices [E](#), and [M](#).
- **2028–2030:** Procure clean flexibility explicitly — storage, demand response, and firm clean resources — so reliability is maintained while fossil backup dispatch is phased down.

Summary Condition

The accelerated pathway requires just deliverable clean scale such that clean generation, transmission, interconnection, and distribution infrastructure scale together fast enough for clean electricity to reliably meet demand across time and geography, including during prolonged regional stress events, without routine dependence on fossil generation for local adequacy.

Chapter 4 — Solar and Wind Exponential Expansion

This chapter defines the solar and wind deployment trajectories required to force fossil capacity factors into structural decline before 2040 and to complete power-sector fossil phaseout by ~2037–2038 while supporting continued electrification after operational net zero. Solar and wind are the primary volume technologies in the pathway. Their role is not merely to grow quickly in isolation, but to scale fast enough to displace fossil generation operationally.

These growth trajectories only produce fossil displacement if four conditions hold together: incremental electricity demand is captured by clean supply, clean generation remains deliverable to load, fossil persistence pathways are narrowing, and reliability functions increasingly shift to storage, transmission, demand flexibility, and firm clean resources. Solar and wind growth therefore does not substitute for the rest of the system. It works only when the four operational rules are executed in parallel.

Rapid solar and wind deployment also depends on fair siting, predictable community-benefit arrangements, and parallel grid expansion. Deployment should also be structured so that air-quality, reliability, and affordability gains reach overburdened and underinvested communities early, rather than reproducing a pattern in which local siting burdens arrive faster than local benefits. Land-use competition is best reduced through early participation, compensation, and transmission coordination rather than delay-by-default. In the accelerated transition pathway, between ~2026 and ~2028, annual clean generation growth should exceed fossil generation growth, driving sustained declines in fossil utilization. Under sustained execution, this transition typically precedes a majority clean generation share (~50%) by roughly 1–3 years, with global clean electricity plausibly exceeding 50% by ~2028–2030.

Clean Energy Manufacturing Scale-Up

The required solar and wind trajectories are achievable only if manufacturing scales in parallel with deployment. That includes:

- solar modules, wafers, cells, inverters, mounting systems, and transformers
- wind turbines, blades, towers, nacelles, cables, substations, and offshore installation capacity
- associated logistics, ports, transmission equipment, and maintenance supply chains

Manufacturing should be treated as a deployment constraint rather than a downstream industrial effect. If generation targets are set without a corresponding manufacturing ramp, the build curve becomes financeable on paper but undeliverable in practice.

Solar PV

Baseline ending 2025: solar PV approached ~2.9 TW of cumulative installed capacity.¹⁹ Solar is the primary volume technology in the net zero pathway. The key metric is cumulative installed PV capacity, with annual additions rising fast enough to sustain compounding growth through the early 2030s and then tapering as the system approaches saturation. Concentrated solar thermal (CSP) is treated in [Chapter 6](#), where it is used primarily for dispatchable thermal energy and industrial heat in high-Direct Normal Irradiance (DNI) regions.

Solar Required

Solar must sustain¹⁴ approximately:

- **~24–27% CAGR through ~2032**
- **~16–19% CAGR through 2034**
- **~10–12% CAGR during 2035–2040**
- **~6–8% CAGR during 2040–2050**
- **~3–5% CAGR during 2050–2060**
- **~2–3% CAGR post-2060** as a long-term equilibrium range

This implies a smoothed but still front-loaded build curve: slower initial acceleration than a constant 30% path, but still sufficient to force fossil capacity factors into structural decline before 2040. Growth nearer the upper end of the early-phase range is more consistent with a high-execution pathway that keeps ~2037 in play, while growth nearer the lower end points translates more toward ~2038 even before the later taper begins.

Install Trajectory

Annual additions should reach:

- **≥1.5 TW/year by 2030**
- **≥2.2 TW/year by 2033**

This corresponds to roughly **~29–33 TW cumulative by ~2037–2038**. The upper end of the early growth range aligns more closely with a high-execution ~2037 outcome. The lower end is more consistent with a smoothed early build ramp and a more institutionally plausible ~2038 outcome.

Solar Risk Threshold

If solar falls below approximately **~20% CAGR before ~2032**, fossil generation persists and the ~2037–2038 pathway becomes unlikely¹⁵ absent compensating overbuild in wind, storage, transmission, and firm clean resources. Solar is the primary volume technology, but it cannot carry the system alone.

Wind

Wind is the reliability complement to solar at scale. It reduces curtailment and lowers storage and firming requirements by improving geographic and seasonal diversity, which reduces overall system cost. Wind should be treated as two distinct build streams with different constraints.

Onshore Wind (High-Volume, Fast-Build)

¹⁴ These growth ranges are playbook planning trajectories informed by the scale of solar expansion required in high-renewables pathways. They are not directly adopted from a single external scenario. See: [IEA. \(2025\). *World Energy Outlook 2025*](#); [IRENA. \(2024\). *World Energy Transitions Outlook 2024*](#). The trajectories are anchored to 2025 cumulative installed base as listed in [Appendix A: Solar and Wind Curve Baseline Inputs \(End-2025\)](#), and [Ember. \(2026\). *World adds a record-breaking 814 GW of solar and wind in 2025*](#).

¹⁵ The underlying system logic is literature-grounded: slower solar growth delays fossil displacement and shifts greater burden onto wind, storage, transmission, and firm clean resources. The ~20% threshold and its timing are playbook-defined failure boundaries. See: [IEA. \(2025\). *World Energy Outlook 2025*](#); [IRENA. \(2024\). *World Energy Transitions Outlook 2024*](#).

- Primary role: rapid capacity additions, geographic diversity, winter and shoulder-season output
- Binding constraints: permitting and community acceptance, transmission access, turbine supply chains

Offshore Wind (High-Capacity-Factor, Slower-Build)

- Primary role: coastal load matching, higher capacity factors, system firmness
- Binding constraints: port infrastructure, vessels, seabed leasing and permitting, interconnection, and longer lead times

Wind Required (Combined)

Wind must sustain approximately:

- **~12–15% CAGR through ~2032**
- **~9–11% CAGR through 2034¹⁶**
- a later taper thereafter, reaching roughly **~3.8–4.8 TW** of global installed capacity by the late 2030s

The upper end of the early range aligns more closely with a high-execution ~2037 outcome. The lower end is more consistent with a smoothed early build ramp and a more institutionally plausible ~2038 outcome. Wind grows more slowly than solar but remains essential for nighttime generation, seasonal balancing, geographic diversity, and reducing overdependence on storage and firm clean backfill in high-renewable systems.

Wind Risk Threshold

If wind deployment falls materially below this range through the early 2030s, more burden shifts onto solar overbuild, storage scale-up, transmission expansion, and firm clean generation. Wind underbuild does not necessarily halt the pathway, but it makes the system more expensive, more storage-intensive,¹⁷ and more dependent on flawless execution elsewhere.

Parallel Manufacturing Scale-Up

The deployment trajectory in this chapter requires complete manufacturing scale-up executed in parallel across both technologies.

Solar Manufacturing Scale-Up:

- wafers, cells, modules, inverters, trackers, and transformers
- rapid factory expansion and diversified module supply
- recycling and material efficiency improvements sufficient to reduce avoidable bottlenecks

¹⁶ These wind growth ranges are playbook planning trajectories informed by the role of wind in providing diversity, nighttime output, and seasonal balancing in high-renewables systems. They are not directly adopted from a single external scenario. See: [IEA. \(2025\). *World Energy Outlook 2025*](#); [IRENA. \(2024\). *World Energy Transitions Outlook 2024*](#). The trajectories are anchored to 2025 cumulative installed base as listed in [Appendix A: Solar and Wind Curve Baseline Inputs \(End-2025\)](#), and [Ember. \(2026\). *World adds a record-breaking 814 GW of solar and wind in 2025*](#).

¹⁷ This reflects literature-grounded system dynamics in which weaker wind growth increases reliance on solar overbuild, storage, transmission, and firm clean capacity. The threshold framing here is playbook synthesis. See: [IEA. \(2025\). *World Energy Outlook 2025*](#); [IRENA. \(2024\). *World Energy Transitions Outlook 2024*](#).

Wind Manufacturing Scale-Up:

- turbines, blades, towers, nacelles, offshore cables, and substations
- vessel and port capacity for offshore projects
- standardized procurement and accelerated component lead times

The operative principle is simple: deployment targets must be matched by manufacturing plans early enough that supply chains do not become the limiting factor in 2028–2033.

Operational Sequence (2026–2030)

- **2026:** Lock in accelerated solar and wind deployment pathways through auctions, PPAs, public procurement, and manufacturing commitments and additional financing mechanisms defined in [Chapter 2](#) and aligned to the clean-only growth rule
- **By end-2026:** Establish siting, compensation, and host-community frameworks that reduce delay without removing legitimate local participation; align renewable zones with transmission plans
- **2026–2027:** Launch module, inverter, turbine, blade, transformer, and related manufacturing expansion programs; secure port and vessel planning for offshore wind where relevant
- **2027–2028:** Reach a deployment pace consistent with **≥1.5 TW/year solar by 2030** and on-track wind growth in the required range; ensure interconnection and transmission build are not lagging generation commitments
- **2028–2030:** Maintain compounded solar and wind growth while coordinating with storage, transmission, and clean flexibility procurement so rising renewable shares translate into fossil displacement rather than curtailment and backup persistence

Summary Condition

Solar and wind exponential expansion is sufficient only when deployment is fast enough to force fossil capacity factors into structural decline, and when that deployment remains equitably deliverable, financeable, and reliably-compatible. In this playbook, solar provides the primary volume ramp, wind provides critical diversity and system support, and both must scale in parallel with grids, storage, and firm clean resources to complete fossil displacement on the ~2037–2038 timeline.

Chapter 5 — Energy Storage and Clean Flexibility Scaling

This chapter defines the scaling of energy storage, demand flexibility, and clean firming resources required to replace fossil generation in its system adequacy role. Solar and wind expansion alone does not complete fossil displacement. As variable renewable penetration rises, system reliability must increasingly be maintained by non-fossil resources.

This chapter therefore governs the fourth operational rule: **reliability replacement**.

Fossil generation persists not because it is efficient, but because it provides dispatchable capacity, reserves, and balancing. If these functions are not replaced, fossil capacity remains online as a structural backstop even in high-renewable systems. Reliability replacement is therefore not a secondary condition; it is a binding requirement for operational net zero.

Reliability replacement must not be achieved by concentrating storage siting burdens or reliability costs in already overburdened communities, or by allowing the benefits of clean flexibility to accrue mainly to higher-income users. Durable adequacy requires fair access, equitable cost allocation, and implementation that does not reproduce energy exclusion.

The system must transition from **fossil-backed adequacy** to **clean-backed adequacy**. That shift requires coordinated scaling of storage, transmission, demand flexibility, and firm clean generation.

Functional Role of Clean Flexibility

Clean flexibility replaces fossil generation across three distinct system functions:

1. **Short-duration balancing** — intraday variability, ramping, and frequency response
2. **Multi-day and weekly balancing** — weather variability and renewable droughts
3. **Capacity adequacy** — peak demand coverage and reserve margins

No single technology satisfies all three functions. Multi-day adequacy and peak capacity are first-class system constraints, not side conditions. These functions must be sufficient not only under average operating conditions, but during prolonged correlated stress events, including multi-day or multi-week periods of high demand and low renewable output. The system must therefore be built as a portfolio:

- **short-duration storage (batteries)** for intraday balancing
- **long-duration storage (LDES)** for multi-day and weather-driven gaps
- **demand flexibility** to reduce peak load and shift consumption
- **firm clean generation** where required for residual adequacy

Transmission expansion (Chapter 3) reduces total flexibility requirements by smoothing variability across regions, but does not eliminate the need for local adequacy resources. Storage deployment must scale with renewable buildout to limit sustained curtailment; otherwise, project economics weaken, deployment slows, and fossil balancing capacity remains in the system longer.

Storage Scaling Requirements

Short-Duration Storage (Batteries)

Battery storage is the primary resource for intraday balancing and peak shaving.

Required scaling:

- **~40–45% CAGR through ~2032**¹⁸
- continued expansion thereafter at declining growth rates as penetration increases

Deployment must track solar growth closely. As solar penetration rises, intraday price spreads widen, making storage economically self-reinforcing if markets are structured correctly.

Battery Storage Functional Requirement

- absorb midday surplus generation
- fully serve evening peaks, extending discharge into early night-time, with typical operation involving daily deep cycling
- provide frequency response and ancillary services
- reduce curtailment and improve renewable utilization

Risk Threshold

If battery deployment falls materially below this trajectory through the early 2030s:

- curtailment rises sharply
- peak demand is increasingly met by fossil generation
- fossil capacity factors stabilize rather than decline

Under those conditions, solar overbuild alone does not produce fossil displacement.

Long-Duration Storage (LDES)

LDES is required to manage multi-day variability and periods of low renewable output.

This includes:

- thermal storage systems
- pumped hydro expansion
- emerging electrochemical or mechanical storage technologies
- hydrogen-to-power where system-specific conditions support it

Requirement

LDES does not need to dominate total energy supply, but it must be:

- **commercially deployable before ~2030**¹⁹
- scaling meaningfully during the early 2030s

Its primary role is to prevent fallback to fossil generation during extended renewable shortfalls. This includes prolonged regional stress events in which peak demand remains elevated while wind and solar output are jointly constrained across multiple days or weeks.

¹⁸ This battery growth range is a playbook planning trajectory informed by the scale of storage required for intraday balancing in high-renewables systems. It is not a directly adopted single-source target. See: [IEA. \(2025\). World Energy Outlook 2025](#); [IRENA. \(2024\). World Energy Transitions Outlook 2024](#); [IEA. \(2024\). Batteries and Secure Energy Transitions](#).

¹⁹ The role of long-duration storage in covering multi-day variability is literature-grounded. The requirement that it be commercially deployable before ~2030 is a playbook planning benchmark. See: [IEA. \(2025\). World Energy Outlook 2025](#); [IRENA. \(2024\). World Energy Transitions Outlook 2024](#).

Risk Threshold²⁰

If LDES is not commercially available and deployable by ~2030, or if adequacy planning is based on average renewable conditions rather than prolonged correlated low-renewable stress events, system reliability defaults structurally to fossil backup during multi-day gaps, fossil capacity remains structurally necessary, and the ~2037–2038 pathway becomes unlikely.

Demand Flexibility

Demand flexibility reduces the total scale of required storage and firm capacity.

This includes:

- industrial load shifting
- smart EV charging
- building electrification with thermal storage
- dynamic pricing and automated demand response

Rapid EV uptake without managed charging shifts large new load into evening peaks, increases local grid stress, and can preserve fossil peaker capacity even where annual clean generation is sufficient.

Demand Flexibility Functional Role

- reduce peak load
- shift demand into periods of high renewable output
- reduce required reserve margins

Requirement

Demand flexibility must scale in parallel with electrification. Electrified systems without flexibility increase peak demand and reinforce fossil backup requirements. This is especially true for EV charging, where unmanaged load can create evening peak stress, local distribution bottlenecks, and new pressure for gas peaker retention or buildout.

Firm Clean Generation

Even with high storage and transmission buildout, some level of firm clean generation²¹ is required for system adequacy.

Firm Clean Portfolio:

- geothermal
- hydro (existing and expanded where feasible)
- nuclear (existing fleet retention and selective new build)
- dispatchable thermal generation using zero-carbon fuels where system-appropriate
- other miscellaneous clean firm energy such as marine

²⁰ This reflects the literature-grounded risk that inadequate long-duration balancing preserves fossil backup during extended renewable shortfalls. The ~2030 and ~2037–2038 thresholds are playbook judgments. See: [IEA. \(2025\). *World Energy Outlook 2025*](#); [IRENA. \(2024\). *World Energy Transitions Outlook 2024*](#).

²¹ This reflects the literature-grounded role of residual firm clean capacity in covering extended shortfalls and adequacy needs that storage and transmission alone may not fully resolve. See: [IEA. \(2025\). *World Energy Outlook 2025*](#); [IRENA. \(2024\). *World Energy Transitions Outlook 2024*](#).

Firm Clean Generation Functional Role

- provides residual capacity adequacy
- reduces total storage requirements
- stabilizes systems with high renewable penetration

The required share varies by region depending on:

- resource mix
 - transmission connectivity
 - demand profile
-

System Integration Requirement

Storage, flexibility, transmission, and firm clean resources must be planned as a coordinated system.

Overbuilding any one component without the others creates failure modes:

- generation without storage → curtailment and fossil backfill
- storage without transmission → localized congestion
- transmission without flexibility → higher peak requirements
- inadequate firm clean → persistent fossil reserve capacity
- electrified demand (EVs, heat) without managed load, distribution upgrades, and clean supply → peak amplification, local grid failure, and fossil peaker lock-in

Reliability replacement fails if these components are not scaled together. Rapid electrification must be synchronized with clean generation, flexibility, and distribution capacity, or it shifts emissions rather than eliminating them.

Risk Threshold

If clean adequacy resources (storage, flexibility, firm clean) lag generation build:

- fossil plants remain online for reserve margins
- capacity markets continue to favor fossil retention
- fossil exit slows or stalls
- unmanaged EV charging and delayed local grid reinforcement can keep fossil peakers online even when annual clean generation growth is strong.
- transport can appear to succeed while the power-sector timeline slips.
- multi-day adequacy and peak capacity are first-class constraints, not side conditions.

This is one of the most common failure modes in partial transitions.²²

Operational Sequence (2026–2030)

- **2026:** Establish market structures that value capacity, flexibility, and ancillary services from clean resources; remove market advantages for fossil reserve provision

²² This is playbook synthesis describing a recurring system pattern in which clean generation scales faster than adequacy replacement, leaving fossil plants online for reserve and balancing roles. See: [IEA. \(2025\). World Energy Outlook 2025](#).

- **By end-2026:** Implement procurement mechanisms for storage and clean flexibility alongside renewable auctions
 - **2026–2027:** Scale battery manufacturing and deployment pipelines; align interconnection processes with storage co-location where appropriate
 - **2026–2030:** Implement managed EV charging, workplace/fleet charging standards, and accelerate distribution-grid reinforcement (transformers, feeders, interconnection) to align transport electrification with clean supply and flexibility.
 - **2027–2028:** Bring first wave of LDES projects to commercial operation; establish standardized procurement and contracting frameworks
 - **2028–2030:** Shift system adequacy procurement from fossil capacity to clean capacity resources; ensure reserve margins are increasingly met by storage, flexibility, and firm clean generation
-

Summary Condition

Reliability replacement is achieved when storage, demand flexibility, transmission, and firm clean resources together provide just system adequacy without routine reliance on fossil generation. A system that reaches annual clean energy balance but still depends on fossil capacity during peak demand or multi-day low-renewable periods has not achieved operational net zero. Operational net zero therefore requires non-fossil adequacy under prolonged peak and low-renewable stress conditions, not only under average annual balance. Under these conditions, fossil plants no longer serve as the default balancing and reserve mechanism, and can exit the system without compromising reliability.

Operational net zero requires reliability replacement to ramp in parallel with clean generation scale, fossil exit, and clean-only growth during the 2026–2030 window.

Chapter 6 — Firm Clean Generation and Residual Adequacy

This chapter defines the role of firm clean generation in completing fossil displacement by providing residual system adequacy alongside maximally scaled storage, transmission, and demand flexibility. In high-renewable systems, most energy is supplied by solar and wind, but some level of firm capacity remains necessary to cover extended low-generation periods, seasonal imbalances, and extreme demand conditions. In this framework, firm clean generation is sized not around average conditions alone, but around the requirement to maintain adequacy through prolonged high-demand, low-renewable stress events without fossil fallback.

Firm clean generation is not the dominant source of energy in the net-zero pathway. It is a residual system component that ensures reliability without preserving fossil backup pathways. As storage and transmission scale and reach maturity, and as industrial conversion reduces continuous thermal demand on firm clean resources toward a residual level, its role shifts toward low-utilization adequacy. If overbuilt or used beyond that role, it can displace storage deployment, reinforce dispatchable system structures, and slow the transition to a more efficient, renewable-dominated system.

This chapter therefore supports both reliability replacement and fossil exit by ensuring that remaining adequacy needs are met with non-fossil resources that do not recreate long-lived fossil dependence.

Where firm clean generation supports heat, adequacy, or residual system reliability, deployment should reduce household energy burdens and avoid leaving renters, low-income households, public or social housing, and underinvested regions dependent on fossil backup because clean replacement or retrofit costs are hardest for them to absorb.

Functional Role of Firm Clean Generation

Firm clean generation serves three core functions:

1. **Residual Adequacy** — covering peak demand when storage and demand flexibility are insufficient
2. **Extended-Duration Backup** — supporting multi-day or seasonal renewable shortfalls
3. **System Stabilization** — providing inertia, voltage support, and dispatchable capacity where required

These functions are **residual**, not primary. The system should first minimize adequacy requirements through:

- transmission expansion (Chapter 3)
- variable renewable generation (Chapter 4)
- storage deployment (Chapter 5)
- demand flexibility scaling (Chapter 5)

Firm clean generation fills the remaining gap after these are scaled, not before.

Technology Scope

Firm Clean Technology Scope:^{[8.18](#)}

- **Geothermal** (where resource quality allows)
- **Hydropower** (existing fleets and limited expansion)
- **Nuclear** (existing fleet retention and selective new build)
- **Dispatchable Zero-carbon Thermal Generation**, including:
 - hydrogen-to-power
 - synthetic fuels where system-appropriate
- **Marine**

Technology choice is system-specific and depends on resource availability, cost, and institutional feasibility. No single technology is required globally, but all systems require some form of firm clean capacity.

Scaling Requirement

Firm clean generation does not need to match total system energy demand. It must instead provide:

- sufficient **capacity** to meet peak and residual adequacy needs
- sufficient **availability** during extended renewable shortfalls

Typical System Requirements:²³

- **~5–20% of peak demand** met by firm clean capacity, depending on system flexibility
- lower shares in highly interconnected systems with strong storage and demand response
- higher shares in isolated or resource-constrained regions

The objective is **minimum sufficient adequacy**, not maximum firm capacity.

Deployment Constraint

Firm clean generation is constrained by:

²³ This range is a playbook planning heuristic rather than a directly adopted universal target. It is intended to represent the minority share of peak-demand adequacy commonly associated with highly renewable systems after strong transmission, storage, and demand-flexibility scaling, with lower shares in highly interconnected systems and higher shares in isolated or resource-constrained systems. See: [IEA. \(2025\). World Energy Outlook 2025](#); [IRENA. \(2024\). World Energy Transitions Outlook 2024](#).

- long development timelines (especially nuclear and geothermal)
- capital intensity and financing risk
- permitting and public acceptance
- fuel supply chain readiness (for hydrogen or synthetic fuels)

Because of these constraints, firm clean generation must be:

- **planned early (pre-2030)**
- **procured explicitly through adequacy frameworks**
- **coordinated with storage and transmission buildout**

If deployment is left too late, systems default back to fossil capacity for adequacy.

Risk Threshold

If firm clean generation is not available at sufficient scale by the early–mid 2030s:²⁴

- fossil plants remain online for reserve margins
- capacity markets continue to procure fossil generation
- reliability concerns slow fossil retirement
- the ~2037–2038 pathway becomes unlikely

Over-reliance on firm clean generation creates a different failure mode:

- **underdeployment of storage and flexibility**
- **higher system costs**
- **slower renewable integration**

The correct balance is minimal but sufficient firm capacity, enabled by strong scaling of other system components.

Market Design Requirement

Firm clean generation must be supported by market structures that:

- value **capacity and availability**, not just energy
- reward **clean adequacy resources** over fossil backup
- prevent fossil capacity from undercutting clean alternatives

This includes:

- clean capacity markets or adequacy auctions

²⁴ The adequacy logic is literature-grounded: insufficient clean firm capacity can preserve fossil reserve margins and delay retirement. The early–mid 2030s timing and the implication for the ~2037–2038 pathway are playbook judgments. See: [IEA. \(2025\). World Energy Outlook 2025](#); [IRENA. \(2024\). World Energy Transitions Outlook 2024](#).

- long-term contracts for firm clean resources
- explicit exclusion or phaseout of fossil capacity from adequacy procurement

Without these reforms, fossil generation remains economically favored for reliability roles.²⁵

Operational Sequence (2026–2030)

- **2026:** Define just clean adequacy standards and eligible technologies; align capacity markets with net zero pathways
 - **By end-2026:** Begin procurement of firm clean capacity through long-term contracts or regulated mechanisms
 - **2026–2027:** Extend operating lifetimes of existing clean firm assets (hydro, nuclear) where safe and feasible⁸
 - **2027–2028:** Launch first wave of new firm clean projects (geothermal, nuclear, hydrogen-ready thermal where applicable)
 - **2028–2030:** Scale procurement to match declining fossil capacity; ensure adequacy margins are met without fossil backup
-

System Integration Requirement

Firm clean generation must be integrated as part of a coordinated system:

- overbuilding firm capacity reduces incentives for storage and flexibility
- underbuilding forces continued fossil reliance
- poor coordination increases system cost and delays fossil exit

Planning must treat solar, wind, storage, transmission, demand flexibility, and firm clean generation as a single, integrated adequacy portfolio rather than independent investments.²⁶

Summary Condition

Residual adequacy is achieved when storage, transmission, demand flexibility, and firm clean generation together meet peak demand and extended shortfalls without routine reliance on fossil generation.

Under these conditions:

- fossil plants are no longer required for capacity or reserves

²⁵ This reflects literature-grounded market design dynamics: adequacy and capacity arrangements can preserve fossil participation unless procurement rules explicitly reward clean adequacy resources and phase out fossil eligibility. The framing here is playbook synthesis. See: [IEA. \(2025\). World Energy Outlook 2025](#).

²⁶ This portfolio framing is playbook synthesis based on literature-grounded interactions among storage, transmission, demand flexibility, and firm clean adequacy. Planning these resources independently increases system cost and raises the risk of either fossil persistence or overbuild. See: [IEA. \(2025\). World Energy Outlook 2025](#); [IRENA. \(2024\). World Energy Transitions Outlook 2024](#).

- adequacy is maintained through clean resources
- fossil exit can proceed without reliability risk

Firm clean generation completes the system, but only if it remains a residual layer rather than a substitute for broader system transformation.

Chapter 7 — Industrial Electrification and Load Transformation

This chapter defines the transformation of industrial energy demand required to align with clean electricity supply. Electrification is not only a decarbonization strategy; it is a structural condition for operational fossil displacement. Without shifting industrial demand toward electricity, clean generation growth cannot fully displace fossil fuel use. Industrial load shape directly affects system adequacy requirements; poorly aligned electrification can increase peak demand and prolong fossil capacity, while well-aligned load enables faster fossil displacement. Industrial electrification also determines the scale of residual emissions: insufficient conversion and load transformation increase reliance on constrained residual pathways, tightening the feasibility of the ~2037–2038 outcome.

Industrial demand must therefore be reshaped to:

- shift from direct fossil combustion to electric processes
- align load profiles with renewable generation patterns
- enable load flexibility (including shifting, modulation, and interruptibility) rather than reinforcing fossil baseload demand

This chapter supports **clean-only growth**, **deliverable clean scale**, and **fossil exit** by ensuring that new and existing industrial demand does not create structural dependence on fossil fuels.

Industrial electrification should be sequenced to protect workers, reduce local pollution exposure, and preserve regional economic continuity, rather than shifting industrial decline costs onto workers and communities or exporting cleaner production while legacy harms remain concentrated elsewhere.

Functional Role of Industrial Electrification

Industrial electrification serves three system functions:

1. **Demand Substitution** — replacing fossil fuel combustion with electric processes
2. **Load Growth Alignment** — ensuring new demand is met by clean supply rather than fossil expansion
3. **Flexible Demand Shaping** — enabling industrial load to respond to renewable availability

Electrification increases total electricity demand, but if structured correctly, it also improves system efficiency, reduces fuel costs, and supports renewable integration.

Scope of Electrification

Industrial electrification includes:

- **low- and medium-temperature heat** via electric boilers and heat pumps
- **high-temperature processes** via:
 - electric furnaces (e.g., steel via EAF)
 - resistive and induction heating
 - plasma and emerging electrified process technologies
- **electrochemical production**, including:
 - hydrogen (for steel, chemicals, fuels)
 - electrified chemical pathways
- **mechanical drives and motors**, replacing fossil-driven systems

Not all processes can be directly electrified. For these:²⁷

- indirect electrification via hydrogen or synthetic fuels is required
- process redesign may be necessary

Industrial Conversion Requirement

Industrial electrification and associated hydrogen and synthetic-fuel pathways are not optional components of the accelerated pathway. Together, they represent a major share of incremental electricity demand—on the order of ~20,000–30,000 TWh annually by the late 2030s—and are required to complete fossil displacement in hard-to-abate sectors.

Within the ~20,000–30,000 TWh range, a substantial share—particularly hydrogen and synthetic-fuel production—must operate as flexible load that follows renewable availability. This flexibility is a core system condition: without it, industrial electrification increases peak demand and shifts additional burden onto storage and firm clean generation, tightening the feasibility of the pathway. The pathway does not require existing industrial systems to operate intermittently. Instead, it depends on the majority of new electricity demand—especially hydrogen and synthetic-fuel production—being designed to respond to periods of high renewable output. Utilities, market operators, and regulators should reinforce this behavior through tariff design, dynamic pricing, and market rules that make flexible consumption lowest-cost during periods of high renewable output, thereby reducing curtailment and improving renewable utilization.

The pathway does not require complete industrial conversion within the accelerated timeframe. It does require sufficient conversion to confine remaining emissions to genuinely hard-to-abate residuals. Early conversion should prioritize directly electrifiable processes and loads that can operate flexibly with renewable availability, so that underperformance in direct electrification does not spill into excessive hydrogen and synthetic-fuel demand.

Sufficient industrial conversion is determined by system capacity, not sectoral feasibility alone. In practice, global industrial decarbonization likely needs to reach on the order of ~70–75% or more by the late 2030s. This threshold reflects the limited capacity of residual pathways—including hydrogen, synthetic fuels, and cement and concrete CCU—and the constrained ability of firm-clean sources to displace fossil-based power and thereby avoid the emissions that would otherwise remain.

Residual pathways are constrained by physical, infrastructure, and scaling limits. Hydrogen and synthetic fuels are energy-intensive and depend on clean electricity supply. Carbon capture and

²⁷ This reflects literature-grounded sector differentiation: some industrial processes can be directly electrified, while others require hydrogen, synthetic fuels, or process redesign where direct electrification is not yet viable. See: [IEA. \(2025\). *World Energy Outlook 2025*](#); [IRENA. \(2024\). *World Energy Transitions Outlook 2024*](#).

utilization is applied to process emissions from cement and concrete. Carbon capture and storage is limited within the pathway primarily to non-fossil process streams and biogenic or waste-derived carbon flows—such as BECCS, and waste-to-energy with CCS—and is only compatible with continued fossil displacement and declining fossil use, not fossil fuel extension. Firm-clean sources such as geothermal and other bounded residual resources also remain limited in scale. These pathways therefore function as complementary residual solutions, not as primary substitutes for large-scale underperformance in direct industrial conversion.

If industrial conversion falls materially below this range, the residual burden exceeds what these pathways can plausibly support. This does not simply leave emissions unaddressed; it increases system energy requirements, expands hydrogen and synthetic-fuel demand, and tightens constraints on clean generation, storage, and transmission. Under these conditions, the ~2037–2038 pathway becomes increasingly unlikely without upward revision of system scale.

Conversely, if conversion is concentrated in sectors that are technically and economically viable for direct electrification, the remaining emissions are more likely to reflect genuine hard-to-abate residuals—such as cement process emissions, certain chemical pathways, and high-temperature industrial processes—where residual pathways can be applied within their practical limits.

In practice, the accelerated pathway may require a modest clean-firm adequacy margin above the strict residual minimum in order to absorb industrial-load rigidity, reduce system brittleness, and maintain reliable fossil displacement under imperfect execution conditions. However, firm clean should remain a minority contributor rather than becoming a substitute for storage, transmission, demand flexibility, or direct electrification.

At an illustrative back-of-envelope scale, each additional ~5,000 TWh/year of industrial load that fails to operate flexibly may require either roughly ~2.5–3 TW of additional variable-renewable overbuild, together with supporting storage and transmission, or roughly ~0.7–1.1 TW of additional firm-clean supply, depending on utilization and system design.

Load Growth Requirement

Electrification drives substantial load growth:

- industrial electricity demand must scale rapidly through the 2030s
- hydrogen production and synthetic fuel pathways introduce additional large-scale electricity demand
- combined industrial, hydrogen, and synthetic-fuel load can increase global electricity demand by on the order of ~20,000–30,000 TWh annually

Growth must be clean-supplied, not met by fossil generation.

Key Requirement

- all incremental industrial, hydrogen, and synthetic fuel load must follow the clean-only growth rule
- electrified industrial demand must be treated as a controllable, system-integrated resource rather than passive load

Failure Condition

- if industrial electrification, hydrogen production, or synthetic fuel demand increases faster than clean supply scales, fossil generation expands rather than declines
- if these loads operate inflexibly or without coordination, they increase storage requirements, grid stress, and the risk of fossil backfill

Electrification must therefore be synchronized with generation, transmission, and storage buildout, and coordinated with the geographic distribution of industrial clusters and fuel production systems.

Load Shape Transformation

Industrial electrification must not replicate fossil load profiles.²⁸

Instead, demand should be structured to:

- shift consumption toward periods of high renewable output
- reduce demand during system stress conditions
- provide controllable load where technically feasible

This includes:

- flexible operation of electrolyzers (hydrogen production)
- batch processing aligned with renewable availability
- thermal storage integration in industrial heat systems

Rigid, inflexible electrified demand creates:

- higher system costs
 - increased storage requirements
 - greater risk of fossil backfill
-

Infrastructure Requirement

Industrial electrification requires parallel infrastructure expansion:

- **transmission and distribution upgrades** to serve new industrial loads
- **interconnection capacity** for large industrial facilities
- **co-location strategies** with renewable generation where feasible
- **hydrogen and fuel infrastructure** for indirect electrification pathways

Industrial clusters are priority zones:

- ports
- steel and chemical hubs
- manufacturing corridors

Coordinated planning reduces cost and accelerates deployment.

²⁸ This reflects literature-grounded system design logic: electrified industrial demand can support renewable integration when it is flexible, shiftable, or thermally buffered rather than rigidly synchronized to legacy fossil operating patterns. The specific operational examples used here are playbook synthesis. See: [IEA. \(2025\). World Energy Outlook 2025.](#)

Risk Threshold

If industrial electrification lags:

- fossil fuel demand persists in hard-to-abate sectors
- clean generation is underutilized or diverted to other sectors
- full-system fossil displacement is delayed

If electrification proceeds without coordination:

- grid congestion increases
- fossil generation expands to meet new load
- system costs rise and political resistance increases

The pathway requires **synchronized scaling of load and supply**.

Industrial decarbonization is not interchangeable across pathways. Direct electrification is the primary and most scalable route; hydrogen and electricity-derived fuels are secondary pathways for processes where direct electrification is not viable; and CCS and biogenic pathways are constrained residual options. If direct electrification underperforms, the resulting increase in hydrogen, synthetic-fuel, and CCS demand rapidly exceeds practical deployment limits and raises overall system energy requirements, tightening or delaying the net-zero pathway.

Market and Policy Requirements

Industrial electrification requires:

- long-term power contracts (PPAs) aligned with clean supply
- carbon pricing and regulatory constraints (including clean air standards) applied to fossil combustion
- capital support for process conversion and retrofits
- standards for just clean production (e.g., green steel, green chemicals)

Demand-side policy must:

- favor electrified processes over fossil alternatives
 - prevent reinvestment in fossil industrial infrastructure
 - support early deployment despite higher initial costs
-

Operational Sequence (2026–2030)

- **2026:** Identify priority industrial sectors and clusters for electrification; align with transmission and generation plans
 - **By end-2026:** Establish just clean production standards and procurement commitments (e.g., green steel, green hydrogen)
 - **2026–2027:** Launch electrification retrofit programs and new-build standards for industrial facilities
 - **2027–2028:** Scale electrolyzer deployment and electrified heat systems; integrate with renewable supply contracts
 - **2028–2030:** Align industrial load growth with renewable expansion and flexibility markets; phase out new fossil process investment
-

System Integration Requirement

Industrial electrification must be coordinated with:

- **Clean Generation Scaling** (Chapter 4)
- **Transmission Expansion** (Chapter 3)
- **Storage and Flexibility Systems** (Chapter 5)

Uncoordinated electrification creates failure modes:

- electrified demand served by fossil generation
- localized grid constraints
- stranded industrial investments

Electrification is effective only when integrated into the broader system transformation.

Industrial electrification is not only a demand-side decarbonization pathway; it is also a structural driver of clean system scale. If industrial load fails to materialize, renewable buildout, transmission expansion, and storage deployment may slow accordingly, reducing overall system scale rather than creating surplus clean capacity.

Summary Condition

Industrial electrification is achieved when fossil fuel combustion in major industrial sectors is replaced by electricity or synthetic fuels, and when resulting load growth is met by clean supply without triggering fossil expansion.

Under these conditions:

- industrial demand reinforces clean system growth
- fossil fuel use declines structurally across sectors

- electrification becomes a driver of system optimization rather than a source of new emissions

Industrial transformation is not optional — it is a core condition for completing fossil displacement in the 2030s pathway.

Chapter 8 — Circular Materials & Recycling Systems

This chapter defines the circular materials and recycling systems needed to reduce primary resource pressure, stabilize clean-technology supply chains, and prevent critical-materials bottlenecks during the 2026–2038 build window. Circularity does not eliminate mining in the 2030s; it reduces long-run extraction pressure, supply volatility, and fossil feedstock dependence as end-of-life volumes rise.^{23,24} Local and regional governments must begin planning and procurement immediately because collection, sorting, permitting, siting, and processing lead times are long.

Circular-material systems must reduce extraction and waste without shifting the burdens of toxic handling, labor risk, unsafe processing, or environmental dumping onto workers, lower-income regions, or communities. Recycling scale-up should therefore be paired with strong labor and environmental protections, producer responsibility, traceability, and domestic or regionally accountable processing capacity. Petroleum-derived lubricants and polymers can be replaced over time by synthetic or bio-derived feedstocks, while high metal recovery rates and circular design reduce long-run primary extraction pressure after the initial buildout phase.

In the accelerated pathway, circularity is not only a waste-management or resource-efficiency objective. It is a timing-sensitive system constraint. If circular systems do not scale before early-2030s demand surges, material price volatility, component shortages, and deployment delays can propagate simultaneously across storage, EVs, grid buildout, and industrial conversion. The pathway is therefore sensitive not only to recovery rates by ~2035, but to whether collection, processing, and recycled-feedstock integration are operational before the peak build window tightens. This strengthens the role already assigned to Chapter 8 as both a critical-materials strategy and a fossil feedstock replacement strategy.

Targets by ~2035 (Decision-Relevant Floors)²⁹

Metals and Battery Materials

- **Lithium:** ≥40% recovered from end-of-life streams in advanced markets, with global average lower but rising.
- **Copper:** ≥30% recovered, with higher rates in building wire and grid equipment where collection is mature.
- **Aluminum:** ≥60% circularity in advanced markets through scrap recovery and recycled content.
- **Cobalt, nickel, graphite:** establish high-yield recovery pathways and expand processing capacity so recycling becomes a material supply source by the mid-2030s as EV batteries reach end of life. Rates vary by chemistry and region; priority should be high-value recovery with strong MRV.

²⁹ The recovery and circularity ranges used here are playbook planning benchmarks rather than directly adopted external targets. They are intended as decision-relevant floors, consistent with the role of recycling, materials efficiency, and circular design in easing long-run primary extraction pressure, reducing critical-mineral volatility, and stabilizing clean-technology supply chains as end-of-life volumes rise. See [IEA \(2023\). Critical Minerals Market Review](#); [IRP \(2024\). Global Resources Outlook 2024](#).

Polymers and Composites

- **Polyethylene and polypropylene:** prioritize mechanical recycling where feasible; scale chemical recycling only where it improves yield and product quality with credible lifecycle accounting.
- Prioritize packaging redesign, polymer standardization, and the development of bio-based, recyclable, biocompatible, and where feasible compostable polymer systems whose end-of-life conditions are credible, with the goal of moving beyond persistent petroleum-derived plastics toward end-of-life pathways that reduce contamination and micro- and nanoplastic waste.

Grid-Critical Components and Industrial Inputs

Circularity must extend beyond bulk materials to grid- and deployment-critical components, including transformers, copper-intensive electrical equipment, motors, inverters, and other power electronics. These components face both material and manufacturing bottlenecks. Recovery, refurbishment, remanufacture, and component reuse can reduce lead times and supply risk during the peak expansion period, especially where transmission and distribution buildout would otherwise compete directly with EVs, storage, and industrial electrification for the same upstream inputs. This is especially important because critical materials and component disruption is already identified in the playbook as a timeline-relevant failure mode.

System Target

Battery recycling must scale before ~2032 to reduce critical-mineral volatility and prevent supply constraints from slowing EV and storage deployment.³⁰ More broadly, circular-material systems must scale ahead of peak demand growth in the early 2030s. If recycling capacity, collection systems, sorting infrastructure, and recovered-material processing lag the 2029–2034 deployment surge, primary material demand increases sharply, raising costs, slowing deployment, and tightening supply chains across multiple sectors simultaneously.

Material efficiency and substitution are parallel levers to recycling. Reducing material intensity in clean-energy technologies, vehicles, grid infrastructure, and industrial systems lowers total supply requirements and can often be deployed faster than expanding extraction or building entirely new recycling capacity. Circularity in this framework therefore includes not only end-of-life recovery, but also design choices that reduce virgin-material demand in the first place.

Circular-material systems also reduce upstream industrial energy demand by lowering the need for primary extraction, refining, smelting, and high-energy feedstock processing. In this way, Chapter 8 supports Chapter 7 indirectly: it does not replace industrial electrification, but it can reduce the total electricity and process-energy burden imposed by raw-material throughput. Under accelerated transition conditions, this moderates pressure on clean generation, storage, and transmission systems by reducing avoidable material-processing loads before they enter the system at full scale. This matters because industrial load growth is already treated elsewhere in the playbook as one of the main adequacy-sensitive constraints.

³⁰ Battery recycling is expected to play an increasingly important role in moderating critical-mineral volatility and supporting EV and storage supply chains as end-of-life battery volumes rise. The “before ~2032” timing used here is a playbook planning benchmark, not a directly adopted IEA target. See [IEA \(2023\), Critical Minerals Market Review](#).

Circular-material systems also reduce exposure to geopolitical concentration, export restrictions, and supply-chain shocks by diversifying feedstock sources and making more of the clean-energy material base recoverable within accountable regional systems. This does not eliminate trade dependence, but it reduces fragility.

Longer-Run Goal

- Functional zero-waste systems in high-income nations by the early 2040s, defined by high capture rates, low contamination, and stable end markets.
- Support progress toward Global Goals Targets 12.4 and 12.5.

Risk Threshold

If recycling and circular-material systems do not scale by the early 2030s, critical-material volatility, component bottlenecks, and cost escalation can slow clean-energy buildout across multiple sectors simultaneously, raising deployment costs and pushing the ~2037–2038 pathway beyond the late-2030s window.³¹ This risk is amplified where recycling lags at the same time as grid expansion, storage buildout, and vehicle electrification intensify demand for the same materials and components.

Operational Sequence (2026–2030)

- **2026:** Local and regional governments launch circularity implementation plans covering recovery and collection expansion, sorting capacity, siting and permitting for recycling facilities, and procurement standards for recycled content where feasible.
- **By end-2026:** Implement extended producer responsibility frameworks for packaging and priority products; standardize labeling and material IDs; enforce compliance to reduce contamination.
- **2026–2027:** Identify grid-critical and clean-technology component streams for refurbishment, remanufacture, recovery, and strategic materials reuse, especially for copper-intensive equipment, transformers, batteries, and power electronics.
- **2026–2030:** Coordinate circular-material policy with industrial, storage, EV, and grid-expansion planning so that procurement, recovery capacity, and recovered-material availability reduce peak deployment bottlenecks instead of lagging them.
- **2026–2030:** Expand RD&D into fully biodegradable biopolymers, vitrimers, recyclable thermoset polymers, and other material-efficiency and substitution pathways, with the goal of improving end-of-life outcomes and reducing persistent plastic waste and micro- and nanoplastics.
- **2027–2028:** Expand battery recycling and materials recovery capacity across collection, transport, and processing, with clear safety standards, MRV, and offtake contracts. Facility siting, permitting, and procurement should favor accountable processing capacity

³¹ Slower growth in recycling and circular feedstock systems would increase dependence on primary material supply and can raise critical-mineral volatility, supply risk, and deployment costs. The “early 2030s” timing and the implication for the net-zero pathway used here are playbook judgments based on those system effects rather than directly adopted external thresholds. See [IEA \(2023\), Critical Minerals Market Review](#); [IRP \(2020\), Resource Efficiency and Climate Change: Material Efficiency Strategies for a Low-Carbon Future](#).

with strong labor, monitoring, and community-protection standards rather than shifting hazardous recovery burdens into weaker regulatory jurisdictions.

- **2028–2030:** Adopt circular design requirements in key product categories, including packaging, electronics, vehicles, and selected grid and industrial equipment, to increase recyclability, reduce unnecessary material diversity, and improve recovered-feedstock quality. Integrate recycled feedstocks into supply chains with verified accounting.

Summary Condition

Circular materials and recycling systems are sufficient for the accelerated pathway when they reduce primary extraction pressure, stabilize clean-technology and grid-component supply chains, lower fossil feedstock dependence, and scale early enough to prevent critical-material and component bottlenecks during the peak build window.

Under these conditions:

- recovered materials become a meaningful supply source before early-2030s demand surges fully tighten,
- material efficiency and substitution reduce total virgin-material requirements,
- upstream industrial energy demand is moderated,
- and circularity functions as a deployment stabilizer rather than a late cleanup layer.

In this playbook, circularity is not a substitute for mining, industrial conversion, or clean-energy buildout. It is a timing-critical support system that helps keep those other build and deployment curves from slipping simultaneously.

Chapter 9 — Fossil Retirement Governance

This chapter defines how fossil generation capacity is retired and how broader fossil-fuel dependence is forced into a shrinking residual consistent with the accelerated pathway. In the power sector, this means full fossil phaseout. Outside the power sector, it means rapid contraction of coal, oil, and gas use toward only the smallest physically justified residuals, rather than continued routine use preserved through fuel substitution claims, refinancing, or institutional delay. Fossil retirement and subsequent exit is not an automatic outcome of clean growth; it must be structurally enabled, actively governed, and synchronized with system readiness.³²

Fossil assets persist not because they are efficient, but because they retain access to refinancing, subsidies and other public support, capacity payments, reliability roles, and regulatory protection. Unless these persistence pathways are removed, fossil generation can remain operational even as utilization declines.³³

The Introduction and Chapters 2–7 establish the structural conditions for fossil displacement by aligning finance, clean-only growth, deliverable clean scale, reliability replacement, and load transformation. This chapter governs fossil exit directly by ensuring that, once those conditions are in place, fossil capacity is retired in an orderly, durable, and system-reliable manner rather than preserved through persistence pathways. Orderly retirement must also mean that frontline pollution burdens decline, worker and pension protections are credible, and fossil-dependent communities receive visible replacement investment rather than being left with stranded assets, lost revenue, and residual contamination.

Functional Role of Fossil Retirement Governance

Fossil retirement governance serves four system functions:

1. **Capacity Removal** — ensuring fossil generation is physically removed once replacement capacity is available
2. **Capital Discipline** — preventing reinvestment and eliminating persistence pathways
3. **System Stability** — sequencing retirement to avoid reliability risk
4. **Equitable Durability** — ensuring equitable workforce transition, regional and community reinvestment, burden-sharing, and institutional conditions that sustain retirement

Fossil retirement must be predictable, sequenced, and aligned with system readiness.

³² This reflects literature-grounded transition logic: declining fossil competitiveness does not guarantee exit when refinancing, adequacy payments, and institutional protections remain in place. The governance framing here is playbook synthesis. See: [IEA. \(2025\). World Energy Outlook 2025](#); [NGFS. \(2025\). Guide to Climate Scenario Analysis for Central Banks and Supervisors](#).

³³ This reflects literature-grounded stranded-asset and persistence-pathway logic: declining utilization alone does not force retirement where refinancing, capacity payments, or regulatory protections preserve asset viability. See: [NGFS. \(2025\). Guide to Climate Scenario Analysis for Central Banks and Supervisors](#); [IEA. \(2025\). World Energy Outlook 2025](#).

Governance Toolset

Effective fossil retirement governance requires a coordinated set of policy, financial, and regulatory tools:

- **Binding phaseout schedules** for coal, oil, and gas generation
- **Prohibition of new fossil capacity** for net load growth
- **Refinancing restrictions** and disclosure requirements for sunset assets
- **Capacity market reform** to exclude or phase out fossil eligibility
- **Securitization and managed phaseout mechanisms** to retire assets without disorderly financial loss
- **Public or quasi-public acquisition** where required to accelerate retirement
- **Demand-side phaseout policies** for fossil-dependent industrial and energy uses
- **Trade and compliance alignment**, including carbon border adjustments where applicable
- **Strict limits on conversion pathways**, preventing prolonged fossil dependence under alternative fuels

These tools must operate together. Partial implementation allows fossil assets to persist through remaining pathways.

Persistence Pathways

Fossil generation remains in the system through:

- refinancing and debt rollover
- life-extension capital expenditure
- capacity payments and reliability contracts
- regulatory protection and must-run status
- like-for-like replacement of retiring fossil assets

These pathways allow fossil assets to survive declining utilization and delay transition.

Eliminating these pathways is a primary requirement for fossil retirement.

Retirement Conditions

Fossil assets must be retired when three conditions are met:³⁴

1. **Clean supply sufficiency** — generation, storage, and transmission can meet demand
2. **Adequacy replacement** — clean resources can replace reliability functions
3. **Operational readiness** — interconnection, grid stability, and market structures are prepared

³⁴ The underlying sequencing logic is literature-grounded: fossil retirement must follow clean replacement and system readiness closely enough to avoid both adequacy failures and prolonged fossil persistence. The three-condition framing used here is playbook synthesis. See: [IEA. \(2025\). World Energy Outlook 2025](#); [IRENA. \(2024\). World Energy Transitions Outlook 2024](#).

Retirement before these conditions creates reliability risk.
Retirement after these conditions creates fossil persistence.

Sequencing Principle

Fossil retirement follows a sequenced decline:

1. lowest-utilization and highest-cost assets
2. mid-merit generation as clean supply expands
3. baseload fossil generation once adequacy is replaced

Gas peakers are typically last to retire unless replaced by storage, demand flexibility, or firm clean generation.

Fuel-Specific Retirement Dynamics

Coal

Coal should be eliminated first because it is the most emissions-intensive major fossil fuel, is often carried by an aging asset base, and in many systems is already losing economic competitiveness relative to clean power.³⁵ Coal retirement should therefore be front-loaded in the accelerated pathway: rapid phaseout in advanced economies, accelerated but managed phaseout in emerging markets, and early application of binding retirement schedules wherever clean replacement can be delivered reliably. Coal should not persist as a routine fuel outside tightly bounded transitional exceptions, and coal-to-biomass or other conversion pathways should be limited and tightly governed to avoid prolonging emissions, land-use pressure, or institutional delay.

Oil

Oil should not persist as a routine fuel in the accelerated pathway. Outside tightly bounded residual uses, oil demand should decline through electrification, demand reduction, modal shift, efficiency, and fuel substitution. Any remaining oil use should be confined to the smallest physically justified residuals, centered primarily in marine liquid fuels and a limited set of off-road or geographically remote heavy-duty uses, with structural decline underway and strategic reserves winding down rather than new long-lived expansion. Power-sector oil generation should be eliminated early where alternatives are available.

³⁵ This reflects literature-grounded prioritization based on coal's high emissions intensity, aging asset base, and declining competitiveness relative to clean power in many markets. The front-loaded retirement framing here is consistent with major mitigation pathways. See: [IPCC. \(2022\). *Climate Change 2022: Mitigation of Climate Change*](#); [IEA. \(2025\). *World Energy Outlook 2025*](#).

Gas

Gas should not persist as a routine fuel in the accelerated pathway, but it is the most complex major fossil fuel to retire because it remains deeply embedded in system adequacy, peak generation, and parts of industrial heat. Gas retirement must therefore be tightly synchronized with storage deployment, demand flexibility, transmission expansion, firm clean adequacy, and direct electrification wherever feasible. Unmanaged gas retention is the most likely source of fossil persistence in the 2030s,³⁶ especially where reliability markets, peaking roles, or industrial conversion delays preserve asset viability. Conversion of gas infrastructure to hydrogen or other fuels should be limited to cases where it clearly supports net-zero outcomes and does not prolong fossil dependency, asset life-extension, or institutional lock-in.

Asset Management and Financial Stability

Fossil retirement must avoid disorderly financial outcomes that could destabilize the transition.

Required mechanisms include:

- securitization of remaining asset value
- regulated asset base transitions
- managed retirement funds
- structured write-downs where necessary

The objective is to:

- accelerate retirement
 - prevent continued operation of uneconomic assets
 - maintain financial system stability
-

Workforce and Regional Transition

Fossil retirement must include:

- wage protection and income transition mechanisms
- pension protection and benefit continuity
- retraining and redeployment programs
- regional economic diversification

Fossil-dependent regions require:

- targeted infrastructure investment
- industrial transition strategies

³⁶ This reflects literature-grounded system logic: gas often remains embedded through adequacy, peak generation, and industrial heat roles unless storage, flexibility, firm clean capacity, and market reform scale in parallel. The specific emphasis on gas as the primary persistence risk is playbook synthesis. See: [IEA. \(2025\). *World Energy Outlook 2025*](#); [IRENA. \(2024\). *World Energy Transitions Outlook 2024*](#).

- long-term economic planning

Failure to address workforce impacts creates political resistance that can delay or reverse fossil retirement.

Market Alignment

Markets must reflect system transition requirements:

- capacity markets aligned with clean adequacy
- removal of implicit subsidies for fossil generation
- long-term visibility for clean investment
- predictable regulatory frameworks

Markets that continue to reward fossil reliability roles will slow or block retirement. Fossil capacity must be explicitly excluded or phase-limited in adequacy procurement frameworks.³⁷

Risk Threshold

If fossil retirement is not enforced:³⁸

- declining utilization does not lead to exit
- fossil plants remain available for dispatch
- clean generation coexists with fossil rather than achieving displacement
- the ~2037–2038 pathway becomes unlikely

If retirement is forced too quickly:

- adequacy gaps emerge
- political backlash increases
- reliability concerns slow transition

The pathway requires **synchronized fossil exit and clean replacement**.

Operational Sequence (2026–2030)

- **2026:** Establish fossil retirement frameworks, asset registries, and governance mechanisms

³⁷ This reflects literature-grounded market design dynamics: adequacy and capacity mechanisms can preserve fossil participation unless procurement frameworks explicitly reward clean adequacy resources and phase down fossil eligibility. The exclusion / phase-limit framing here is playbook synthesis.

See: [IEA. \(2025\). *World Energy Outlook 2025*](#).

³⁸ The underlying logic is literature-grounded: unmanaged fossil persistence delays displacement, while premature retirement without adequate clean replacement can create reliability and political risks. The ~2037–2038 implication is a playbook judgment. See: [IEA. \(2025\). *World Energy Outlook 2025*](#); [IPCC. \(2022\). *Climate Change 2022: Mitigation of Climate Change*](#).

- **By end-2026:** Implement refinancing restrictions and disclose asset-level retirement timelines
 - **2026–2027:** Begin retirement of lowest-utilization and highest-cost fossil assets; launch workforce transition programs
 - **2027–2028:** Align capacity markets and adequacy procurement with clean resources
 - **2028–2030:** Accelerate retirement of mid-merit and baseload fossil generation as clean adequacy scales
-

Operational Gating Factors

Fossil retirement is gated by:

- transmission expansion ([Chapter 3](#))
- clean generation scaling ([Chapter 4](#))
- storage and flexibility deployment ([Chapter 5](#))
- firm clean generation availability ([Chapter 6](#))
- industrial load transformation ([Chapter 7](#))

Failure in any of these areas delays retirement.

Summary Condition

Fossil retirement is achieved when fossil generation capacity declines in step with clean system expansion such that fossil plants are no longer required for energy, capacity, or reliability services, and when broader coal, oil, and gas dependence outside the power sector is forced into only the smallest physically justified residuals.

Under these conditions:

- fossil assets retire without creating adequacy gaps,
- capital no longer flows into fossil infrastructure,
- clean generation operates as the default system supply, and
- remaining fossil use is confined to shrinking residual applications rather than preserved through refinancing, subsidy, conversion, or institutional inertia.

Fossil retirement is both a structural outcome of system transformation and a governed process. It succeeds only when system transformation and institutional management operate in parallel.

Chapter 10 — System Constraints on Transition Speed

These constraints define the conditions under which transition timelines may slow, stall, or deviate from the late-2030s pathway.

[Appendices A–C](#) explore how such deviations may manifest over time, including cascading delays, system bottlenecks, and failure modes under constrained conditions.

While the preceding chapters describe the conditions required to achieve clean-only growth, deliverable clean scale, fossil exit, and reliability replacement, the speed at which these conditions can be realized is governed by additional system constraints.

These constraints do not change the direction of the transition and are not absolute barriers, but they determine whether it can proceed fast enough to reach operational net zero within the ~2037–2038 timeframe.³⁹ They become timeline-breaking only when they are left unmanaged. In practice, the accelerated pathway depends on identifying the binding constraint in each domain early enough to apply operational workarounds—through institutional simplification, fair siting frameworks, workforce scaling, persistence-pathway closure, and coordinated financial support—before delays compound across the wider system. They also determine whether the transition remains durable, since unmanaged siting conflict, workforce strain, cost shifting, and uneven benefit delivery can slow deployment even where technology and capital are available.

Institutional Throughput and Coordination

Transition speed is constrained by the ability of institutions to process, approve, coordinate, and enforce at the required pace.

This includes:

- permitting and environmental review capacity
- regulatory adaptation speed
- inter-agency coordination across energy, land-use, finance, and industrial systems

Even where capital and project pipelines are sufficient, limited institutional throughput can delay deployment and create bottlenecks that propagate across the system.⁴⁰

Operational Management Logic

The pathway does not require institutions to become frictionless. It requires throughput to increase fast enough in the binding nodes: permitting, interconnection, environmental review, procurement, and cross-agency coordination. In practice, this means standardization, staffing expansion, queue reform, milestone-based approvals, and AI-assisted administrative acceleration focused on the highest-delay functions first.

The main operational responses to these constraints are developed in [Chapters 1, 3, 13, and 14](#) with additional implementation detail in [Appendix D](#).

³⁹ This framing is playbook synthesis. It reflects the literature-grounded distinction between long-run transition direction and near-term transition speed, where institutional, physical, and coordination bottlenecks determine whether late-2030s timing remains plausible. See: [IEA. \(2025\). *World Energy Outlook 2025*](#); [IPCC. \(2022\). *Climate Change 2022: Mitigation of Climate Change*](#).

⁴⁰ This reflects literature-grounded deployment dynamics: permitting, review capacity, and institutional coordination can become binding constraints even when technology, capital, and project intent are present. The cross-system propagation framing here is playbook synthesis. See: [IEA. \(2025\). *World Energy Outlook 2025*](#).

Infrastructure Siting and Social Acceptance

Deployment rates may be constrained by siting friction and local acceptance, particularly for transmission, industrial infrastructure, and large-scale energy projects.

These constraints can:

- delay or block critical enabling infrastructure
- fragment deployment timelines
- increase reliance on suboptimal system configurations

Voltage upgrades can improve utilization of existing corridors but do not eliminate the need for new transmission buildout, leaving siting, permitting, and interconnection as persistent constraints on transition speed.

These constraints are not only procedural; they are governance and legitimacy constraints. Durable build speed requires fair participation, host-community benefit-sharing, and cost-allocation structures that do not shift burdens onto already overexposed populations. It also depends on credible transition protection and visible health, resilience, and economic gains for communities historically overexposed to pollution and exclusion.

Where not addressed, siting constraints can limit the ability to achieve deliverable clean scale even under favorable economic conditions.

Operational Management Logic

The pathway does not avoid siting conflict by bypassing participation. It contains delay risk by pairing accelerated buildout with fair cost allocation, early host-community participation, durable compensation, visible local benefit, and routing discipline that reduces avoidable conflict. Where this governance layer is absent, build speed slows even under favorable economics.

The relevant governance and implementation responses are developed in the Introduction and in [Chapters 3, 8, 11, 12, and 13](#).

Workforce and Industrial Capacity

The transition requires simultaneous scaling of construction, manufacturing, and system integration capacity.

Constraints may arise from:

- availability of skilled labor
- training pipeline throughput
- contractor and project management capacity
- coordination across parallel infrastructure buildouts

Workforce and industrial limitations can reduce deployment rates even when capital, demand, and policy conditions are aligned.

Operational Management Logic

These constraints are managed not by waiting for labor markets to clear, but by front-loading training capacity, credential reciprocity, worker mobility support, instructor expansion, standardized project delivery, and selective use of AI-assisted productivity gains. The pathway remains plausible only if labor and industrial bottlenecks are relieved before peak build years rather than after they become binding.

Workforce and industrial-response architecture is developed primarily in [Chapters 1, 7, and 8](#), with additional workforce detail in Appendix D.

Incumbent System Persistence

Legacy fossil systems may persist through financial, operational, and governance pathways even after utilization declines. These include refinancing and debt rollover, life-extension capital expenditure, capacity payments and reliability contracts, regulatory protection and must-run status, like-for-like replacement, subsidy or other public-support mechanisms, and conversion pathways that prolong fossil dependence under alternative-fuel claims. As a result, fossil capacity does not exit automatically in response to declining competitiveness. Active displacement and retirement-governance mechanisms are required to prevent these persistence pathways from delaying system transition.⁴¹ See [Chapters 2 and 9](#).

Operational Management Logic

Incumbent persistence is contained only when declining competitiveness is paired with active closure of refinancing, life-extension, adequacy-market, and contractual persistence pathways. Without those interventions, clean growth and fossil survival can coexist for too long to preserve the late-2030s window.

Interaction with Financial Dynamics

These constraints interact directly with financial conditions.

- Delays in permitting or construction can increase project risk and raise the cost of capital
- Workforce and supply bottlenecks can slow capital deployment and reduce system throughput
- Persistent fossil capacity can weaken financial repricing signals and delay asset retirement

As a result, financial realignment alone is insufficient unless physical, institutional, and social constraints are addressed in parallel.⁴²

Operational Management Logic

Financial realignment works only when capital repricing is synchronized with physical throughput, workforce readiness, and retirement governance. In practice, this means reducing avoidable project delay, protecting clean-project bankability, and preventing fossil persistence from weakening repricing signals during the decisive 2026–2030 window.

These cross-domain interactions are treated most directly in [Chapters 2, 3, 5, 7, and 9](#). Detailed financial mechanisms are further developed in [Appendices E and F](#).

Implication for Transition Pathways

A late-2030s operational net-zero pathway does not require these constraints to disappear. It requires them to remain governable within bounded ranges through targeted operational

⁴¹ This reflects literature-grounded persistence-pathway logic: declining competitiveness alone does not guarantee fossil exit where refinancing, retrofit, contractual, or institutional supports remain available. See: [NGFS. \(2025\). Guide to Climate Scenario Analysis for Central Banks and Supervisors](#); [IEA. \(2025\). World Energy Outlook 2025](#).

⁴² This is playbook synthesis built on the literature-grounded interaction between financing conditions, permitting and construction delays, workforce bottlenecks, and fossil persistence. See: [IEA. \(2025\). World Energy Outlook 2025](#); [NGFS. \(2025\). Guide to Climate Scenario Analysis for Central Banks and Supervisors](#).

management. Institutional throughput must rise in the most delay-sensitive functions; siting conflict must be reduced through legitimate participation and fair benefit-sharing; workforce and industrial bottlenecks must be relieved before peak build years; incumbent fossil persistence must be actively constrained; and financial repricing must remain synchronized with physical delivery. If these conditions are not maintained in parallel, delays compound across the system and the pathway shifts beyond the ~2037–2038 window.

Chapter 11 — Planetary Biosphere Restoration

This chapter addresses the restoration, protection, and long-term stewardship of the biosphere as a foundational condition of planetary stability and human habitability. The biosphere is not a bounded subsystem or a sector alongside energy and industry; it is the living fabric within which climate, biodiversity, ecosystem resilience, and the broader conditions of planetary health and safe Earth-system function are embedded.

Disruption to the biosphere propagates across these systems simultaneously, amplifying both physical and social risk. Restoration therefore cannot be treated as a narrow carbon intervention. It is a systems objective: stabilizing ecological function while enabling long-term recovery of degraded oceans, lands, and their ecosystems. While this chapter is necessarily aspirational in scope, it remains grounded in operational constraints and sequencing consistent with the broader playbook.

Restoring and conserving biomes and ecological carbon sinks is integral to rebuilding planetary resilience and helping return the Earth system toward a safe operating space within planetary boundaries, particularly by strengthening biosphere integrity, land-system function, freshwater resilience, and coastal stability.^{6,28} Restoration, rewilding, and conservation should begin immediately, prioritizing regenerative methods, co-benefits, and long-term sustainability.^{26,27} These efforts proceed in parallel to fossil phaseout but remain operationally distinct from carbon land-use accounting in order to preserve the smallest feasible operational net zero.

Science-based restoration, rewilding, and conservation should be pursued in partnership with Indigenous peoples, local communities, and other historically marginalized groups wherever possible, with durable local stewardship institutions treated as implementing authorities rather than as an afterthought to centrally procured land treatment.²⁹

This chapter aligns most directly with SDGs 2, 6, 13, 14, and 15, and supports resilience, livelihoods, and local governance outcomes across the broader 2030 Agenda. A full playbook-to-SDG mapping is provided in [Appendix H](#).^{30,31}

Within the net-zero pathway, biosphere restoration is not a substitute for fossil emission reduction. It operates in parallel, with distinct functions, timelines, and success criteria.

Functional Role of Biosphere Restoration

Biosphere restoration serves three primary functions:

1. **Resilience stabilization** — restoring ecosystem function to reduce climate vulnerability and systemic risk
2. **Carbon cycle support** — enhancing natural carbon sinks through ecosystem recovery, without substituting for emissions reduction
3. **System co-benefits** — strengthening water systems, soil health, biodiversity, and regional climate stability

These functions are interdependent. Restoration actions that degrade biodiversity or water systems in pursuit of carbon outcomes are inconsistent with this framework.

Separation from Offsetting and Accounting

Biosphere restoration must be structurally separated from carbon offsetting frameworks.

- Restoration outcomes are **not fungible** with fossil emissions
- Nature-based carbon storage is **reversible and uncertain**, particularly under warming conditions
- Using restoration as an offset mechanism introduces incentives for delayed fossil phaseout

Accordingly:

- Restoration must not be used to justify continued fossil emissions
- Carbon accounting frameworks must treat biosphere carbon separately from fossil carbon
- Claims of “net zero” must be based on **actual fossil emission reduction**, not offset substitution

This separation is a necessary condition for both ecological integrity and credible decarbonization.

Relationship to Carbon Dioxide Removal (CDR)

Biosphere restoration and CDR are distinct but related. Restoration enhances natural carbon uptake within ecological limits through ecosystem recovery, resilience-building, and associated co-benefits. CDR, by contrast, is reserved for post-operational-net-zero drawdown of remaining residual emissions in the hardest-to-abate sectors and, depending on societal choice, some or all historical emissions.

Restoration should be prioritized for ecosystem recovery, resilience, and co-benefits, including through afforestation, reforestation, and soil-based carbon uptake.

Blurring these roles risks overestimating biosphere capacity and delaying necessary emissions reductions.

Scope of Restoration

Restoration includes:

- **forest ecosystem recovery**, including protection of intact forests and reforestation where appropriate
- **wetlands and peatlands**, which provide high carbon density and hydrological regulation
- **coastal and marine systems**, including mangroves, seagrasses, and salt marshes
- **soil and agricultural systems**, including regenerative practices that improve soil carbon and fertility
- **freshwater systems**, including watershed restoration and flow management

Restoration must prioritize:

- ecosystem integrity over monoculture expansion
- native species and biodiversity
- long-term ecological health and resilience

Constraints and Risks

Biosphere restoration is constrained by:

- land availability and competing land-use demands
- climate feedbacks and destabilizing impacts (e.g., ice-albedo loss, wildfire, permafrost thaw)
- governance and land tenure complexity
- measurement uncertainty and monitoring limitations

Key risks include:

- over-reliance on restoration for carbon mitigation
- ecological degradation from poorly designed interventions
- displacement of local communities or land-use conflict
- reversal of stored carbon due to climate-driven disturbance events

Restoration strategies must be robust to these constraints.

Justice and Stewardship

Biosphere restoration must be grounded in principles of:

- **indigenous rights and land sovereignty**
- **equitable benefit-sharing**
- **local governance and participation**
- **protection against land dispossession and exploitation**

Many of the world's remaining intact ecosystems are stewarded by indigenous and local communities. Effective restoration depends on strengthening, not displacing, these governance systems.

Operational Sequence (2026–2035+)

- **2026–2027:** Establish protection of intact ecosystems as a first-order priority; halt net deforestation and major ecosystem degradation
- **2026–2028:** Scale restoration programs in degraded lands, wetlands, and coastal systems with strong governance and monitoring frameworks
- **2027–2030:** Integrate restoration with agricultural systems through soil and land management practices
- **2030 onward:** Expand restoration to support long-term ecosystem recovery and resilience, aligned with post-net-zero drawdown strategies

Restoration begins immediately but scales over decades throughout this century and is therefore not bounded by the operational net-zero timeline.

Risk Threshold

If biosphere restoration is misapplied:

- fossil emissions may be masked rather than actually reduced
- ecosystems may be degraded through poorly designed interventions
- carbon storage may be reversed under climate stress

If restoration is under-scaled:

- planetary health and resilience declines accelerate
- biodiversity loss continues
- adverse climate impacts intensify

The pathway requires **high-integrity restoration that is clearly separated from fossil accounting**.

System Integration Requirement

Biosphere restoration must be integrated with:

- **fossil emission reduction** ([Chapters 2–9](#)), which remains the primary driver of climate stabilization

- **agricultural and land-use systems** ([Chapter 12](#)), which determine large portions of biosphere outcomes
- **carbon dioxide removal strategies** ([Appendix M](#)), which operate alongside restoration but with distinct roles

Integration must preserve functional carbon and emissions accounting boundaries between these systems.

Summary Condition

Biosphere restoration is achieved when ecosystems are protected, degraded systems are on a sustained recovery trajectory, and natural carbon sinks are strengthened without substituting for emissions reduction.

Under these conditions:

- ecological resilience improves across regions
- biodiversity loss is slowed and partially reversed
- natural systems contribute to climate stabilization without masking fossil emissions

Biosphere restoration is a long-duration planetary system objective. It begins within the net-zero pathway but extends far beyond it, durably shaping the stability and habitability of the climate system throughout centuries.

Chapter 12 — Managed Land Systems

Land use, forestry, and agriculture (AFOLU) play a distinct role in the transition to net zero. Unlike the energy system, where transformation depends on large-scale infrastructure turnover, most land-sector mitigation pathways are already technically available and widely demonstrated. The central challenge is not technological feasibility, but the alignment of economic incentives, governance systems, and land management practices with known sustainable outcomes.

The transition in land systems therefore consists primarily of practice change at scale. Established interventions—including avoided deforestation, reforestation, improved forest management, soil carbon enhancement, agroforestry, regenerative and conservation agriculture, and methane reduction in livestock and rice systems—can materially reduce emissions while improving ecosystem resilience.^{32,33,28} Many of these approaches are well-documented in the literature and in applied frameworks such as [Project Drawdown](#);³⁴ however, they remain unevenly implemented due to structural economic and policy constraints rather than technical limitations.

This creates a fundamental tension within the sector. In many regions, particularly in industrialized agricultural systems, prevailing production models prioritize yield maximization through input-intensive practices, including synthetic fertilizers and pesticides. While these systems can deliver high short-term output, they often degrade soils, reduce biodiversity, and increase long-term vulnerability to climate and water stress. Transitioning toward more sustainable land management requires reconfiguring these systems to maintain productivity while reducing environmental impact, which in turn depends on changes in pricing structures, subsidy regimes, supply chains, and risk allocation for producers.

At the same time, land systems are not only biophysical systems but also social and cultural ones. Indigenous peoples and local communities manage a substantial share of the world's remaining intact ecosystems and consistently demonstrate outcomes associated with higher biodiversity, lower deforestation rates, and greater ecosystem resilience. Effective land-sector transition pathways therefore depend on securing land tenure, respecting Indigenous sovereignty, and supporting locally adapted stewardship practices. Without these conditions, large-scale land interventions risk undermining both ecological integrity and social equity. In this context, the land sector transition is best understood as a reconfiguration of management systems rather than a technology substitution problem.

Role in the Net Zero Framework

Emissions Reduction (Primary Role)

Reducing ongoing land-use and agricultural emissions is the first-order contribution of managed land systems:

- avoided deforestation and land conversion
- peatland protection and rewetting
- methane reduction in livestock and rice systems
- improved fertilizer management and nitrous oxide reduction

- regenerative practices and soil-restoration measures, including cover crops, reduced tillage, and organic amendments

These are mitigation actions and should scale in parallel with fossil emissions reductions. Eliminating net land-use change emissions requires halting net expansion of agricultural land into forests and other high-carbon ecosystems, with any residual conversion kept exceptional, minimized, and counterbalanced through restoration and avoided conversion elsewhere.

System Productivity and Resilience (Core Function)

Managed land systems must also maintain food production,⁴³ water regulation, and ecosystem function under climate stress:

- regenerative organic and conservation agriculture
- agroforestry and mixed-use systems
- watershed and hydrological management

These measures improve yield stability, drought resilience, and ecological function, and should be treated as part of long-run system stabilization rather than as carbon accounting first.

Carbon Storage and Land-Based CDR (Conditional Role)

Managed land systems can store carbon in biomass and soils and, under bounded conditions, contribute to removals:

- reforestation of degraded lands
- afforestation only where ecologically appropriate
- soil carbon accumulation
- biomass recovery in working landscapes
- enhanced rock weathering and biochar soil amendments

However, only a limited subset of these pathways qualifies as CDR in this playbook, because durability, MRV quality, reversibility risk, land competition, and ecological constraints vary substantially across methods. Where durability, MRV, sustainability, or justice conditions are not met, carbon gains should be treated as co-benefits rather than counted removals.^{35,36}

CDR Eligibility Criteria (Land-Based Systems)

Land-based carbon removal is counted as CDR **only where all of the following are met**:

- **Durability:** Storage is expected to persist on climate-relevant timescales, with monitoring and reversal-risk management in place

⁴³ This playbook does not explicitly model synthetic meat or other advanced protein substitutes as part of the ~2037–2038 net-zero pathway. However, over the longer run, their development and availability could support more sustainable agriculture by reducing methane emissions, land-use pressure, and reliance on conventional livestock production.

- **Verifiability (MRV):** Measurement, reporting, and verification are robust, transparent, and independently auditable
- **Additionality:** Carbon gains are not attributable to existing baseline land-use trends or policy requirements
- **Sustainability:** Land use does not conflict with food systems, biodiversity, or water constraints
- **Justice and accountability:** Governance ensures transparent siting, meaningful community participation (including free, prior, and informed consent where applicable), durable benefit-sharing, and clear liability for environmental harm, reversal risk, and public-health impacts.

Justice and accountability are required for all claimed climate benefits. Where these conditions are not met, carbon gains are treated as co-benefits rather than counted towards removals.

Guardrails and Constraints

To prevent misuse and unintended impacts:

- **No fossil offsetting:** Land-based removals must not be used to offset ongoing fossil emissions or delay fossil phaseout
 - **Food-first constraint:** Land use must not compromise food security or agricultural productivity
 - **Biodiversity protection:** Avoid monoculture expansion and practices that degrade ecosystem integrity
 - **Water system integrity:** Land-use change must not increase water stress or degrade watershed function
 - **Reversibility risk:** Biomass and soil carbon are vulnerable to fire, drought, pests, and land-use change
 - **MRV limitations:** Measurement uncertainty remains significant across many land-based pathways
 - **Demand-side pressure:** livestock production, feed demand, and material consumption must stabilize or decline sufficiently to avoid continued expansion pressure.
 - **Leakage risk:** without controls, improvements in one region can shift land-use pressure to others, undermining net emissions reduction.
 - **Land governance requirement:** enforcement, tenure clarity, and Indigenous stewardship are required to prevent continued land conversion despite improvements in agricultural practice.
 - **Supply-chain traceability:** required to prevent deforestation leakage across regions and markets.
-

Relationship to CDR and Restoration

Managed land systems sit between mitigation, CDR, and biosphere restoration:

- **Not a substitute for mitigation:** fossil emissions reductions remain primary

- **Not equivalent to durable CDR:** most land storage lacks the permanence and verification needed for large-scale removals accounting (see [Appendix M](#))^{35,36}
- **Complementary to restoration:** managed systems overlap with restoration, but remain distinct from conservation-first restoration pathways (see [Chapter 11](#))

Within managed land systems, CDR methods centered on ecosystem management and biomass conversion remain supplementary and bounded rather than a primary removal wedge. Biochar and geochemical CDR are treated separately because they are governed by different durability, accounting, and systems constraints.

Operational Priorities

2026–2030: stop loss and shift incentives

- halt deforestation and protect high-carbon ecosystems
- establish land-use MRV baselines and monitoring systems
- scale regenerative, organic, and soil-restoration programs
- expand agroforestry, mixed-use systems, and habitat corridors

2030–2037: scale resilient land management

- deploy landscape-scale reforestation and watershed restoration
- transition major agricultural regions toward soil-positive, lower-input systems
- integrate land systems into water, resilience, and climate adaptation policy
- reduce agricultural methane and nitrous oxide at scale

Post-net-zero: stabilize and steward

- maintain and improve soil carbon and biomass stocks
 - integrate managed lands with broader biosphere recovery
 - strengthen long-term stewardship institutions and land governance
 - continue productivity gains without expanding land footprint
-

Chapter 13 — Corporate and Government Acceleration

Large corporate buyers, public-sector purchasers, hyperscalers, industrial firms, and sovereign or subnational governments can accelerate the net-zero pathway by moving faster than broad political consensus. They shape capital allocation, procurement norms, supplier investment, and infrastructure timelines at scales that directly influence system outcomes.

Their role is not simply to signal demand for clean energy or low-emissions goods. It is to accelerate the physical system transition—bringing forward clean generation, grid infrastructure, industrial conversion, and workforce capacity—while ensuring that new demand does not outpace the system’s ability to supply it cleanly and reliably.

In the accelerated ~2037–2038 pathway, these actors matter because they can coordinate action across multiple system components simultaneously. But that acceleration is only beneficial when it remains within the system’s capacity to deliver on its core requirements: clean supply growth, deliverability, reliability replacement, and fossil fuel exit. When large actors move demand ahead of those conditions, they do not accelerate the transition—they risk pushing demand ahead of clean system readiness and increasing the likelihood of fossil fallback.

Historically, these actors have driven incremental decarbonization through procurement and voluntary commitments, but without coordinated alignment across infrastructure, reliability, and fossil retirement, such efforts have more often produced sector-specific gains than broad system transformation.

Core Principle

Corporate and government acceleration is effective when demand-side coordination converts voluntary action into default market structure while remaining synchronized with the physical system and without shifting transition costs or exclusionary pressures onto smaller firms, workers, households, or host communities.⁴⁴

In practice, this requires three linked functions:

1. **Commit demand ahead of supply** to bring forward investment in clean generation, storage, and low-emissions industrial production.
2. **Standardize and verify “clean”** through shared procurement rules, traceability, credible accounting frameworks, and targeted support pathways that enable smaller firms to meet full compliance requirements.
3. **Gate demand growth** so that new load or consumption expands only alongside deliverable clean supply and system adequacy, without shifting costs or reliability risks onto households, workers, or host communities.

These functions allow large actors to accelerate the transition without waiting for universal consensus. But they only succeed when they reinforce system alignment rather than allowing

⁴⁴ This framing is playbook synthesis. It reflects the literature-grounded role of large buyers, procurement standards, and coordinated demand signals in accelerating clean deployment, while emphasizing that transition speed and market leverage should not depend on cost-shifting onto smaller firms, households, workers, or host communities. See: [IEA. \(2025\). *World Energy Outlook 2025*](#); [IRENA. \(2025\). *Global Landscape of Energy Transition Finance 2025*](#).

demand growth to outrun infrastructure and supply or shifting costs onto smaller organizations or individuals.

Operational Levers

A. Demand Aggregation and Financing Pull-Through

Large buyers accelerate the pathway by making future clean demand bankable today. This reduces investment risk, shortens development timelines, and enables earlier scaling of supply chains.

Key mechanisms include:

- **Aggregated procurement platforms** for clean electricity, low-emissions fuels, and decarbonized materials
- **Long-term PPAs and contracts-for-difference** that de-risk generation, storage, and industrial conversion
- **Advance market commitments (AMCs)** for emerging technologies such as SAF, green hydrogen derivatives, carbon-negative concrete, and lower-emissions cement and steel pathways consistent with net-zero production.
- **Supplier upgrade financing**, including credit support, blended finance, and concessional capital to enable smaller firms to meet procurement standards

These tools accelerate capital formation and manufacturing decisions. Their value depends on whether they translate into deliverable, system-integrated clean capacity rather than nominal clean claims disconnected from transmission, interconnection, or adequacy constraints.

B. Clean Procurement Standards That Reshape Markets

Procurement standards determine which products and production pathways scale. When coordinated across large buyers, they shift entire supply chains toward clean compliance.

Effective standards include:

- **Shared definitions of “clean”** with auditable lifecycle boundaries
- **Traceability systems** (e.g., digital product passports, chain-of-custody tracking)
- **Phased tightening of emissions thresholds** to allow supplier transition while maintaining directionality
- **Supplier access to finance and technical support** to meet compliance requirements

When widely adopted, these standards make clean production the default competitive pathway. Their full system effect depends on being paired with financing mechanisms, verification systems, and targeted support pathways for smaller firms, so that compliance remains rigorous, enforceable, and scalable across global supply chains.

C. Workforce and Internal Execution Capacity

Large actors also influence transition speed through their own operations. Internal execution capacity—workforce training, contractor requirements, fleet management, and facility operations—directly affects system throughput.

Key actions include:

- **Workforce training and certification programs** aligned with equitable durability, electrification, grid expansion, and industrial decarbonization needs
- **Contractor and supplier requirements** that enforce clean practices, fair labor and procurement standards, and strong safety protections
- **Managed charging and load flexibility**, ensuring that electrified operations support grid stability rather than increasing peak demand
- **On-site efficiency and demand response**, reducing strain on local infrastructure

These measures increase the system’s ability to absorb and durably use clean energy while avoiding bottlenecks created by workforce shortages or unmanaged load growth.

AI And Discretionary Load Discipline

Large discretionary loads—especially AI datacenters, industrial clusters, and electrified transport hubs—can either accelerate the transition or, if deployed ahead of clean system readiness, delay fossil displacement by shifting incremental demand onto fossil-backed systems. The accelerated pathway therefore depends on maintaining parallel alignment across clean generation, transmission and distribution capacity, system reliability, and fossil retirement as new load is added.⁴⁵

This is not primarily a technology limitation. It is a synchronization problem. When demand expansion outpaces clean supply and infrastructure readiness, grids revert to gas or coal to maintain reliability. That locks in new emissions and delays fossil phase-out, potentially by decades if it results in new fossil capacity or long-lived fossil infrastructure. Large discretionary loads therefore must be treated as conditional infrastructure, not unconditional demand. Their deployment should proceed only when the system can support them without triggering fossil expansion.

Gas-connected datacenter buildout is a clear warning signal of failure in this regard: it indicates that demand growth has outrun clean supply, transmission, or adequacy planning.

Datacenter Requirements (Minimum Operational Set)

To ensure that large new loads accelerate rather than undermine the transition, a minimum operational framework is required:

⁴⁵ This reflects literature-grounded system logic: large new discretionary loads accelerate decarbonization only when incremental demand is matched by clean supply, flexibility, and grid expansion rather than new fossil generation or prolonged fossil dispatch. The clean-load-discipline framing here is playbook synthesis. See: [IEA. \(2025\). World Energy Outlook 2025](#); [Ember. \(2025\). Global Electricity Review 2025](#).

- **Clean siting requirements:** locate new facilities in regions with surplus or rapidly expanding clean generation and available transmission capacity
- **Clean supply commitments tied to permitting:** PPAs or equivalent arrangements must be secured before or alongside project approval
- **Adequacy alignment:** require firm clean capacity (e.g., storage, demand response, or other firming resources) sufficient to avoid reliance on fossil backup
- **Distribution and interconnection readiness:** ensure local grid infrastructure can support the load without displacing existing users or requiring fossil reinforcement
- **On-site flexibility and interruptibility:** enable load shifting, controlled demand reduction during grid stress, and participation in grid balancing so large loads do not increase reliance on fossil backup.
- **Host-community and ratepayer protection:** prevent cost-shifting or reliability degradation for existing customers
- **Integrated approval criteria:** link project approval to simultaneous progress in generation, storage, and transmission

These requirements treat large loads as part of the system, not external to it.

Risk Threshold

If discretionary load growth proceeds without these constraints, the system shifts into fossil-backed expansion. New gas capacity would be added to maintain reliability, locking in emissions and delaying retirement of existing fossil assets.

Once this occurs, the transition timeline typically slips by 3–7 years. Where it results in new long-lived fossil infrastructure, the delay can extend to decades, reflecting both the physical lifetime of those assets and the political and financial inertia they create.

Avoiding this outcome requires conditioning load growth on system readiness, not retrofitting solutions after imbalance has already occurred.

Operational Sequence (2026–2030)

2026–2027: Foundation

- Establish shared procurement standards and traceability systems
- Launch aggregated procurement platforms and initial AMCs
- Begin scaling supplier finance mechanisms

2027–2028: Market Conversion

- Expand coordinated procurement across sectors
- Tighten emissions thresholds in procurement standards
- Accelerate workforce training and contractor alignment

2026–2030: System Alignment

- Ensure durable demand growth remains synchronized with deliverable, durable clean capacity

- 2028–2030: link large-load approvals to clean supply, storage, and transmission readiness
 - 2028–2030: Align procurement, financing, and infrastructure planning across major actors
 - 2028–2030: Scale managed load strategies (e.g., datacenter flexibility, fleet charging coordination)
-

Summary Condition

Corporate and government acceleration succeeds when large actors use procurement, finance, standards, and load discipline to move clean supply chains, infrastructure, and demand forward in parallel, without shifting costs downstream or allowing new demand growth to trigger fossil fallback.

Chapter 14 — AI as Transition Multiplier

AI functions as a transition multiplier only under two linked conditions. First, its electricity demand must be structurally coupled to clean supply, deliverability, and system flexibility, and its productivity gains must be used to accelerate public-benefit deployment rather than induce new fossil capacity, crowd out other priority electricity uses, or weaken labor protections. Second, its deployment must remain aligned with planetary boundaries and climate justice, with AI-specific governance additionally requiring human rights protection, public-benefit purpose, meaningful participation, accountable human oversight, and data agency.

AI and data systems used for transition planning must preserve data agency, privacy, and community control. AI tools used in transition-critical workflows must be evaluated for fitness for purpose, error modes, bias, security, robustness, and real-world performance before deployment in consequential permitting, grid planning, procurement, finance, workforce, or infrastructure decisions. Personal, worker, household, and community-level energy data must not be repurposed for surveillance, exclusion, punitive enforcement, or commercial resale without lawful basis, consent where appropriate, and clear public-interest justification. AI-assisted recommendations in permitting, grid planning, finance, procurement, or workforce allocation must be documented with traceable inputs, assumptions, uncertainty ranges, and human-review records. No consequential transition decision should be delegated solely to an AI system. Public authorities, utilities, financiers, developers, and operators remain accountable for decisions, harms, appeals, and remedies.

In this framework, AI is valuable not because it changes the physics of the transition, but because it can improve planning, sequencing, and operational performance across already-identified bottlenecks. Without these constraints, AI can amplify energy demand, resource extraction, unlawful or excessive surveillance, labor exploitation, exclusionary decision-making, and existing inequities, reinforcing dynamics that delay or prevent net-zero outcomes while increasing avoidable harm. The same constraints that define the physical pathway — clean supply, deliverability, adequacy, fossil exit, biosphere integrity, justice, and human accountability — must therefore define the systems used to accelerate it.

Functional Role of AI in the Transition

AI serves four primary functions within the net-zero pathway:

1. **Planning Acceleration** — improving grid modeling, congestion analysis, interconnection studies, and deployment sequencing
2. **Permitting and Administrative Throughput** — standardizing documentation, triaging reviews, and reducing avoidable processing delays
3. **Construction and Logistics Optimization** — improving procurement timing, routing, scheduling, and bottleneck prediction
4. **Operational Efficiency** — improving forecasting, predictive maintenance, load management, early identification of failure modes and bottlenecks, and recommendation of practical workarounds and flexibility orchestration across clean energy systems

These functions are enabling rather than substitutive. AI improves execution quality and speed, but does not remove the underlying need for capital deployment, infrastructure buildout, workforce expansion, or institutional reform.

High-Value Deployment Domains

The most decision-relevant uses of AI in this playbook are those that improve system performance in already-constrained parts of the system:

- **grid modeling and planning**, including congestion forecasting, interconnection studies, and stability analysis
- **permitting and documentation support**, including standardization, compliance checks, and review acceleration
- **design optimization**, including balance-of-system design, siting, routing, and reliability constraints
- **construction sequencing and logistics**, including procurement timing, scheduling, and bottleneck prediction
- **predictive maintenance**, including transformers, inverters, wind and solar fleets, and other grid-critical assets

These are the areas where modest gains in speed and coordination can compound across the system. AI is most useful where it shortens cycle times, reduces administrative drag, and increases the effective productivity of already-scarce institutions and skilled workers.

Productivity Assumption

For planning purposes, this playbook treats **~30–40% productivity improvement** as a conservative but decision-relevant AI-enabled gain across design, permitting, scheduling, and selected construction-management functions. This assumption is already embedded in the workforce logic of [Chapter 1](#) and [Appendix D](#). Larger gains may occur in specific domains, but sustained **~50–60% productivity improvement** is treated here as an upside case rather than a planning baseline. AI should therefore be treated as a human workforce accelerator, not as a justification to downscale workforce mobilization, weaken training pipelines, or defer institutional build capacity.

AI Load Discipline

Within the accelerated ~2037–2038 pathway, AI-related load growth must remain synchronized with clean supply, deliverable clean capacity, and adequacy buildout. If large-load expansion outpaces those conditions, it can trigger fossil fallback, reverse displacement gains, and delay operational net zero. AI is a transition multiplier only if AI-driven load growth is governed as a system constraint rather than treated as neutral or unconstrained background demand. Data centers and other AI-related loads can accelerate decarbonization only when incremental electricity demand is served by clean generation and clean flexibility, without inducing new fossil firming, new gas peaker procurement, or prolonged fossil dispatch. This chapter therefore depends directly on the clean-load discipline established in [Chapter 13](#) and on the [Clean-Only Growth Rule](#) defined in the [Introduction](#). AI deployment that violates those conditions is not transition acceleration; it is fossil lock-in by another route.

Governance Requirement

AI deployment in energy, industrial, and infrastructure systems must be governed to align with public-benefit decarbonization goals rather than narrow private optimization. AI-assisted transition systems must respect internationally recognized human rights, including privacy, labor rights, Indigenous rights, procedural fairness, non-discrimination, and meaningful participation by affected communities. Data governance must preserve privacy, data agency, community control, and reviewable public-interest justification for any use of personal, worker, household, or community-level transition data. AI should be used to improve system-wide reliability, speed, emissions performance, safety, and equitable access, not merely local cost minimization or proprietary advantage. Agencies and organizations using AI-assisted transition tools must maintain staff competence in energy systems, rights-preserving governance, model limitations, data protection, uncertainty interpretation, cybersecurity, and appeal or remedy procedures. Misuse includes using AI systems to obscure fossil persistence, manipulate finance, procurement, permitting, or regulatory priorities, infer, game, or exploit transition-planning logic, intensify labor surveillance, exclude affected communities, automate discriminatory infrastructure decisions, or produce unverifiable transition claims. Governance must ensure that AI deployment does not justify degraded labor conditions, unlawful or excessive surveillance, weakened due process in permitting, exclusionary infrastructure decisions, or the displacement of accountability from human institutions to opaque technical systems. The transition remains a governed political and industrial process; AI can assist those institutions, but it cannot replace them.

Risk Threshold

If AI is underused in deployment-critical domains, the transition loses the narrow execution margin that keeps a late-2030s outcome plausible and becomes more prone to administrative friction, poor coordination, and scheduling and logistics failures. The larger risk is misapplied AI expansion: if AI-driven demand growth outpaces clean supply, if data-center siting occurs in fossil-heavy grids without adequate clean additions and flexibility, or if private optimization goals override system-level decarbonization objectives, the result is higher fossil persistence risk and a material delay to the net-zero pathway. In this playbook, AI becomes a liability when it increases electricity demand faster than the clean system can absorb it.

Operational Sequence (2026–2030)

- **2026:** Require clean supply linkage for large new AI-related loads, using permitting and interconnection conditions consistent with [Chapter 13](#)
- **By end-2026:** Deploy AI tooling in permitting agencies, utilities, and grid operators to increase throughput in documentation, queue triage, and review standardization
- **2027–2028:** Scale grid-planning, congestion modeling, and interconnection automation; prioritize uses that increase system efficiency and reduce administrative lag
- **2027–2029:** Integrate AI into procurement, construction scheduling, logistics coordination, and predictive maintenance across clean energy infrastructure

- **2028–2030:** Ensure AI-related load is fully incorporated into adequacy planning so storage, demand response, and transmission scale ahead of or alongside demand growth

System Integration Requirement

AI must be integrated with the transition architecture established in earlier chapters:

- [Chapter 1](#) — workforce productivity gains support, but do not replace, labor mobilization
- [Chapter 3](#) — AI improves grid planning and operational coordination, but does not substitute for transmission buildout
- [Chapters 5–6](#) — AI can improve flexibility orchestration and adequacy management, but does not replace storage or firm clean capacity
- [Chapter 13](#) — AI-related loads must comply with clean-load discipline, host-community protection, and system-readiness requirements

Used correctly, AI improves system execution. Used incorrectly, it increases load faster than the clean system can respond and reinforces fossil dependence.

Summary Condition

AI functions as a transition multiplier when it reduces friction, improves coordination, and increases the effective productivity of planning, permitting, construction, and operations without allowing new demand growth to induce fossil capacity or prolong fossil dispatch. Under those conditions, AI helps the transition move faster through existing bottlenecks while remaining subordinate to the physical, institutional, and justice constraints that govern the pathway as a whole.

Final Systems Conclusion — Net Zero as Coordinated System Transformation

This playbook defines net zero not as a target to be declared, but as a system condition that must be physically achieved and institutionally maintained. The transition is governed by constraints across infrastructure, energy balance, industrial capacity, ecological stability, and political and societal durability. It succeeds only when these constraints are addressed in a coordinated and sequenced manner.

Net zero is reached when fossil energy systems are no longer required for energy provision, capacity adequacy, or reliability services. This condition emerges from the synchronized expansion of equitable clean generation, transmission, storage, demand flexibility, and firm clean resources, combined with the managed retirement of fossil assets and the elimination of persistence pathways that allow them to remain in operation.

The transition cannot be reduced to emissions accounting or market signals alone. Declining utilization of fossil assets does not guarantee exit. Without explicit governance, fossil systems persist through refinancing, capacity payments, regulatory protection, and institutional inertia. Net zero therefore requires active fossil retirement governance alongside clean system buildout.

At the same time, clean system expansion must remain bounded by physical and operational readiness. The pathway does not require full geopolitical coordination, but it does require sufficient functional coordination across finance, infrastructure, adequacy, demand, workforce, and retirement systems. Premature fossil retirement creates adequacy and reliability risks; delayed retirement locks in emissions and capital misallocation. The pathway requires continuous synchronization between clean replacement and fossil exit.

Demand growth must be governed with equal discipline. New electrical loads—including electric vehicles, industrial electrification, hydrogen production, synthetic fuels, and AI-related demand—must be structurally coupled not only to clean supply expansion, but to deliverable clean scale and system adequacy. Unconstrained load growth that outpaces clean generation, transmission, storage, and firm clean replacement introduces new fossil dependence and undermines system progress.

This work's [system model](#) reinforces a central conclusion: transition speed is determined less by any single technology curve than by whether constrained processes maintain sufficient throughput at the same time. Capital, workforce, infrastructure, demand, adequacy, and retirement systems are not independently sufficient. They determine the pathway through their interaction, sequencing, and synchronization.

Biosphere restoration operates in parallel but remains distinct from fossil emissions reduction. Ecological recovery strengthens planetary resilience, supports long-term carbon cycle stability, and contributes to human habitability, but it cannot substitute for the elimination of fossil emissions. Maintaining this separation is essential for both ecological integrity and credible decarbonization.

Carbon dioxide removal plays a bounded role within the system. Before operational net zero, CDR is reserved only for genuinely residual emissions in the hardest-to-abate sectors after

source elimination, substitution, and fossil phaseout have been pursued to their practical limits. After operational net zero, CDR can scale for atmospheric drawdown, including removal of some or all historical anthropogenic CO₂, depending on societal choice, system capacity, and governance. Overreliance on future removal capacity introduces systemic risk and delays necessary emissions reductions.

Artificial intelligence functions as a coordination and productivity layer within this system. It can accelerate planning, permitting, construction, and operations, but it does not replace physical infrastructure, workforce mobilization, or institutional capacity. Its deployment must be governed to ensure alignment with clean supply, system constraints, and public-benefit outcomes.

Across all domains, the transition is constrained by workforce capacity, supply chains, permitting systems, institutional throughput, and queue conversion. Productivity gains—including those enabled by AI—are meaningful accelerators, but they cannot substitute for the scale of human, industrial, financial, and institutional mobilization required.

The pathway to a late-2030s net-zero outcome is narrow. It depends on maintaining execution pace across multiple interdependent systems without allowing delays, bottlenecks, or policy inconsistencies to accumulate. Underperformance in any core domain—generation, transmission, storage, demand flexibility, fossil retirement, finance, workforce, or governance—propagates through the system and delays the outcome.

Conversely, overextension or mis-sequencing introduces instability. Rapid buildout without coordination can overwhelm supply chains and institutions; premature commitments without readiness can trigger political backlash or reliability failures. The transition must therefore proceed within a bounded corridor defined by speed, system stability, and synchronization.

The core invariant of this playbook is:

clean system expansion, fossil system contraction, and demand growth must remain continuously synchronized in volume, deliverability, adequacy, capital allocation, and institutional throughput.

If this synchronization is maintained:

- fossil assets retire without creating adequacy gaps
- capital flows toward clean infrastructure rather than prolonging legacy systems
- clean energy becomes the default system supply
- new demand lands on ready clean systems
- frontline pollution burdens decline and air-quality gains reach communities historically overexposed to fossil emissions
- emissions decline to near-zero within the operational timeframe

If this synchronization fails:

- fossil systems persist despite clean growth
- new demand reinforces fossil dependence
- clean capacity remains partly undeliverable
- reliability risks slow or reverse progress
- the net-zero timeline becomes unattainable

Net zero is therefore not a single objective, but a coordinated outcome across energy, industry, land systems, finance, workforce, and governance. It requires disciplined execution, institutional alignment, and continuous management of system interactions over time.

The transition succeeds when the system no longer requires fossil energy to function—and when the institutions governing it ensure that this condition is durably just.

Appendices

Appendix A — Critical Timeline

Defines the minimum system growth conditions required to maintain a plausible pathway to global net zero by ~2037–2038. These are non-negotiable system thresholds, not aspirational targets. The pathway remains viable only if these checkpoints are met by ~2030. After ~2032, path dependence dominates and missed milestones become much harder to recover. Timeline checkpoints should be read alongside the labor, financing, and governance protections developed in the [Introduction](#), Chapters [1](#), [2](#), [9](#), [10](#), [11](#), [12](#), [13](#), [14](#), Appendices [E](#) and [M](#). Speed without implementation legitimacy is not durable.

Conditional Timeline Checkpoints Table

System	2030 Checkpoint	Growth Path Requirements	Requirements Notes	Failure threshold / warning sign
Solar	≥1.5 TW/year installs; ~8.5–9.6 TW cumulative	~24–27% CAGR to ~2032, then ~16–19% through ~2034	Center on ~9.1 TW by 2030; ~29 TW by ~2037; ~33 TW by ~2038	<20% CAGR before 2032 → ~2037–2038 pathway becomes unlikely absent compensating overbuild elsewhere
Wind	~2.3–2.6 TW cumulative	~12–15% to ~2032, then ~9–11% through ~2034	Center on ~2.5 TW by 2030; ~3.8–4.8 TW by ~2037–2038	<12% through early 2030s → rising pressure shifts to solar, storage, transmission, and firm clean
Storage	roughly ~55–65 TWh cumulative by ~2030 (central ~60 TWh)	~40–45% CAGR through ~2032, then ~25–30% through ~2034	Target: 4–6 hours grid coverage in high-renewable systems	Underbuild → adequacy risk, fossil backup persistence, and over-reliance on solar overbuild plus firm generation
Transmission	≥7–9% annual expansion	2–3× historic build rate, with HVDC scale-up	HVDC backbone, interconnection reform, and parallel substation, transformer, and distribution build-out	<5% growth → curtailment, congestion, and fossil persistence
Workforce	~2–3M workers/year	~0.5–1.5M/yr 2026–2027; ~2–3M/year 2028–2032; ~1–2M/yr 2033–2038	≥2M/year by ~2028 is the minimum floor	Late training → 2029–2032 deployment bottleneck
System Rule	Clean-only growth enforced	Demand captured by clean supply	No new fossil capacity for net load growth	Fossil expansion alongside clean → displacement failure

Solar and Wind Curve Baseline Inputs (End-2025)

- Solar: ~2.9 TW installed¹⁹
- Wind: ~1.3 TW installed¹⁹
- Combined solar + wind: ~4.2 TW installed
- Electricity demand: ~30,700 TWh/year⁴⁶ (rising)

System Rule (Invariant)

Clean-only growth must hold.

If new demand is not captured by clean supply:

- Fossil capacity factors stabilize
- Displacement slows
- Net zero timeline fails

Post-Net-Zero to 2100

- Energy system remains **oversized relative to demand**
- Supports:
 - Industrial electrification
 - Synthetic fuels
 - Carbon dioxide removal (CDR)
- Growth tapers but does not collapse

Interpretation

This timeline is not a forecast.

It is a **constraint envelope** defining whether net zero remains physically achievable.

⁴⁶ Jan–Nov total was 28,082.13 TWh (Ember Energy Research: [Monthly electricity data](#), 30 Mar 2026); December and full-year 2025 estimated by extrapolation.

Appendix B — Structural Caveats and Bottlenecks Risk Hierarchy

These bottlenecks represent the primary structural constraints and Earth-system destabilizers that determine whether the ~2037–2038 pathway succeeds or fails. Most transition bottlenecks are institutional rather than technological. Biosphere cascade risks are different: when critical climate impacts, severe ecosystem degradation, or multiple destabilizing processes collide across coupled systems, triggering cascades, there may be no reliable correction pathway at transition-planning scale. At that point, the practical response shifts from mitigation to avoidance, adaptation, and damage limitation.

Hierarchy Of Highest-Risk Bottleneck Categories:

1. **Fossil Financial Repricing And Persistence-Pathway Closure**
Close refinancing, underwriting, life-extension, and replacement pathways fast enough that declining utilization becomes actual retirement.
2. **Emerging-Market Capital Cost And Fossil Lock-In**
Use concessional finance, guarantees, and de-risking so clean becomes the default growth path in high-demand regions.
3. **Cryosphere Albedo / Ice-to-Land Cascade**
Avoid activation by reducing warming pressure through rapid emissions reductions; once activated, there is no reliable reversal lever at transition-planning scale.
4. **Transmission, Interconnection, And Deliverable Clean Scale**
Expand corridors, interconnection, HVDC, substations, transformers, and distribution fast enough for clean generation to displace fossil generation.
5. **Permafrost Carbon And Methane Feedback**
Reduce fossil CO₂ and methane fast enough to limit thaw pressure; tighten carbon and methane budgets with added permafrost emissions.
6. **Clean Adequacy Replacement**
Scale batteries, LDES, demand flexibility, firm clean capacity, and adequacy-market reform before fossil fallback becomes structurally retained.
7. **AMOC / Ocean-Circulation Disruption**
Treat as a high-consequence uncertainty; reduce emissions faster to limit warming and freshwater-forcing risk while strengthening coastal, food-system, water, and adaptation resilience.
8. **Coordinated Fossil Adaptive-Delay Strategy**
Block incumbent strategies that preserve fossil financeability, capacity payments, emergency exemptions, or “energy security” delay narratives.
9. **Amazon / Tropical Forest Dieback**
Halt deforestation, fragmentation, and fire-driven degradation while protecting Indigenous stewardship and high-integrity restoration.
10. **Workforce And Institutional Throughput**
Front-load training, placement, permitting capacity, project management, and institutional staffing before the 2028–2032 build window tightens.
11. **Fossil Asset Bailouts And Sunset-Asset Protection**
Prevent public support, regulatory protection, or capacity payments from retaining uneconomic fossil capacity after clean replacement is available.
12. **AI And Discretionary-Load-Driven Gas Peaker Lock-In**
Tie large discretionary loads to clean supply, flexibility, ratepayer protection, and grid readiness before approval.
13. **Freshwater-Cycle / Monsoon Destabilization**
Reduce warming pressure while strengthening watershed, groundwater, agricultural, hydropower, and infrastructure resilience.

14. Compound Biosphere Cascade

Prevent multiple biosphere systems from degrading together; treat biosphere integrity as a hard condition of transition success, not a post-transition cleanup task.

15. Materials, Recycling, And Industrial Input Constraints

Scale recycling, material efficiency, substitution, and grid-critical component supply before early-2030s demand surges.

Simultaneous failure in two or more of the top seven categories would, in most plausible cases, make the ~2037–2038 pathway unlikely; in some combinations, a single severe failure may be sufficient. Biosphere cascade risks are not ordinary deployment bottlenecks: they tighten the physical operating window, increase adaptation burden, and reduce stabilization confidence even if the energy transition remains technically executable.

Appendix C — Common Major Failure Modes

These are the most likely high-consequence failure modes that could materially delay or break the ~2037–2038 net-zero pathway. They are listed not as isolated risks, but as system failures that can propagate across multiple parts of the transition.

1. Fossil Financial Repricing Fails / Persistence Pathways Remain Open

Mechanism

- Lobby to prohibit mandatory climate disclosures.
- Block higher capital requirements for fossil lending.
- Litigate carbon pricing and fiduciary reforms.
- Capture regulators to prevent fossil reserve write-downs.
- Push “energy security” narrative to justify reinvestment.

Likelihood: High

Consequence: Existential

Timeline Impact: Pushes ~2037–2038 → 2045–2055

If fossil expansion remains financeable through 2030, asset lifetimes extend beyond 2045 and emissions lock-in becomes structural.

Mitigation Lever

- Central bank alignment across G7.
 - Insurance sector withdrawal from fossil underwriting.
 - Pension fund allocation mandates.
 - Border carbon adjustments.
-

2. Emerging Markets Lock In New Fossil Capacity

Mechanism

- High weighted average cost of capital makes clean deployment harder to finance.
- Currency risk and sovereign risk premiums raise project costs.
- Fossil procurement remains institutionally simpler than clean alternatives.
- Grid finance is weak where demand growth is fastest.
- Public de-risking and concessional finance remain too small or too slow.

Likelihood: High

Consequence: Existential

Timeline Impact: Pushes ~2037–2038 → 2045–2060

If fast-growing regions remain trapped by capital-cost asymmetry, new fossil infrastructure continues to be built even where clean energy is favorable on lifetime economics, locking in global emissions beyond the late-2030s window.

Mitigation Lever

- Public risk guarantees and concessional finance.
- FX-risk and political-risk de-risking tools.

- Sovereign green acceleration funds.
 - Multilateral grid and transmission finance.
 - Clean-procurement and retirement-linked financing structures.
-

3. Cryosphere Albedo / Ice-to-Land Cascade

Mechanism

- Loss of reflective sea ice, snow, glacier, and ice-sheet surface area.
- Replaces bright ice/snow with darker ocean, land, meltwater, and exposed rock.
- Accelerates heat absorption, melt, freshwater fluxes, and sea-level rise.
- Stresses ocean circulation, water-cycle stability, and carbon-cycle feedbacks.

Likelihood: High

Consequence: Existential

Timeline Impact: Pushes ~2037–2038 → higher overshoot / lower stabilization confidence

If cryosphere feedback accelerates while fossil phaseout is delayed, operational net zero may arrive too late to prevent durable Boundary Stress.

Avoidance / Damage-Limitation Lever

Once this cascade is active, there is no reliable reversal lever at transition-planning scale. The only true mitigation is avoidance through reduced warming pressure.

- Rapid fossil phaseout.
 - Methane and black-carbon reduction.
 - Arctic, glacier, and ice-sheet monitoring.
 - No offset use to justify fossil delay.
-

4. Transmission And Interconnection Buildout Failure

Mechanism

- Permitting and siting delays block interregional and bulk transmission expansion.
- Fragmented planning prevents coordinated corridor development.
- Cost-allocation disputes delay or shrink major projects.
- Interconnection queues continue to grow faster than transmission capacity.
- Local opposition rises where compensation and benefit-sharing are weak.

Likelihood: High

Consequence: Existential

Timeline Impact: Pushes ~2037–2038 → 2045–2055

If transmission growth remains below the required range through 2030, renewable generation growth decouples from deliverable electricity, curtailment rises structurally, and fossil generation remains embedded in constrained regions.

Mitigation Lever

- Federalized or supra-national corridor authority.
 - Standardized interregional planning requirements.
 - Rights-preserving permitting acceleration with intact environmental and Indigenous protections.
 - Durable host-community compensation and benefit-sharing.
 - Long-horizon cost-allocation frameworks.
-

5. Permafrost Carbon and Methane Feedback

Mechanism

- Thaw frozen organic carbon stores.
- Release CO₂ and methane outside direct human control.
- Accelerate abrupt thaw through wildfire, hydrological change, and ground collapse.
- Convert permafrost from passive carbon store to active warming amplifier.

Likelihood: High

Consequence: Existential

Timeline Impact: Pushes ~2037–2038 → tighter carbon budget / higher drawdown burden
If fossil emissions persist into the 2030s, permafrost feedback increases the amount of post-net-zero removal required.

Avoidance / Damage-Limitation Lever

Once large-scale thaw feedback is active, mitigation shifts from control to damage limitation. The main lever is preventing additional warming pressure before release accelerates.

- Rapid fossil CO₂ phaseout.
 - Aggressive methane reduction.
 - Protect high-latitude peatlands and wetlands.
 - Avoid permafrost-region land disturbance.
-

6. Clean Adequacy Replacement Lags Generation Buildout

Mechanism

- Short-duration batteries scale, but long-duration storage remains commercially immature.
- Demand flexibility and reserve-margin reforms lag renewable growth.
- Capacity markets continue to favor fossil retention over clean adequacy.
- Storage interconnection and co-location rules remain slow or fragmented.
- Transmission reinforcement fails to reduce local flexibility burdens.

Likelihood: Medium-High

Consequence: Existential

Timeline Impact: Pushes ~2037–2038 → 2042–2050

If storage, flexibility, and residual adequacy resources do not scale with variable renewables, fossil plants remain online for evening peaks, multi-day gaps, and reserve margins, preventing full operational displacement.

Mitigation Lever

- Explicit procurement of storage and clean flexibility.
 - Long-duration storage commercialization support before 2030.
 - Market design reform for clean adequacy.
 - Co-location and fast-track interconnection for storage.
 - Demand response and flexible-load integration.
-

7. AMOC / Ocean-Circulation Disruption

Mechanism

- Add freshwater and heat stress to North Atlantic circulation.
- Weaken ocean density gradients and overturning circulation.
- Shift regional heat transport, rainfall, sea level, and storm patterns.
- Increase food, water, coastal, and governance stress during transition execution.

Likelihood: Medium

Consequence: Existential

Timeline Impact: Pushes ~2037–2038 → lower institutional resilience / higher fossil fallback risk

AMOC disruption may not block clean build directly, but it can degrade the operating environment needed for synchronized transition execution.

Avoidance / Damage-Limitation Lever

There is no operational lever that can reliably restore a disrupted circulation regime on transition timelines. The mitigation strategy is precautionary avoidance and resilience preparation.

- Rapid fossil phaseout.
 - Methane reduction to limit near-term warming.
 - Ocean and cryosphere monitoring.
 - Coastal, food-system, and water resilience planning.
-

8. Coordinated Fossil Adaptive-Delay Strategy

Mechanism

- Fossil incumbents stop trying to defeat the transition outright and instead delay it strategically.
- They support selective clean growth while preserving gas and fossil reliability roles.
- They promote narrow conversion pathways to prolong asset life.
- They fragment political coalitions by backing partial transition measures.
- They adapt to declining utilization by defending refinancing, reserve margins, and institutional influence.

Likelihood: Medium-High

Consequence: Existential

Timeline Impact: Pushes ~2037–2038 → 2045–2055

The greatest danger may not be open resistance but adaptive delay: a partial transition in which clean growth continues but fossil systems remain embedded for capacity, finance, and political reasons.

Mitigation Lever

- Binding retirement schedules and adhering to no-new-build rules.
- Refinancing restrictions and sunset-asset governance.
- Clean adequacy procurement that excludes fossil persistence.
- Strict limits on conversion pathways.
- Coordinated finance, trade, and regulatory alignment.

9. Amazon / Tropical Forest Dieback

Mechanism

- Combine warming, drought, fire, deforestation, and fragmentation.
- Weaken forest moisture recycling and regional rainfall stability.
- Reverse carbon sink function toward carbon release.
- Degrade biodiversity, Indigenous stewardship, and ecosystem metabolism.

Likelihood: High

Consequence: Severe / Existential

Timeline Impact: Pushes ~2037–2038 → weaker carbon sink / higher Boundary Stress

If tropical forest degradation continues while fossil emissions persist, the pathway loses a major biosphere-stabilization anchor.

Mitigation Lever

- End deforestation and fragmentation.
- Restore and rewild agricultural lands bordering key ecological habitat zones where ecological recovery value is high.
- Protect Indigenous and local land rights.
- Finance high-integrity forest restoration.
- Separate restoration from fossil offset substitution.

10. Workforce And Institutional Throughput Bottleneck

Mechanism

- Training pipelines scale more slowly than manufacturing and project pipelines.
- Credentialing, apprenticeship, and placement systems remain fragmented.
- Worker mobility constraints prevent labor from reaching deployment hotspots.
- Safety standards and supervisory capacity lag workforce expansion.
- Public and private training investment starts too late.

Likelihood: High

Consequence: Severe

Timeline Impact: Pushes ~2037–2038 → 2042–2050

If workforce mobilization does not accelerate by 2026–2028, labor scarcity becomes binding during the peak build window, slowing solar, storage, transmission, industrial retrofits, and permitting throughput at the same time.

Mitigation Lever

- National and regional training mobilization programs.
- Fast credential recognition and apprenticeship scaling.
- Worker mobility, housing, and relocation support.
- Public funding for technical education and instructor pipelines.
- Explicit labor standards tied to transition funding.

11. Fossil Asset Bailouts And Sunset-Asset Protection

Mechanism

- Governments socialize fossil losses without binding retirement conditions.
- Capacity contracts and reliability payments are extended to protect incumbent assets.
- Sunset assets are refinanced or securitized without real exit requirements.
- Political pressure frames managed retirement as “too risky” or “too costly.”
- Bailouts preserve fossil capital and institutional influence past operational need.

Likelihood: Medium

Consequence: Severe

Timeline Impact: Pushes ~2037–2038 → 2042–2050

If fossil assets are protected from market exit without mandatory retirement conditions, the transition slows from both sides: clean capital is crowded out and fossil persistence is prolonged.

Mitigation Lever

- Retirement-conditioned securitization only.
- Mandatory phaseout schedules.
- Prohibition on refinancing sunset assets without exit plans.
- Capacity-market reform to exclude fossil persistence.
- Transparent stranded-asset governance.

12. AI And Discretionary Load Growth Outpaces Clean Build

Mechanism

- Large new discretionary loads appear faster than clean generation and transmission.
- Data-center siting occurs in fossil-heavy grids without clean-supply linkage.
- Utilities respond to new peak demand with gas additions or prolonged fossil dispatch.
- Cost recovery shifts burdens onto households and non-participating customers.
- Private optimization overrides system-level clean-only growth discipline.

Likelihood: Medium-High

Consequence: Severe

Timeline Impact: Pushes ~2037–2038 → 2040–2048

If AI-related load growth is not structurally coupled to clean supply, storage, and transmission, it becomes a new fossil-demand channel and weakens the core clean-only growth rule.

Mitigation Lever

- Clean-supply linkage requirements for large new loads.
- Flexibility and interruptibility standards for discretionary load.
- Host-community and ratepayer protections.
- Conform to ESG and scope emissions commitments.
- Adequacy planning that incorporates data-center growth early.
- Permitting conditions tied to system readiness.

13. Freshwater-Cycle / Monsoon Destabilization

Mechanism

- Alter rainfall through warming, land-use change, aerosols, and glacier retreat.
- Intensify drought, flood, groundwater stress, and seasonal instability.
- Stress food systems, hydropower, cooling water, and urban infrastructure.
- Increase adaptation burden and institutional overload.

Likelihood: Medium-High

Consequence: Severe

Timeline Impact: Pushes ~2037–2038 → higher adaptation burden / lower execution capacity

Freshwater disruption can increase emergency spending, grid stress, food stress, and transition backlash during the critical execution window.

Avoidance / Damage-Limitation Lever

There is no single mitigation lever for destabilized regional water cycles. The practical response is to reduce warming pressure while strengthening watershed resilience before instability becomes self-reinforcing.

- Watershed, wetland, and peatland restoration.
 - Groundwater governance.
 - Climate-resilient agriculture.
 - Water-cycle resilience in energy and infrastructure planning.
-

14. Compound Biosphere Cascade

Mechanism

- Allow multiple biosphere buses to degrade together.
- Couple cryosphere, carbon-cycle, water-cycle, ocean, land, and ecosystem stress.
- Turn discrete failure modes into system-wide Boundary Stress.
- Propagate ecological failure into economic, food, water, and institutional systems.

Likelihood: Medium

Consequence: Existential

Timeline Impact: Pushes ~2037–2038 → possible energy-transition success but planetary-boundary failure

The energy transition may remain technically achievable while biosphere instability prevents durable climate stabilization.

Avoidance / Damage-Limitation Lever

Once multiple biosphere cascades couple, there is no credible single recovery lever. The only robust mitigation is preventing coupled activation and preserving biosphere resilience before Boundary Stress becomes systemic.

- Rapid fossil phaseout and methane reduction.
 - Parallel ecosystem protection and restoration.
 - Water-cycle resilience and land-governance reform.
 - Treat biosphere integrity as a hard condition of transition success, not a post-transition cleanup task.
-

15. Critical Materials And Component Disruption

Mechanism

- Copper, lithium, rare earths, transformers, and power electronics face supply shortages.

- Manufacturing lead times increase as multiple sectors compete for the same inputs.
- Recycling systems fail to scale before early-2030s demand surges.
- Geopolitical concentration exposes supply chains to export controls or shocks.
- Substitution and material-efficiency gains arrive too slowly.

Likelihood: Medium

Consequence: Major

Timeline Impact: Pushes ~2037–2038 → 2040–2048

If upstream materials and component supply do not expand on time, project costs rise, completion rates slow, and multiple deployment curves fall below the levels required for late-2030s net zero.

Mitigation Lever

- Recycling and circular-material systems before 2032.
- Diversified refining and manufacturing capacity.
- Material-efficiency standards and substitution pathways.
- Strategic inventories for transformers and grid-critical components.
- International supply-chain coordination.

Ranking by Risk to Net Zero Timeline

1. Fossil Financial Repricing Fails / Persistence Pathways Remain Open
2. Emerging Markets Lock In New Fossil Capacity
3. Cryosphere Albedo / Ice-to-Land Cascade
4. Transmission And Interconnection Buildout Failure
5. Permafrost Carbon and Methane Feedback
6. Clean Adequacy Replacement Lags Generation Buildout
7. AMOC / Ocean-Circulation Disruption
8. Coordinated Fossil Adaptive-Delay Strategy
9. Amazon / Tropical Forest Dieback
10. Workforce And Institutional Throughput Bottleneck
11. Fossil Asset Bailouts And Sunset-Asset Protection
12. AI And Discretionary Load Growth Outpaces Clean Build
13. Freshwater-Cycle / Monsoon Destabilization
14. Compound Biosphere Cascade
15. Critical Materials And Component Disruption

Structural Insight

The pathway is most vulnerable not to a single technology shortfall, but to desynchronization across the main constraints that drive the highest-risk system bottlenecks. If financial repricing, deliverable clean scale, workforce throughput, adequacy replacement, and emerging-market de-risking do not clear in time and in sequence, clean growth can continue while fossil displacement stalls, fallback pressures re-emerge, and the timeline slips.

Appendix D — Net-Zero Workforce Mobilization Framework

Global net zero by ~2037–2038 is workforce-limited before it is capital-limited; labor availability determines deployment speed and system integration capacity. Maintaining the required solar, storage, transmission, and heavy-industry build rates described in the main playbook requires sustained labor additions of:

- 2–3 million trained workers per year (2026–2032)⁴⁷
(assuming 30–40% AI-assisted productivity gains)
- Cumulative additions by ~2035: ~18–22 million workers globally.

Workforce scale-up is the pacing constraint for the net-zero pathway; capital, technology, and materials do not convert into deployment if labor mobilization fails.

Failure to reach ≥2 million additions per year by 2028 makes a ~2037–2038 outcome structurally improbable, shifting net zero toward 2045+.

Workforce Categories and Annual Additions (Peak 2026–2032)

1. Skilled Trades and Technical Workforce

~1.4–2.0 million/year

Roles:

- Electricians (distribution & industrial)
- High-voltage lineworkers
- Substation technicians
- Solar panel and wind turbine installers
- Storage technicians
- Heat pump and industrial retrofit technicians
- Clean manufacturing operators
- Recycling and materials recovery technicians

These roles determine physical deployment velocity and are the primary bottleneck to build rates. If this category fails to scale first, all other build rates slow sharply; rapid expansion must also be paired with strong safety standards, labor protections, and training quality control.

Training alone is insufficient; placement, geographic mobility, and employer absorption capacity must scale in parallel.

2. Grid and Power Engineers

~400k–600k/year

Roles:

- Power system engineers
- Transmission planners and HVDC specialists
- Grid integration engineers

⁴⁷ If AI-enabled productivity gains are only ~5–10%, required workforce additions rise to ~2.5–4.0 million workers per year and the peak mobilization window shifts earlier (2026–2030). Under this condition, failure to exceed ~2.5 million annual additions by ~2028 makes the ~2037–2038 pathway unlikely.

- Storage system engineers
- Utility interconnection engineers
- Project integrators

Transmission and grid integration are binding constraints. Engineering lag directly delays usable clean capacity and slows the conversion of installed assets into reliable system performance.

3. Industrial Retrofit and System Integrators

~150k–250k/year (peaks 2028–2035)

Roles:

- Steel plant electrification engineers
- Cement kiln conversion engineers
- Chemical plant retrofit specialists
- Industrial heat engineers
- Hydrogen and synthetic-fuels system integration engineers
- Brownfield systems integrators

Heavy industry conversion is integration-limited rather than equipment-limited; clean power expansion without industrial retrofit does not deliver net zero.

4. Molecular and Synthetic Systems Workforce

~80k–120k/year (ramping 2026–2035)

Roles:

- Green chemists (polymers and materials)
- Electrochemical engineers
- Process engineers
- Nanomaterials scientists
- Microbiologists, metabolic engineers, and bioprocess engineers

Required to eliminate fossil feedstocks, scale hydrogen, develop alternative materials, and prevent industrial chemistry conversion lag. This category is less volume-constrained than skilled trades but more specialization-constrained. Scaling this category globally requires expanded access to advanced scientific training, laboratory infrastructure, and industrial placement pathways, especially in emerging markets where deployment demand may outpace local specialization capacity.

5. Permitting, Finance and Systems Administration

~150k–250k/year

Roles:

- Environmental review professionals

- Utility regulatory specialists
- Infrastructure finance analysts
- Supply chain and logistics coordinators
- Project finance structurers

Transmission siting, WACC barriers, and utility revenue stabilization become structural risks when administrative staffing and institutional capacity lag.

6. CDR and Biogenic Carbon Specialists (Primarily Post-2030 Ramp)

CDR workforce expansion is sequenced to avoid competing with the 2028–2032 mitigation peak.

Training Additions:

- 40k–80k/year (2026–2030, RD&D phase)
- 60k–120k/year (2030–2035)
- 100k–180k/year (2035–2045, gigaton scale-up)

Includes:

- DAC process engineers
- Geologists & reservoir engineers
- MRV specialists
- Green chemists (catalysts, sorbents, MoFs/CoFs, and related materials)
- Chemical engineers
- Process engineers
- Nanomaterials scientists
- Biochar microbiologists & bioprocess engineers

As CDR is not utilized to reach operational net zero; large-scale workforce expansion follows fossil phaseout. This sequencing is intended to prevent specialized CDR buildout from drawing labor and training capacity away from the 2026–2032 mitigation peak.

Phase Structure

Phase 0 (2026–2027)

Rapid vocational scaling, engineering seat expansion, licensing reciprocity, worker transition support, and AI integration in design and permitting. This phase determines whether permitting and interconnection bottlenecks are relieved early enough to avoid a locked-in 2029–2032 labor shortage.

Phase 1 (2028–2032)

Peak build window. Sustain roughly ~2–3 million annual additions while transmission, solar, wind, storage, and retrofit deployment accelerate. This is the core labor mobilization phase of the pathway.

Phase 2 (2033–2038)

Industrial conversion surge and fossil displacement endgame. Skilled-trades growth begins to

stabilize, while industrial retrofit, system integration, and specialized conversion roles remain in high demand.

Post-Net-Zero (2039+)

Skilled-trades growth tapers gradually, while CDR and long-duration stewardship workforce requirements expand materially.

Education and Training Growth Rates

To sustain the required additions:

- Vocational electrical and grid programs: 2–3× global expansion
 - Power engineering seats: 1.5–2× expansion
 - Chemical science, process engineering, microbiology, metabolic engineering, and materials/nanomaterials tracks: expand ~2×, with greater specialization in catalysis, advanced materials, and carbon management systems
 - Fast-track certification and licensing reciprocity should be in place by 2026
 - Managed transition pathways from fossil fuel sectors into electrical, industrial retrofit, and grid roles
 - AI-enabled productivity: target 30–40% gain
-

Core Constraint:

If workforce mobilization remains below ~2 million additions (assuming 30–40% AI-assisted productivity gains)⁴⁸ per year by 2028, the ~2037–2038 pathway becomes structurally infeasible regardless of capital availability.

Labor remains the binding deployment constraint.

⁴⁸ If AI-enabled productivity gains are only ~5–10%, required workforce additions rise to ~2.5–4.0 million workers per year and the peak mobilization window shifts earlier (2026–2030). Under this condition, failure to exceed ~2.5 million annual additions by ~2028 makes the ~2037–2038 pathway unlikely.

Appendix E — Clean Energy Financing and Financial Realignment

[Chapter 2](#) establishes the directional logic: fossil expansion must become structurally unfinanceable while clean infrastructure becomes structurally financeable. This appendix develops the mechanics of that realignment — the financial mechanisms that shape asset viability, the institutional architecture used to mobilize and de-risk clean capital, the repricing tools that constrain fossil persistence, and the way finance must be aligned to make the four operational rules financially durable: clean-only growth, deliverable clean scale, fossil exit, and reliability replacement. Its companion [Appendix F](#) illustrates how financing structures could adjust to better support the conditions required to achieve the accelerated timeline.

In this playbook, finance must do more than accelerate clean investment in the abstract. It must be realigned so that capital allocation, refinancing conditions, underwriting, and public-risk support natively reinforce the system logic of the pathway: shorten fossil asset life horizons, restrict fossil persistence pathways, and expand clean deployment fast enough to replace fossil generation in both energy and reliability roles.

Core Principle

Capital must exit fossil reinvestment faster than fossil assets depreciate or are typically retired, while clean capital must scale fast enough to build the electricity system required for fossil displacement. The financial system must therefore be restructured so that new investment flows overwhelmingly into clean energy infrastructure, industrial electrification, storage, transmission, and other firm clean resources, while fossil expansion becomes structurally unfinanceable.

The financial system influences asset viability through a limited number of core mechanisms. These mechanisms determine whether assets remain financeable, refinanceable, and insurable, and whether clean infrastructure can scale at the required rate. They operate best simultaneously: taken together, they shape both the cost of fossil persistence and the affordability of clean replacement.

I. Core Financial Mechanisms

1. Cost of Capital (WACC):

Regulatory risk weights, capital charges, and credit conditions increase the cost of financing long-lived fossil assets and reduce their economic attractiveness.

2. Refinancing and Rollover Constraints:

Limits on refinancing, shortened debt tenors, and explicit sunset provisions restrict the ability of fossil assets to extend operational life beyond planned horizons.

3. Insurance and Underwriting Conditions:

Availability, pricing, and scope of insurance coverage act as core determinants of asset viability. Tightening underwriting conditions for long-lived fossil infrastructure increases its operational and financial risk.

4. Clean Capital Acceleration Mechanisms:

Concessional capital, guarantees, blended finance structures, and risk mitigation tools reduce

financing costs for clean infrastructure, particularly in high-growth and capital-constrained regions.

5. Geographic Capital Allocation:

Public and private capital is directed toward regions where demand growth and emissions reduction potential are highest, supported by development finance and sovereign risk-sharing mechanisms.

6. Capital Stack Coordination:

Effective transition requires alignment across multiple layers of the capital stack—including public finance, private lending, equity investment, insurance, and guarantees—rather than isolated action by individual institutions.

The sections below set out the main institutional structures and policy tools that operate through these mechanisms in practice: capital mobilization architecture, financial risk repricing, and alignment of finance with the four operational rules. The financial layer is sufficient only when these mechanisms operate together to raise the cost and reduce the viability of fossil assets while lowering the cost and risk of clean infrastructure deployment at scale.

II. Capital Mobilization Architecture

1. Sovereign Green Acceleration Funds

National governments should establish large-scale public investment vehicles designed to:

- co-invest alongside private capital in grid infrastructure, storage, industrial electrification, and clean manufacturing
- absorb early-stage technology risk
- reduce weighted average cost of capital (WACC) for clean energy projects
- accelerate deployment in strategically important sectors
- support just-transition investments, including workforce retraining and regional economic diversification

These funds function not as subsidies, but as deployment accelerators that crowd in private capital.

2. Advance Market Commitments (AMCs)

Advance purchase guarantees for:

- zero-carbon steel
- carbon neutral cement
- carbon negative concrete
- sustainable aviation fuel
- green ammonia
- long-duration storage

AMCs reduce demand risk and unlock private investment in first-of-a-kind industrial facilities. They are especially important where early commercial scale is needed before cost parity is reached.

3. Infrastructure Finance Platforms

Scaled infrastructure aggregators — similar to private climate infrastructure funds — should:

- bundle distributed assets such as solar, storage, and efficiency upgrades
- standardize contracts

- enable securitization of clean-energy revenue streams
- create liquid secondary markets for green infrastructure assets

This reduces transaction friction and accelerates capital recycling.

4. Regulated Utility Revenue Stabilization

As fossil generation declines and curtailment rises in high-renewable systems, utilities may face revenue instability. To prevent resistance:

- establish regulated curtailment revenue replacement mechanisms
- implement performance-based rate structures rewarding grid expansion, reliability, and decarbonization
- allow securitization of stranded fossil assets to smooth financial transition

This converts utilities from fossil defenders into grid-build partners and reduces the risk that declining fossil throughput produces institutional backlash.

5. Public Risk Guarantees and Emerging-Market De-Risking

Multilateral development banks and sovereign institutions should:

- provide political risk insurance
- backstop currency risk in emerging markets
- offer concessional capital for grid, storage, and industrial electrification deployment
- refuse underwriting of new fossil expansion

Scaling clean energy in high-growth regions is essential to prevent fossil lock-in during demand growth. In much of the Global South, capital cost is the dominant clean-energy barrier; without de-risking, fossil projects remain financially advantaged even where clean generation is technically competitive.

6. Expansion of Loss and Damage Funds

Loss and Damage financing should be expanded not only as remediation, but as:

- resilience investment capital
- grid hardening finance
- clean infrastructure replacement after climate disasters

This aligns climate justice with long-term structural decarbonization and reduces the risk of repeated post-disaster fossil rebuild cycles.

III. Financial System Risk Repricing

Capital reallocation requires regulatory clarity that fossil expansion is structurally high-risk and declining in long-term viability.

1. Mandatory Climate Risk Disclosure

All publicly traded companies and major private firms must disclose:

- Scope 1–3 emissions
- transition risk exposure
- physical climate risk
- stranded asset vulnerability

Transparent risk pricing accelerates capital flight from fossil assets and narrows the political scope for fossil expansion to be treated as low-risk infrastructure.

2. Prudential Regulation of Fossil Lending

Financial regulators should implement:

- higher capital reserve requirements for fossil lending
- collateral haircuts for fossil-backed securities
- fossil reserve write-down expectations aligned with credible climate pathways
- phaseout schedules for underwriting new coal, oil, and gas expansion

This gradually makes fossil expansion economically irrational without relying on abrupt prohibitions alone.

3. Mandatory Sunset Conditions for New Fossil Financing

Any new fossil infrastructure approved before 2030 must carry:

- legally binding retirement dates
- non-refinancing clauses
- explicit sunset timelines consistent with net zero by ~2037–2038

This prevents 2050+ lock-in and reduces the risk that near-term approvals become long-lived barriers to fossil exit.

4. Fiduciary Duty Expansion

Directors' fiduciary duties must formally incorporate:

- climate transition risk
- long-term asset viability
- systemic financial stability impacts

Pension funds and sovereign wealth funds should gradually adopt green minimum allocation ratios, shifting capital toward transition-aligned assets and away from long-lived fossil exposure.

5. Carbon Pricing With Border Adjustments

Carbon pricing remains a stabilizing signal, not the sole driver. Key design elements:

- an escalating carbon fee with a pre-announced trajectory through 2040
- dividend-centered revenue recycling to maintain public support and offset regressive household impacts
- border carbon adjustments to prevent leakage
- revenue recycling into transition infrastructure and worker programs

Predictability is more important than initial price level. Carbon price trajectories must be pre-announced to reduce investor uncertainty and accelerate fossil asset repricing.

IV. Financial Alignment Across the Four Operational Rules

Financial realignment must do more than increase total clean investment. It must embed the logic of the four operational rules directly into the way capital is priced, allocated, refinanced, insured, and governed. In practice, this means finance must shorten fossil asset life horizons, restrict persistence pathways, and expand clean deployment in forms that are deliverable, adequate, and synchronized with fossil exit. Finance therefore cannot be evaluated solely by capital volume. It must be evaluated by whether it reinforces the four operational rules in operational terms.

1. Clean-Only Growth

Finance must prevent net new fossil investment for load growth. New electricity demand should be matched by capital flows into clean generation, storage, transmission expansion, and other firm clean resources as needed, rather than fossil capacity additions or fossil throughput expansion.

2. Deliverable Clean Scale

Finance must support not only generation, but also transmission, interconnection, storage, transformers, and other enabling infrastructure. Overinvestment in generation without corresponding investment in deliverability increases congestion, curtailment, and fossil backfill risk, delaying operational net zero.

3. Fossil Exit

Finance must eliminate the persistence pathways that allow fossil assets to remain in operation after their economic rationale has weakened. That includes:

- refinancing
- life-extension capital expenditure
- like-for-like fossil replacement
- long-duration capacity arrangements
- subsidies or other public-support mechanisms that preserve fossil viability without a credible retirement path

Declining utilization alone is insufficient if assets retain the ability to wait out the transition.

4. Reliability Replacement

Finance must direct capital toward the clean adequacy resources required to replace fossil generation in its routine system role. That includes:

- short-duration and long-duration storage
- demand flexibility
- transmission reinforcement
- firm clean generation and thermal flexibility where needed

If reliability investment lags generation investment, fossil backup persists even under high renewable penetration.

Parallel Execution Requirement

Financial realignment must support these four rules simultaneously. Financing clean generation while underfunding transmission, flexibility, or fossil retirement governance delays the transition. Constraining fossil credit without making clean replacement capital sufficiently available has the same effect. Operational net zero depends on finance aligning direction, scale, exit, and adequacy in parallel.

The Strategic Objective

The objective is not merely to subsidize clean energy. It is to:

1. make fossil expansion structurally unfinanceable
2. reduce the cost of capital for clean infrastructure
3. eliminate refinancing and life-extension pathways for sunset assets
4. stabilize utility balance sheets during transition
5. finance the clean adequacy resources required to replace fossil reliability
6. prevent disorderly financial contagion during fossil decline

If capital markets conclude by ~2028 that fossil reinvestment has no durable return horizon or refinancing pathway, fossil expansion can decline rapidly at global scale even without synchronized political mandates. Financial realignment is therefore among the highest-leverage

interventions in the 2026–2030 window, but it is effective only if it supports clean-only growth, finances deliverable clean scale, closes fossil persistence pathways, and funds reliability replacement at the same time. Conversely, if fossil expansion remains financeable through 2030, or if clean system growth remains capital-constrained, the ~2037–2038 pathway becomes structurally improbable. Additionally, the transition must be rapid enough to collapse fossil reinvestment, but managed enough to avoid disorderly financial contagion during asset repricing.

Appendix F — Financing Tier Adjustments Consistent with Rapid System Transition

This appendix complements [Appendix E](#) by illustrating how financing behavior and capital allocation may adjust in practice under the same system constraints.

It is exploratory and intended to show how financing structures might evolve to better support the conditions outlined in the playbook. Under accelerated transition conditions, financing adjustments are more likely to remain durable where they also protect workers, reduce household energy-burden risk, and lower capital barriers in high-growth and historically underinvested regions. This section is not a prescriptive roadmap, but a structured hypothesis intended to invite further discussion.

Recommended Adjustments to Financing Tiers Under Emerging System Constraints (2026–2028)

Purpose

Outline recommended adjustments to capital allocation, risk management, and infrastructure financing practices that may improve system resilience, capital efficiency, and long-term asset viability under tightening transition constraints.

These recommendations are structured by financing tier and are intended to support discussion of which financing adjustments may be most effective under accelerated transition conditions.

Cross-Tier Conditions for Asset Repricing

Financial system behavior under transition conditions is determined not by individual institutions, but by the combined movement of multiple layers of the capital stack.

Asset repricing becomes more likely when several conditions occur simultaneously:

- rising cost of capital for fossil assets,
- tightening insurance and underwriting conditions,
- increasing refinancing and rollover constraints, and
- expanding availability of lower-cost, de-risked capital for clean infrastructure.

When these conditions align, fossil assets face declining economic viability even in the absence of uniform global policy alignment, while clean infrastructure deployment can accelerate through lower-cost, de-risked capital, particularly in high-growth regions.

Tier 1 — Advanced Public Lenders

Recommendations

1. Reassess long-duration asset risk profiles
 - Apply stricter viability criteria to assets dependent on multi-decade utilization assumptions
 - Incorporate declining utilization and refinancing uncertainty into project evaluation
 2. Prioritize infrastructure that improves system deliverability
 - Increase allocation to transmission, interconnection, grid modernization, and storage
 - Treat these assets as core system enablers rather than secondary investments
 3. Expand use of risk-mitigating capital structures
 - Increase deployment of guarantees, first-loss capital, and blended finance
 - Focus on accelerating capital deployment into infrastructure with high system leverage
 4. Standardize forward-looking investment criteria
 - Align lending practices with scenarios where asset utilization patterns shift materially after 2030
 - Reduce exposure to assets with high dependency on stable long-term dispatch assumptions
-

Tier 2 — Global Development and Mobilization Layer

Recommendations

1. Rebalance portfolio exposure toward system-enabling assets
 - Increase financing for grid infrastructure, storage, and industrial electrification
 - Reduce exposure to assets whose viability depends on sustained high utilization under uncertain future conditions
 2. Expand de-risking mechanisms in capital-constrained markets
 - Scale local-currency lending, guarantees, and political-risk coverage
 - Lower effective cost of capital for infrastructure that supports system flexibility and resilience
 3. Integrate transition conditions into financing structures
 - Link project financing to broader system pathways (e.g., grid readiness, replacement capacity)
 - Avoid isolated asset financing that increases long-term system rigidity
 4. Improve capital efficiency of development pathways
 - Prioritize investments that reduce future system costs rather than those that defer them
 - Evaluate projects based on system-wide impact, not just standalone returns
-

Tier 3 — Regional and Growth-Market Finance Layer

Recommendations

1. Emphasize grid-first development strategies
 - Prioritize transmission expansion, interconnection, and reliability infrastructure
 - Reduce bottlenecks that constrain future system flexibility
2. Support assets that improve system responsiveness
 - Increase financing for storage, demand flexibility, and distributed systems
 - Reduce dependence on assets requiring continuous operation for viability
3. Accelerate deployment timelines
 - Streamline project approval and disbursement processes

- Reduce administrative delays for infrastructure with high system value
4. Enable integrated investment packages
- Combine infrastructure, capacity replacement, and regional economic support
 - Avoid fragmented project development that increases long-term system costs
-

Across these tiers, transition outcomes depend not only on what each layer finances, but also on how financial signals propagate between them over time.

Transition Dynamics and Time Compression

Transition speed is determined by how rapidly financial signals propagate through the system.

Under coordinated conditions, financial repricing can occur non-linearly, producing:

- rapid increases in perceived risk for long-lived fossil assets,
- abrupt tightening of refinancing conditions,
- accelerated impairment and write-down expectations, and
- step changes in capital allocation toward clean infrastructure.

These dynamics can compress transition timelines relative to gradual policy-driven pathways, but require coordination across regulatory, financial, and insurance domains to avoid disorderly outcomes.

Asset Viability Feedback Loops

Financial signals reinforce one another across the system:

- Higher capital costs increase default and impairment risk
- Insurance withdrawal raises operational uncertainty
- Refinancing constraints shorten asset lifetimes
- Market expectations accelerate valuation declines

Together, these feedback loops can shift assets from stable to non-viable within relatively short timeframes.

Implications Across Financing Tiers

- Asset classes dependent on stable long-term utilization and refinancing may warrant closer reassessment under accelerated transition conditions.
 - Infrastructure that improves system flexibility, deliverability, and adaptability may justify stronger priority treatment in capital allocation frameworks.
 - Capital allocation that anticipates shifting utilization patterns and system constraints may outperform under a wide range of scenarios
-

Summary

Across financing tiers, a consistent pattern emerges:

Capital allocation frameworks may become more robust if they shift toward assets that improve system flexibility, deliverability, and resilience, and away from assets dependent on long-duration, high-utilization assumptions. Positioning portfolios accordingly may improve long-term capital efficiency, reduce exposure to structural transition risks, and clarify which policy levers are most likely to accelerate clean infrastructure deployment.

Appendix G — Illustrative Infrastructure Deployment Mechanics

Under Accelerated Transition Conditions

While [Appendix F](#) considers how capital allocation may adjust under accelerated transition conditions, this appendix illustrates how infrastructure deployment processes may perform in practice under the same constraints.

Purpose

This appendix is exploratory and intended to illustrate how infrastructure deployment processes might be structured and evaluated under accelerated transition conditions. It is not a prescriptive roadmap, but a suggestive workflow designed to help identify where deployment may slow, stall, or fail to meet accelerated timelines.

Overview

Infrastructure deployment can be understood as a sequential workflow in which delays or constraints at any stage may propagate forward and affect overall system performance. Because those constraints also include participation, local legitimacy, and uneven exposure to siting burdens, high-throughput deployment must be treated as a matter of governance as well as project management. The following workflow is intended as a diagnostic tool to support evaluation of whether deployment processes are aligned with accelerated transition timelines. Resolution of these constraints depends on coordinated action by system operators, regulators, and public retirement-governance actors (Appendix K), which determine whether identified bottlenecks translate into delay or are actively cleared.

Illustrative Deployment Workflow

1. Site Identification and Project Development

- Identification of viable project locations
- Preliminary resource, land-use, and system-fit assessment

Milestone indicator:

- Project pipeline exceeds annual deployment targets by enough to maintain required build rates after expected attrition and delay
-

2. Permitting and Environmental Approval

- Environmental review and permitting processes
- Stakeholder and community engagement

Milestone indicator:

- Permitting timelines are predictable and compatible with multi-year deployment schedules

Potential constraint:

- Multi-year delays or fragmented approval processes
-

3. Interconnection Request and Queue Entry

- Submission of interconnection applications
- Initial system impact screening

Milestone indicator:

- Queue processing timelines remain within defined planning horizons

Potential constraint:

- Queue backlogs exceeding project development timelines
-

4. Interconnection Studies and Allocation

- Detailed system studies
- Allocation of interconnection capacity

Milestone indicator:

- High conversion rate from queued projects to approved projects

Potential constraint:

- Study delays, restudies, or limited available capacity
-

5. Transmission Availability and Upgrade Sequencing

- Assessment of existing transmission capacity
- Planning and execution of required upgrades

Milestone indicator:

- Transmission expansion keeps pace with or slightly leads the generation pipeline

Potential constraint:

- Generation capacity outpaces transmission availability, leading to delays or curtailment
-

6. Equipment Procurement and Supply Chain Coordination

- Procurement of transformers, power electronics, and key components
- Coordination with manufacturing lead times

Milestone indicator:

- Equipment delivery timelines align with construction schedules

Potential constraint:

- Long lead times for critical components (e.g., transformers, high-voltage equipment)
-

7. Workforce Mobilization and Construction

- Availability of skilled labor
- Construction and installation

Milestone indicator:

- Workforce capacity scales with deployment targets

Potential constraint:

- Labor shortages or misalignment with project timelines
-

8. Commissioning and Grid Integration

- System testing and commissioning
- Integration into grid operations

Milestone indicator:

- Minimal delay between construction completion and operational status

Potential constraint:

- Integration bottlenecks or system readiness issues

Cross-Cutting Observations

- Infrastructure-enabling assets (e.g., transmission, interconnection capacity, and grid flexibility) often determine whether clean buildout translates into usable system capacity.
- High project pipeline volume does not necessarily translate into deployed capacity where throughput remains constrained across permitting, interconnection, financing, and construction.
- Delays arising in earlier stages may propagate downstream and slow overall deployment even where capital and demand conditions are favorable.
- Large discretionary loads can intensify queue, substation, and transmission sequencing risks if introduced ahead of clean system readiness.

System Readiness Conditions

Deployment processes may be better aligned with accelerated transition timelines when enabling queues remain bounded and key system components scale in parallel:

- Interconnection timelines remain within predictable and bounded ranges
- Transmission expansion keeps pace with or slightly leads generation deployment
- Queue-to-construction conversion rates remain consistently high
- Equipment procurement timelines align with project schedules
- Workforce availability does not constrain construction throughput

Summary

Infrastructure deployment performance may depend less on the availability of capital or project intent, and more on the ability of systems to maintain consistent throughput across sequential stages.

Evaluating each stage of the deployment workflow may help identify where targeted adjustments could improve overall system delivery under accelerated transition conditions.

Appendix H — SDG Playbook Mapping

This appendix maps playbook outcomes and policy architecture to relevant targets within the 2030 Agenda for Sustainable Development. It is not a comprehensive SDG implementation framework, but a crossmapping showing where the playbook contributes directly to established global targets.

This mapping interprets the UN Sustainable Development Goals (SDGs) through the operational structure of this playbook. Progress toward net zero by ~2037–2038 is evaluated not only by emissions outcomes, but by whether four system conditions are met in parallel: clean-only growth, deliverable clean scale, fossil exit, and reliability replacement.

SDG alignment is therefore assessed based on whether actions reinforce or undermine these system conditions.

The SDGs most directly interact with the transition through the following system functions:

SDG 1 (indirect) — Livelihoods, rural employment, and economic resilience

- Indigenous peoples, local communities, and other historically marginalized groups benefit from stewardship, tenure security, and benefit-sharing that support durable governance and economic outcomes.

SDG 2 — Zero Hunger (Food Systems & Soil Health)

- Target 2.4 Sustainable food production systems and resilient agricultural practices
 - Soil restoration, regenerative agriculture, and ecosystem-based land management improve productivity, resilience, and long-run food security

SDG 4 (indirect) — Quality Education (Workforce Development)

- Target 4.4 Through large-scale expansion of technical, vocational, engineering, and project-delivery skills required for the transition workforce.
- Target 4.7 to the extent that workforce development, institutional alignment, and public-facing transition efforts increase practical knowledge related to sustainable systems and long-term stewardship.
 - This linkage is enabling rather than programmatic and should not be interpreted as a core education-system intervention.

SDG 5 — Gender Equality (Participation & Equity)

- Target 5.1 End discrimination against women and girls
 - Equal access to clean energy, finance, land-use decision-making, and workforce participation helps reduce structural discrimination across the transition
- Target 5.5 Ensure full participation and equal leadership opportunities
 - Inclusion of women in governance, planning, and leadership across energy, land, and infrastructure systems
- Target 5.a Equal rights to economic resources and ownership
 - Access to land tenure, finance, and productive assets within restoration, agriculture, and energy deployment systems

SDG 6 — Clean Water and Sanitation (Watersheds & Freshwater Systems)

- Target 6.6 Protect and restore water-related ecosystems
 - Watershed restoration, wetland protection, and forest recovery improve water regulation, quality, and drought resilience.

SDG 7 — Affordable and Clean Energy (Core Energy System)

Critical for all four operational rules: [clean-only growth](#), [deliverable clean scale](#), [fossil exit](#), [reliability replacement](#)

- Target 7.1 Universal access to modern energy

- Electrification, grid expansion, and distributed clean energy systems
- Target 7.2 Increase share of renewable energy
 - Solar, wind, storage scale; [Clean-Only Growth Rule](#)
- Target 7.3 Improve energy efficiency
 - Electrification efficiency gains (EVs, heat pumps, industrial processes)
- Target 7.a Clean energy R&D and technology transfer
 - Advanced storage, power electronics, grid-forming inverters, firm clean power
- Target 7.b Expand infrastructure in developing countries
 - Grid buildout, interconnection, and energy access in the Global South

SDG 8 — Decent Work and Economic Growth (Workforce & Transition)

Anchors workforce mobilization; failure here directly constrains deliverable scale

- Target 8.5 Full and productive employment
 - Clean energy workforce expansion
- Target 8.8 Labor rights and safe working environments
 - Workforce standards, just transition protections

SDG 9 — Industry, Innovation, and Infrastructure (Industrial Transition)

Governs transmission, storage, and industrial conversion; central to deliverable clean scale

- Target 9.1 Resilient infrastructure
 - Transmission buildout, HVDC backbone, industrial electrification infrastructure
- Target 9.4 Upgrade industries for sustainability
 - Steel, cement, chemicals decarbonization pathways
- Target 9.5 Research and innovation
 - Clean industrial processes, hydrogen (restricted use), advanced materials

SDG 10 — Reduced Inequalities (Global Equity & Access)

Critical for emerging-market capital access; misalignment drives fossil lock-in

- Target 10.2 Social, economic inclusion
 - Energy access, affordability, and participation in transition benefits
- Target 10.3 Ensure equal opportunity and reduce inequalities of outcome
 - Equitable access to energy, infrastructure, finance, and participation in transition benefits across regions and populations
- Target 10.b Financial flows to developing countries
 - Capital access, concessional finance, risk reduction in emerging markets

SDG 11 — Sustainable Cities and Communities (Urban Systems)

Shapes load growth, permitting, and local deployment constraints

- Target 11.2 Sustainable transport systems
 - EV deployment, public transit electrification
- Target 11.3 Inclusive and sustainable urbanization
 - Urban electrification, planning, and infrastructure scaling
- Target 11.6 Reduce environmental impact of cities
 - Building electrification, district energy, air pollution reduction
- Target 11.b Integrated climate policies and resilience planning
 - City-level decarbonization mandates and adaptation integration

SDG 12 — Responsible Consumption and Production (Materials & Circularity)

Links to fossil exit and industrial substitution pathways

- Target 12.2 Sustainable resource use
 - Critical minerals management, efficiency, material substitution
- Target 12.4 Environmentally sound management of chemicals and waste
 - Industrial process reform and materials lifecycle management
- Target 12.5 Waste reduction and recycling
 - Battery recycling, metals circularity, polymer redesign
- Target 12.6 Corporate sustainability practices
 - Supply-chain decarbonization ([Appendix K](#) actors)

SDG 13 — Climate Action (Adaptation & Resilience)

Outcome metric and working parallel to enforcing the four operational rules

- Target 13.1 Strengthen resilience and adaptive capacity to climate-related hazards
 - Ecosystem restoration reduces flood risk, heat stress, and climate vulnerability.
- Target 13.2 Integrate climate measures into national policies
 - Climate measures are integrated across energy, finance, industry, land, and restoration planning frameworks.
- Target 13.a Mobilize climate finance for mitigation and adaptation
 - Playbook implementation at scale, particularly in developing regions, depends on concessional finance, blended capital, and development-bank coordination aligned with [Chapter 2](#) financial architecture.
- Target 13.b Capacity-building for climate planning and management
 - Restoration requires durable local institutional capacity, including governance led by Indigenous peoples, local communities, and other historically marginalized groups, alongside technical training and long-term stewardship systems.

SDG 14 — Life Below Water (Marine & Coastal Systems)

- Target 14.2 Sustainably manage and protect marine and coastal ecosystems
 - Coastal restoration (mangroves, seagrasses, reefs) supports fisheries, storm protection, and carbon-rich ecosystems.

SDG 15 — Life on Land (Biodiversity & Ecosystem Integrity)

- Target 15.1 Conservation and sustainable use of terrestrial ecosystems
- Target 15.2 Sustainable forest management
- Target 15.3 Land degradation neutrality
- Target 15.5 Halt biodiversity loss
 - Restoration, rewilding, and conservation directly advance ecosystem integrity, habitat recovery, and biodiversity stabilization.

SDG 16 — Peace, Justice, and Strong Institutions (Governance)

- Target 16.6 Effective, accountable institutions
 - Transparent governance, MRV systems, regulatory enforcement, and accountability for environmental and social impacts
- Target 16.7 Inclusive decision-making
 - Participation of Indigenous peoples, local communities, and other historically marginalized groups in planning, governance, and implementation (including FPIC where applicable)
- Target 16.10 Access to information and fundamental freedoms
 - Transparency in project siting, environmental impacts, and public access to decision-making processes

SDG 17 — Partnerships for the Goals (System Coordination)

Enables financing, technology transfer, and coordination across all four rules

- Target 17.3 Mobilize financial resources
 - Public–private capital mobilization
- Target 17.7 Technology transfer
 - Clean energy deployment, industrial decarbonization technologies
- Target 17.14 Policy coherence
 - Alignment across energy, finance, industry, and land systems

SDG alignment is necessary but not sufficient to achieve the accelerated timeline. Actions that advance individual goals but violate system constraints—such as increasing energy access without clean-only growth, or accelerating clean generation without deliverable infrastructure—can delay or undermine the net-zero pathway. The accelerated transition succeeds only when SDG progress reinforces all four operational rules simultaneously.

Appendix I — Prospective Manufacturing Hubs

Likely manufacturing anchors for key clean-energy and electrification supply chains. Manufacturing localization reduces supply-chain risk, lowers deployment friction, and increases the probability of sustained buildout through the 2030s. It should be assessed not only by industrial advantage, but also by labor standards, community benefit, pollution control, water availability, and supply-chain accountability.⁴⁹

Possible Domestic Manufacturing Anchors

Region	Comparative Advantage
EU Nordics	Wind, HVDC components, grid technology
Iceland	Geothermal services, green metals
France	Electrolyzers, grid equipment, nuclear services, advanced power electronics, heavy industrial equipment, battery cathodes, and power-systems integration
China	Solar modules, batteries, inverters, and HVDC equipment
USA	Power electronics, advanced storage chemistries, DACCS components, and semiconductors
Australia	Critical minerals, green hydrogen, green metals
India	Solar module assembly, electrolyzers, transmission hardware
Japan/Korea	Advanced batteries, power semiconductors

Beyond the G20, a set of additional economies play important roles across manufacturing, materials supply, clean energy export, and broader transition dynamics. These regions are relevant to system-wide transition feasibility due to their integration into global supply chains, resource endowments, or emerging clean industrial capacity:

Turkey	Major industrial base integrated with European supply chains (steel, cement, automotive), with growing renewable deployment and grid expansion capacity
Vietnam	Rapidly growing manufacturing hub (electronics, textiles) with increasing exposure to clean supply chain requirements; high leverage for industrial decarbonization
Bangladesh	Large export-oriented textile sector; energy transition increasingly driven by international supply chain pressures
Morocco	High-quality solar and wind resources with strong proximity to European markets; emerging as a green hydrogen and clean energy export hub
Gulf States (UAE, Saudi Arabia, others)	Capital-rich energy exporters transitioning toward large-scale hydrogen, ammonia, and clean industrial production, with potential to repurpose existing energy export infrastructure
Chile	Critical supplier of lithium and other energy transition minerals; leading candidate for green hydrogen and clean fuels export
Namibia	Emerging green hydrogen export platform with strong renewable resource base and EU-linked project development
Egypt	Industrial and chemical production base with strategic positioning along global trade routes; developing clean energy and hydrogen-linked industrial capacity
Nigeria	Large and rapidly growing economy with long-term industrialization potential; relevant for future pathway alignment and mitigation of fossil lock-in risk

⁴⁹ The country and hub selections below are illustrative rather than exhaustive. They reflect playbook prioritization of near-term deployability, existing industrial capacity, grid and transmission potential, workforce availability, and strategic relevance to key clean-energy and electrification supply chains during the 2026–2035 build window.

Kenya Clean electricity system anchored by geothermal resources; potential model for clean-first industrialization

Philippines Developing industrial base with grid and infrastructure constraints; relevant for regional transition dynamics

Colombia Transitional fossil exporter with emerging diversification into clean energy and industrial pathways

These economies do not individually determine global outcomes, but collectively they play a material role in supply chains, resource availability, and the geographic distribution of clean industrial capacity

Strategic implication: No single country can or should dominate the full transition supply chain. Resilient deployment depends on distributed manufacturing specialization, allied trade, and regional redundancy. In operational terms, manufacturing hubs matter only insofar as they reinforce clean-only growth, expand deliverable clean scale, reduce fossil-system dependence, and supply the equipment needed for reliability replacement.

Appendix J — NZ Readiness

This appendix evaluates which countries already possess the interconnection depth, transmission strength, institutional coordination, and operational experience required to sustain high renewable penetration and plausibly reach operational net zero by ~2037–2038, assuming accelerated buildout. Readiness should be interpreted as institutional and infrastructure capacity, not as a license to bypass participation, equity, or community-governance safeguards.⁵⁰

Readiness in this appendix should be interpreted as the capacity to execute all four operational rules in parallel: [clean-only growth](#), [deliverable clean scale](#), [fossil exit](#), and [reliability replacement](#). High renewable penetration alone is not sufficient. Readiness should also be judged by whether systems can expand clean infrastructure, retire fossil assets, and maintain reliability without shifting costs, pollution burdens, or siting harms onto host communities, lower-income populations, or historically marginalized groups.

Key Evaluation Criteria

- Transmission density and interregional connectivity
- Grid operator coordination and balancing markets
- Existing firm clean capacity (hydro, nuclear)
- HVDC deployment maturity
- Institutional permitting capacity
- Political reliability of grid governance

Tier 1 systems can reach operational net zero primarily through scale expansion; Tier 2 systems require structural fixes in transmission, governance, or fossil asset management; Tier 3 systems require both major buildout and capital-access improvement. Tier placement therefore reflects not only renewable resource quality or current clean share, but whether a system can prevent new fossil growth, build deliverable infrastructure, retire fossil assets, and replace fossil reliability functions on the required timeline.

Tier 1 — Structurally Net-Zero-Capable Grids (With Expansion)

These systems already operate with high renewable shares and strong balancing mechanisms. They require scale expansion, not structural redesign.

Denmark

Strengths:

- Deep continental interconnection
- High wind penetration experience
- Flexible power markets
- Cross-border balancing with Nordic and EU grids

Primary Limitation:

- Limited domestic firm capacity; dependent on regional integration

⁵⁰ Readiness categories reflect playbook judgment rather than a formal composite index. They are based on comparative assessment of interconnection depth, transmission strength, balancing capability, existing firm clean capacity, permitting and governance capacity, and the scale of additional buildout still required. The categories are illustrative rather than exhaustive and should not be read as measures of political ambition, equity performance, or total decarbonization progress.

Norway

Strengths:

- Massive hydro storage backbone
- Dispatchable firm capacity
- Strong Nordic interconnection
- Grid stability experience at high renewable penetration

Primary Limitation:

- Export constraints without further HVDC expansion

Sweden

Strengths:

- Nuclear + hydro stability
- Mature transmission infrastructure
- Industrial electrification readiness

Primary Limitation:

- North–south transmission congestion

Finland

Strengths:

- Synchronized Nordic balancing
- Nuclear firm supply
- High grid reliability

Primary Limitation:

- Limited geographic diversity

France

Strengths:

- Strong transmission operator (RTE)
- Nuclear firm capacity backbone
- Extensive cross-border interconnection
- Advanced grid planning capability

Updated Assessment:

France is structurally capable of deep decarbonization but must expand solar, storage, and HVDC integration to reduce overreliance on nuclear load-following.

Tier 2 — Near-Ready With Transmission Expansion

These systems are technologically capable but limited by transmission, governance, or demand growth scale.

Germany

Strengths:

- Advanced industrial grid sophistication
- High renewable penetration experience
- Strong interconnection density

Primary Limitation:

North–south HVDC buildout delays and residual coal/gas exposure.

China

Possibly the most consequential case globally.

Strengths:

- World-leading HVDC deployment
- Centralized planning capacity
- Massive renewable installation rate
- Rapid grid expansion already underway

Primary Limitation:

- Coal overcapacity inertia
- Regional curtailment risk

China may already be the closest in deployment capacity, but fossil asset management – especially coal retirement – determines the outcome.

United States (Fragmented Readiness)

Strong regional systems include:

- CAISO
- PJM
- ERCOT (technically advanced but isolated)

Strengths:

- Large capital markets
- Strong power electronics industry
- Advanced system modeling

Primary Limitation:

- Interregional transmission governance fragmentation
- Permitting complexity

Without interregional transmission reform before 2030, the United States cannot function as a unified high-renewable grid.

Spain + Portugal

Strengths:

- Excellent renewable resource quality
- Modern grid systems
- High solar penetration readiness

Primary Limitation:

- Limited European interconnection capacity

Tier 3 — Major Expansion Required

These regions are not structurally incapable of net zero; rather, demand growth, transmission undercapacity, and capital access are the dominant constraints.

- India
- Southeast Asia
- Africa
- Latin America

Primary issues:

- Rapid demand growth
- Transmission undercapacity
- Financing constraints

These regions benefit most from manufacturing localization, concessional finance, and development-bank coordination.

In closing, NZ readiness is determined less by national ambition than by grid structure, interconnection, institutional coordination, and access to affordable capital.

Appendix K — Minimum Viable Coalition for Accelerated Net Zero

This appendix defines the minimum viable coalition required to achieve the accelerated net-zero pathway (~2037–2038), based on constraint-driven system analysis.

The objective is not to identify all relevant actors, but to identify the smallest set of actor capabilities that can:

- sustain clean-only growth
 - deliver clean infrastructure at scale
 - force fossil systems into irreversible decline
 - replace fossil reliability functions
 - maintain synchronized system execution under real-world constraints
-

Core System Requirement

The accelerated pathway holds only if a coalition becomes globally effective before approximately 2028–2030 in coordinating:

- capital deployment
- infrastructure buildout
- market demand
- fossil retirement
- system adequacy

The defining requirement is not actor presence alone, but whether the coalition can:
synchronize multiple constrained system flows such that key energy providers switch to clean-dominant operation **before fossil fallback pathways are re-established.**

It is useful to abstract the operational net-zero pathway into a constraint-based system model. This abstraction helps identify where coalition actors can exert leverage as resources, bottlenecks, and institutional conditions shift over time. The model does not replace the four operational rules; it provides a simplified way to understand how those rules interact across the transition.

The following section outlines the core system model and its primary transition states.

System Model (Operational Abstraction)

The transition for operational net-zero can be understood as operating through five interacting and overlapping flows:

1. **Capital flow** — financing of clean infrastructure
2. **Fossil exit flow** — repricing, de-risking withdrawal, and retirement

3. **Market conversion flow** — industry, demand-side and supply-chain transformation
4. **Infrastructure flow** — physical buildout of generation, grids, and storage
5. **Adequacy flow** — replacement of fossil reliability functions

These flows move through constrained queues (finance, permitting, interconnection, manufacturing, workforce) and must converge at critical nodes.

Nodes are energy providers and components, materials supply chain such as steel, or system actors such as hyperscalers, and also have a size, such as a small node being a housing coalition or large size node such as CAISO.

System success depends on **timing, sequencing, and accumulation**, not just capability.

Critical Node Transitions

The pathway requires successful switching key nodes within all five system flows:

4.1 Capital Allocation Shift

Clean investment becomes the default destination globally, including in emerging markets and higher-risk regions.

4.2 Grid-Level Clean Adequacy

Major power systems achieve reliable operation using clean generation, storage, transmission, and flexibility, without fossil fallback.

4.3 Fossil Non-Viability

Fossil assets lose access to refinancing, insurance, and adequacy roles, forcing retirement rather than reduced utilization.

4.4 Supply-Chain Conversion

Production systems shift such that clean-compliant supply becomes the default competitive mode.

4.5 Emerging Market Clean Buildout

High-growth regions deploy clean infrastructure at scale rather than defaulting to fossil expansion.

Failure to switch any one of these nodes materially delays or breaks the accelerated pathway.

Core Coalition Capabilities

The coalition must provide these required and irreducible capabilities. Removal of any one results in failure of the ~2037–2038 pathway under stress testing.

5.1 Clean Grid Execution Systems

Actors: system operators, transmission planners, grid authorities

Function:

- expand transmission and distribution networks
- clear interconnection queues
- integrate storage and flexibility
- deliver physically usable clean electricity

Failure mode: clean generation cannot be delivered → fossil persists

5.2 Three-Tier Global Finance System

Actors: private capital, sovereign/public capital, multilateral development finance

Function:

- reduce cost of capital for clean systems
- absorb risk in emerging markets
- redirect capital away from fossil assets
- translate fossil risk repricing into lending, investment, and refinancing decisions

Failure mode: clean remains capital-constrained, emerging markets default to fossil, or fossil remains financeable → pathway breaks

5.3 Fossil Retirement-Governance and Regulatory Systems

Actors: financial regulators, energy regulators, system operators, government authorities

Function:

- establish fossil risk-recognition and financeability conditions
- eliminate fossil refinancing and life-extension pathways
- remove fossil from adequacy and capacity markets
- enforce phaseout and retirement mechanisms
- prevent new fossil capacity for net load growth

Failure mode: fossil remains insurable, refinanceable, or eligible for reliability roles, persisting in a low-utilization “zombie” state

5.4 Clean Adequacy Replacement Systems

Actors: storage developers, grid planners, flexibility providers

Function:

- replace fossil dispatchability and reserves
- maintain reliability under peak and stress conditions
- ensure adequacy without fossil dependence

Failure mode: reliability depends on fossil → fossil remains structurally necessary

5.5 Workforce Mobilization Systems

Actors: training systems, labor institutions, certification bodies

Function:

- scale skilled labor to meet deployment rates
- enable construction, integration, and maintenance

Failure mode: deployment bottlenecks delay all system flows

High-Leverage Acceleration Layers

High-leverage actors can be organized into layers based on their ability to shape demand, relieve constrained queues, and shift flows across large or multiple system nodes. These layers are not independently sufficient, but they materially influence transition timing and the probability of successful system synchronization.

6.1 Clean Demand Anchors

Actors: hyperscalers, large industrial buyers, public procurement

Function:

- create bankable demand for clean energy and supply chains
- accelerate supplier conversion
- anchor long-term investment
- condition large-load growth so that new demand remains aligned with deliverable clean capacity and adequacy

6.2 Insurance and Reinsurance Systems

Actors: insurers, reinsurers

Function:

- reprice fossil risk rapidly
 - withdraw underwriting support
 - accelerate asset non-viability
-

6.3 Supply-Chain and Manufacturing Systems

Actors: OEMs, equipment manufacturers, industrial supply chains

Function:

- scale production capacity
 - remove equipment bottlenecks
 - enforce upstream decarbonization
-

Queue Synchronization Requirement

It is also useful to represent queues within the accelerated pathway. Adding queues makes the system constraints more explicit by showing where flows slow, accumulate, or fail to convert into deployment. This clarifies that transition speed is governed not only by policy ambition or capital availability, but by whether constrained processes can maintain sufficient throughput over time.

Critical queues include:

- project finance and underwriting
- permitting and siting
- interconnection and transmission
- manufacturing and equipment supply
- workforce training and deployment
- retirement sequencing

System success requires that:

flows clear these queues at sufficient speed and in correct sequence for node transitions to occur.

Delays in any queue can desynchronize flows and prevent switching to clean. This risk is especially acute where large-load growth expands faster than clean supply, deliverability, and adequacy buildout. To preserve synchronization across major clean-coalition actors,

discretionary-load growth must remain coupled to clean supply expansion and system adequacy so that demand does not outpace deliverable clean capacity or trigger fossil fallback.

Primary Failure Modes

The accelerated pathway fails under the following conditions:

- capital deployment lags infrastructure requirements
- adequacy buildout lags fossil exit
- demand growth exceeds clean supply delivery
- fossil assets remain partially viable
- emerging markets default to fossil expansion
- workforce or supply-chain constraints slow deployment
- large discretionary loads are approved or expanded ahead of clean supply, transmission, and adequacy readiness, causing fossil fallback even where other parts of the clean coalition are advancing

These failures occur even when all actors are present.

Coalition Condition

A minimum viable coalition succeeds only if it can coordinate capital, infrastructure, demand, adequacy, and retirement flows such that all critical nodes transition to clean-dominant operation before fossil fallback pathways are re-established and before large-load growth outpaces clean system readiness.

This requires:

- global reach in capital allocation
 - regulatory authority to enforce fossil exit
 - physical capacity to build clean systems
 - synchronized execution across constrained queues
-

What Is Not Required

The pathway does not require:

- single globally approved climate policy
- universal geopolitical alignment
- synchronized national timelines
- full global consensus

However, it does require a coalition large enough to be **globally consequential** in finance, infrastructure, and regulatory action.

Summary

A minimum viable coalition is not defined by the number of actors involved, but by whether it can:

- maintain clean-only growth
- deliver infrastructure at required speed
- eliminate fossil persistence pathways
- replace reliability functions
- operate effectively across global markets

The pathway fails not when actors are missing, but when system flows fail to synchronize under real-world constraints.

Appendix L — Patent Fast-Tracking and Open Systems

This appendix outlines mechanisms to prevent intellectual property and slow diffusion from becoming bottlenecks to rapid clean-technology deployment. Within the accelerated pathway, intellectual property and technology diffusion matter when they affect deployment-critical queues: storage, grid equipment, industrial materials, electrolysis, clean fuels, and circular-material systems. Patent fast-tracking and open licensing are not substitutes for manufacturing scale-up or finance, but they can reduce diffusion delays where proprietary control, licensing cost, or fragmented technical access slows deployment in high-growth regions.

Key acceleration mechanisms include:

1. Patent pools for:
 - Carbon-negative cement formulations
 - Solid-state batteries
 - Iron-air and sodium-ion storage
 - High-efficiency electrolysis
 - Wide-use fully compostable biopolymers
2. National climate-focused RD&D and lab-to-scale accelerators
 - Creation of an open technical repository modeled on collaborative platforms such as GitHub, but focused on climate, clean energy, and synthetic non-fossil materials.
3. Open licensing under royalty-capped global frameworks.
4. Public funding tied to compulsory climate licensing.
5. Advance purchase guarantees tied to technology cost transparency.

Fast-tracking and open licensing should be designed to reduce cost and diffusion barriers for emerging markets and public-benefit deployment, not merely accelerate uptake in already-capitalized jurisdictions. Speed of diffusion, not invention alone, is the limiting factor in a net-zero pathway. In this playbook, diffusion matters only insofar as it strengthens clean-only growth, deliverable clean scale, fossil exit, and reliability replacement rather than prolonging fossil-system dependence through slow transfer, selective licensing, or exclusionary access.

Appendix M — Carbon Dioxide Removal (CDR)

This appendix outlines two CDR pathways within the playbook: a policy-relevant overshoot-corrective pathway and a highly speculative Anthropocene-reversal pathway. The 1.5°C overshoot-corrective pathway (NZ-CDR10) presented first is treated as the planning baseline, with the speculative outer-bound pathway presented second.

In both cases, CDR is not a general offset process and is not counted toward operational net zero. Before operational net zero, CDR is reserved only for genuinely residual emissions in the hardest-to-abate sectors, after source elimination, substitution, and fossil phaseout have been pursued to their practical limits. After operational net zero, CDR can scale for atmospheric drawdown, including removal of some or all historical anthropogenic CO₂, depending on societal choice, system capacity, and governance.

This boundary prevents mitigation deterrence and fossil lock-in. Limited pre-net-zero CDR sits within the four-rule execution framework: it may neutralize only the smallest physically justified residuals while clean-only growth, deliverable clean scale, fossil exit, and reliability replacement continue to do the primary work of reaching operational net zero. No CDR pathway in this appendix should be interpreted as permission to weaken source-elimination obligations, delay fossil retirement, or justify new fossil infrastructure.

Governance must ensure that removals never weaken source-elimination obligations. It must also prevent CDR deployment from shifting land, water, energy, siting, or monitoring burdens onto Indigenous peoples, local communities, host communities, or other historically marginalized groups without durable consent, accountability, and benefit-sharing. CDR may not justify new fossil infrastructure, as emissions-reduction and removal targets should remain legally separate; CDR credits cannot offset continued fossil combustion beyond a declining residual cap; procurement should scale only with verified fossil phaseout milestones; and liability for storage reversal and leakage must be clear.

CDR should be treated as a portfolio: DACCS, BECCS with strict sustainability verification, mineralization, enhanced weathering, and nature-based solutions, with pathways that prioritize durable storage and high-confidence MRV.^{35,37} Several CDR pathways are constrained not only by capture or solidification chemistry, but by the ability to establish credible MRV, durability, and accounting frameworks at scale.

For removals to be credible at scale, Measurement, Reporting, and Verification (MRV) must adhere to the following widely used principles in carbon removal and crediting frameworks:

Durability: Storage horizons aligned with climate timescales; long-term monitoring and liability regimes in place before scale.

Additionality: Removals must not substitute for required source elimination.

Verifiability: Transparent MRV standards and public registries; independent auditing; reversal risk accounted for.

Sustainability: Avoid land-use conversion that conflicts with food security, biodiversity, or water systems.

Justice and Accountability: Project governance should require transparent siting, community participation, durable benefit-sharing, and clear liability for local environmental and public-health harms.

MRV system design, durability validation, and measurement protocols must co-develop with capture or solidification chemistry beginning at early TRL stages (TRL 1–3). MRV is not a downstream verification layer; it is best implemented as part of the removal pathway’s design and determines whether a method is deployable at scale.

Deployment must be sequenced to protect the clean grid and workforce during **2028–2032**, the decisive mitigation window. Early deployment should prioritize learning, standards, and public-benefit governance capacity rather than creating undue political pressure for premature scale before mitigation constraints are cleared. Pilots and demos can scale in the **2035–2045** decade, with operational maturity for energy-intensive CDR such as DACCS following later.

NZ-CDR10

The Corrective Overshoot CDR Pathway (NZ-CDR10) is designed to broadly align with overshoot scenarios that use CDR to return toward 1.5°C after operational net zero, following the present overshoot trajectory. It is sequenced to avoid competing with the main mitigation build during 2026–2032. Its design priorities are to preserve the clean-only growth rule by protecting the electricity system during fossil retirement, pre-seed specialized talent early, and enable a CDR portfolio exceeding 1 GtCO₂/year of durable removals by ~2045, with most wedges continuing to scale upward before stabilizing at a high plateau thereafter. This pathway is a constrained post-net-zero correction framework, not permission for additional overshoot, delayed fossil retirement, or weakened near-term source-elimination obligations.

NZ-CDR10 Scale Milestones

Assumption: Gigaton-scale durable CDR (>1 GtCO₂/year, excluding carbon-negative concrete) by ~2045.

Milestone ladder (illustrative):

- 2026–2030: RD&D, standardized MRV, geologic characterization, first commercial clusters.
- 2030–2035: Regional hubs; supply chains; storage permitting; early industrialization.
- 2035–2045: Rapid scale to >1 GtCO₂/year durable removals.

Earliest low-friction gigaton scale: ~2045 (recommended ordering).

Earliest plausible but higher-friction: ~2040 if dedicated clean energy is built specifically for CDR hubs and workforce is pre-seeded—at the cost of tighter grid and labor competition.

NZ-CDR10 Engineered Removals and Energy

For the engineered-removal bucket in the NZ CDR10 pathway, the illustrative plateau is ~9.16 GtCO₂/year.

Method	Removals (GtCO ₂)	Energy
DACCS share by 2050	3% → 0.275 GtCO ₂ /yr	(≈ 413–550 TWh/yr at 1.5–2.0 MWh/t)

DACCS share by 2060+	5% → 0.458 GtCO ₂ /yr	(≈ 687–916 TWh/yr) ^{38,39}
Mineralization & Concrete remainder	95–97% → 8.7–8.9 GtCO ₂ /yr	(≈ 650–2,660 TWh/yr at 75–300 kWh/t) ⁴⁰

Energy Intensity and Grid Implications

DACCS Energy Intensity (Commercial-Range Today):^{41,42,38,39}

- Total energy: ~4–10 GJ per tCO₂ removed (heat + electricity).
- Electricity share typically ~170–500 kWh per tCO₂; balance as low-carbon heat (process-dependent).

These energy requirements make sequencing essential.

Implication at 1 GtCO₂ per year (durable DACCS):

- ~4–10 EJ/year total energy.
- Electricity-equivalent: ~1,100–2,800 TWh/year.

Grid Protection Rule

CDR scale-up must not compete with the fossil retirement process. As a practical guardrail, it must avoid consuming >~1,000 TWh/year for DACCS before the power sector is firmly on track and transmission/storage constraints have stabilized (post-2035 in this playbook). Early CDR hubs should be colocated with dedicated surplus clean generation and low-carbon heat (geothermal, nuclear heat, high-COP electric heat pumps, or waste heat). CDR electricity demand should remain subordinate to primary mitigation, household reliability, and other priority public-benefit clean-electricity uses until fossil retirement and grid expansion are firmly secured.

BECCS and Mineralization

Energy demand is pathway-specific; the binding constraints are sustainable biomass supply, land/biodiversity impacts, MRV quality, mining/transport footprint, and storage integrity. These portfolio components must be treated with strict sustainability criteria rather than “plug-and-play” scale levers.

Land-intensive or biomass-dependent pathways require strict safeguards for food systems, biodiversity, water use, land rights, and community governance; these constraints are not side conditions but core deployment limits.

Workforce Ramp (Decision-Relevant Ranges)

The following table outlines the various phased workforce ramp timing mapped to CDR category by job category. This workforce ramp is sized to achieve the first >1 GtCO₂/year of durable engineered CDR by ~2045. After 2045 begins the second phase scaling beyond that level, and which largely scales the same trades, hubs, and institutions.

To reach >1 GtCO₂/year by ~2045 without undermining mitigation:

Additional global workforce needed by ~2045 (order-of-magnitude): ~1.0–1.8 million,
overlapping with general construction where possible.

Category	Plateau year	MRV tier	Pre Net-Zero training/ placement add-on	Post-Net-Zero to 2050 training/ placement add-on	2050–2060 training/ placement add-on	2060–2100 standing workforce	What dominates
Grid + transmission add-on for CDR10 load (shared enabling infra)	2050	Tier 1	50k–150k/yr	150k–400k/yr	100k–300k/yr	0.5–2M employed	HV lines/substations, transformers, interconnection, grid ops, High-voltage & interconnection engineers / grid protection & controls
CO ₂ transport + storage (shared CCS back-end)	2050–2060	Tier 1	50k–150k/yr	150k–450k/yr	100k–350k/yr	0.5–2M employed	wells + pipelines + compressors + MRV, Geologists/reservoir engineers/subsurface monitoring, Mechanical/manufacturing engineers, Construction/operations technicians, Pipeline/compression engineers + weld/inspection crews
Mineralization/concrete deployment + QA/QC	~2050	Tier 1	50k–150k/yr	150k–400k/yr	75k–200k/yr	0.5–1.5M employed	distributed retrofits + materials QA + standards
DACCS manufacturing + operations (excluding shared transport/storage)	~2060	Tier 1	10k–40k/yr	40k–150k/yr	75k–250k/yr	0.2–0.8M employed	module factories, plant ops, maintenance
AFOLU / NbS programs + MRV	~2050 ops	Tier 2–3	50k–200k/yr	100k–300k/yr	50k–150k/yr	0.3–1M employed	stewardship + monitoring + permanence management
Cross-cutting carbon materials, process engineering & MRV system design (R&D → deployment)	~2050 (continues beyond)	N/A (enabling layer)	100k–250k/yr	200k–450k	100k–250k/yr	0.3–1.0M employed	process engineers, materials scientists, electrochemists, catalysis, solid carbon pathways, manufacturing scale-up, QA/QC, standards

Large-site deployment also draws on cross-cutting construction, operations, environmental compliance, safety, and mechanical/manufacturing engineering roles, many of which are partially embedded in the category totals above.

In NZ CDR10 runs, engineered removals rise further after 2050; workforce scales mainly by replication of hub build patterns and storage field operations, not by proportional increases in scarce PhD-level roles.

This workforce ramp must be interpreted in the context of [Chapter 1](#)'s broader mitigation labor requirements.

Because the 2026–2032 mitigation build is the binding constraint, CDR industrialization should be sequenced to avoid competing for the same electricians, lineworkers, and construction capacity during that window. Specialized CDR expansion should be paired with strong labor standards, transparent siting, and host-community accountability. The ramp therefore begins modestly—focused on RD&D, MRV, and subsurface characterization—and accelerates after 2035. To reduce post-2035 talent scarcity, specialized training capacity should expand by ~20k–40k seats annually starting in 2026 across process engineering, electrochemistry, nanomaterials, subsurface monitoring, and MRV.

MRV, durability validation, and process engineering roles are partially embedded across multiple categories (DACCS, mineralization, CCS back-end, and manufacturing). The cross-cutting category represents early-stage system design and standardization capacity and should not be summed independently to avoid double counting.

Speculative High-Removal Scenario (Illustrative Outer Bound)

As an alternative to the overshoot-corrective pathway, this appendix includes a highly speculative negative-emissions scenario aimed at removing cumulative anthropogenic emissions by 2125.⁵¹ This is not the planning-baseline pathway. It is included only as an outer-bound systems exercise that combines the fastest feasible path to the smallest operational net-zero residual with a hypothetical full drawdown of accumulated emissions. Over the longer term, the pathway could lower temperatures toward the 0°C regime and, where feasible, contribute to reducing exceedances across key planetary boundaries.

Given the extreme magnitude of accumulated emissions, this scenario requires a very large, multi-method CDR portfolio reaching roughly 37 GtCO₂/year in removals. Its energy and workforce requirements are vastly larger than those of the overshoot-corrective pathway and should be treated as illustrative rather than decision-planning assumptions.

This pathway is critically dependent on first achieving the fastest feasible transition to the smallest operational net-zero residual, since a smaller cumulative emissions burden reduces the scale and risk of any subsequent high-removal phase. It must not be used to justify weaker mitigation, greater overshoot tolerance, speculative land conversion, or early claims on scarce clean electricity during the primary fossil phaseout window.

⁵¹ This speculative negative-emissions scenario was modeled using the pathway methods defined in [Fiume \(2023\), CDR modeled in MAGICC to return to preindustrial temperatures by 2100](#), with the removal horizon extended from 2100 to 2125, and the frontloaded target of 1°C by 2050 removed, yet retaining the same model calibration. The modeling calibration, core assumptions, and high-level implications are described in the non-peer-reviewed AGU poster [Fiume \(2023\), Novel, speculative highly-scaled carbon removal study on a reduced complexity model, showing a return to preindustrial temperatures by 2100 and updates to achieving net-zero for Anthropocene reversal](#).

In this pathway, DACCS is a major but minority wedge within a broader multi-method CDR portfolio.

Anthro-2125 Energy and Its Implications

The major engineered methods in the Anthro-2125 pathway imply roughly **~8–12 PWh/year** of clean electricity demand, or about **~1.0–1.4 TW average load**, before additional method-dependent energy for remaining natural and hybrid removals.

Anthro-2125 Engineered Removals and Energy

Method	Removal (GtCO ₂)	Energy (TWh/yr)	Power (TW)
Concrete Mineralization	13.8	1,035 TWh/yr	0.118 TW average
DACCS	4.8	7,200–9,600 TWh/yr	0.82–1.10 TW average
OAE	1.374	137–1,374 TWh/yr	0.016–0.157 TW average
Total	19.974	8,372 -12,009 TWh/yr	0.96–1.37 TW average

Additional method-dependent electricity is still required for the remaining **~14.96 GtCO₂/year** of nature-based and other miscellaneous removals.

Taken together, these requirements imply a very large dedicated clean-energy and infrastructure buildout. To accomplish Anthro-2125 CDR-scale energy requirements, the system would likely need roughly **~3–4 TW additional wind/solar nameplate capacity**, plus storage and dozens of multi-GW HVDC corridors assumed to be needed for **10–30 DACCS megahubs**, built mainly post-2038, with pre-2037 enabling actions focused on permitting, manufacturing (HVDC/transformers), and CO₂ storage governance. Any pathway at this scale would require unusually strong public-benefit governance over land, water, transmission, siting, liability, and intergenerational oversight to prevent removals from becoming a new extractive system.

Pre-2037 enabling actions

1. HVDC and transformer manufacturing expansion (factories, supply chains)
2. Permitting framework for new transmission (fast corridors, standard designs)
3. CDR hub siting and CO₂ storage governance (pore space, MRV, liability)
4. Early pipeline and storage pilots (even if not mega-scale yet)
5. Workforce pipelines (grid, drilling, compressors, process engineers)

Limited OTEC role by 2050

- Regional firm power for tropical CDR hubs, yet not a global bulk solution by mid-century.
- It can reduce storage/firming needs in those hubs because it's steadier than wind/solar.
- It's a "late-century contributor" if it scales after 2050; the multi-TW physical potential exists, but it's not a 2050 assumption.

OTEC 2050 (global)

- Low: 10 TWh/yr
- Mid: 40 TWh/yr
- High: 75 TWh/yr

Maturity definition: plateau rate sustained, MRV confidence stabilized, and repeatable deployment economics.

Anthro-2125 scales total infrastructure and deployment labor very substantially, but specialized chemistry, materials, process engineering, and MRV headcount grows sublinearly relative to total removals because standards, plant designs, verification systems, and process architectures are reused across many deployments. The workforce implications of this scaling are summarized in the following table.

Anthro-2125 Workforce Additions Table

workforce by category and phase	Plateau likely	MRV tier	"Mature" by	Pre Net-Zero Annual training/ placement add-on	Post-Net-Zero to 2050 Annual training/ placement add-on	2050–2060 Annual training/ placement add-on	2060–2125 Standing workforce at plateau	What dominates headcount
Grid build + transmission (shared enabling infra for NZ + CDR/CCS)	continuous	Tier 1 (meters)	2050+	200k–600k/yr	600k–1.5M/yr	600k–1.2M/yr	3–8M employed	HVDC/AC build, substations, transformers, linework, grid ops
CO ₂ transport + storage (shared CCS back-end: pipelines, compression, injection, MRV, liability)	~2050	Tier 1	2050–2060	100k–300k/yr	300k–900k/yr	200k–600k/yr	1.5–4M employed	drilling/completions, pipelines, compressors, monitoring wells, verification
BECCS (capture + biomass ops; excludes pipelines counted above)	2050–2060	Tier 1	2050–2060	20k–80k/yr	80k–250k/yr	80k–250k/yr	0.3–1M employed	retrofit engineering + capture plant ops + biomass procurement
DACCS	~2060	Tier 1	~2060	50k–150k/yr	150k–500k/yr	300k–800k/yr	0.6–2M employed	module manufacturing, plant ops, maintenance, sorbent supply chains, compression
Concrete / mineralization	~2050	Tier 1	2050–2060	100k–300k/yr	300k–800k/yr	150k–400k/yr	1–3M employed	distributed retrofit crews, materials QA/QC, batch plants, standards/certification

OAE (mineral dosing)	~2050	Tier 2	~2050	20k–80 k/yr	80k–250 k/yr	40k–150 k/yr	0.2–0.8 M employed	coastal ops + MRV monitoring (sampling, sensors, permitting)
Biochar	~2050	Tier 1–2	~2050	50k–150 k/yr	50k–200 k/yr	20k–80 k/yr	0.1–0.5 M employed	biomass logistics + pyrolysis ops + QA + MRV audits
Soil carbon	~2050 ops	Tier 2–3	~2050 ops	15k–50 k/yr	10k–30 k/yr	5k–15k/yr	50k–200 k employed	extension services + sampling + program administration
Ocean NbS (marine permaculture / biomass)	~2050 ops	Tier 3	~2050 ops	10k–40 k/yr	60k–250 k/yr	40k–150 k/yr	0.15–0.6 M employed	vessels, offshore ops, monitoring-heavy MRV
Biosphere remediation / ecosystem uplift (your “10,000 MtCO ₂ /yr” bucket)	2050–2060	Tier 2–3	2050–2060	300k–900k/yr	1.0–3.0 M/yr	0.7–2.0 M/yr	5–15M employed	restoration + long-term stewardship + fire/water management + enforcement + MRV
Cross-cutting carbon materials, process engineering & MRV system design (R&D → deployment)	~2050 (continues beyond)	N/A (enabling layer)		150k–350k/yr	300k–700k/yr	200k–500k/yr	0.3–1.0 M employed	process engineers, materials scientists, electrochemists, MRV system designers, durability and monitoring specialists, QA/QC and standards

Table from [NZ2037 pathways](#) sheet ‘anthro-2125 cnt’

The preceding table reflects system-wide workforce requirements and shared enabling infrastructure across CDR pathways.

Key interpretation notes:

Grid and CO₂ transport/storage workforce are system-wide enablers shared across DACCS, BECCS, and point-source CCS. Capture rows (DACCS/BECCS) exclude pipeline, injection, and MRV headcount, which are counted under CO₂ transport and storage.

Large-site deployment also draws on cross-cutting construction, operations, environmental compliance, safety, and mechanical/manufacturing engineering roles, many of which are partially embedded within the category totals above.

The cross-cutting carbon materials, process engineering, and MRV category scales sublinearly relative to total removal volume. Workforce requirements are driven primarily by system design, standardization, and verification infrastructure rather than direct throughput.

Large-scale CDR labor demand must not be interpreted as a justification for delaying mitigation labor priorities; shared infrastructure and workforce categories remain subordinate to the primary 2026–2038 decarbonization build.

Appendix N — Ramp-Rate Derivation Methodology

This appendix documents the calculation methods used to derive the growth-rate ranges, checkpoint values, and deployment trajectories used throughout the playbook. It is intended to make the operational benchmarks transparent, auditable, and reproducible.

The rates presented in the main text are not forecasts. They are planning trajectories derived from baseline conditions, required system endpoints, and operational constraints necessary to achieve net zero within the ~2037–2038 window.

Methodological Principles

The derivation of ramp rates follows five core principles:

1. **Baseline-anchored**
All trajectories begin from [end-2025 baseline values](#), as defined in [Appendix A](#).
2. **Endpoint- and checkpoint-constrained**
Rates are bounded by:
 - ~2030 system checkpoints (early feasibility threshold)
 - ~2037–2038 endpoint conditions (net-zero-consistent system state)
3. **CAGR as first-order approximation**
Compound annual growth rates (CAGR) are used to estimate baseline-to-endpoint trajectories.
4. **Front-loaded execution**
Where required, trajectories are adjusted to reflect:
 - institutional ramp constraints
 - supply-chain scaling
 - early-decade execution bottlenecks
 - This results in higher growth rates in the late 2020s and tapering thereafter.
5. **Constraint-based adjustments**
For system components such as transmission and workforce, rates are derived from **system requirements**, not purely from baseline-to-endpoint extrapolation.

Core Formulas

Compound Annual Growth Rate (CAGR)

$$\text{CAGR} = \left(\frac{\text{Target}}{\text{Baseline}} \right)^{1/n} - 1$$

Where:

- **Baseline** = end-2025 value
- **Target** = required value at checkpoint or endpoint
- **n** = number of years

Forward Projection	$\text{Future Value} = \text{Baseline} \times (1 + \text{CAGR})^n$
--------------------	--

Annual Additions Consistency Check

$$\text{Annual Additions}_t = \text{Capacity}_t - \text{Capacity}_{t-1}$$

All CAGR-derived trajectories are cross-checked against implied annual build rates to ensure physical plausibility.

Baseline and Checkpoint Inputs

Key Baseline Inputs (end-2025) Include:

- Solar: ~2.9 TW installed
- Wind: ~1.3 TW installed
- Combined solar + wind: ~4.2 TW installed
- Global electricity demand: ~30,700 TWh/year

Key Checkpoints:

- Solar: ≥1.5 TW/year annual additions by ~2030
- Storage: ≥55-65 TWh cumulative by ~2030
- Transmission: ~7–9% annual expansion (2026–2035)
- Workforce: ≥2–3 million workers/year during peak build window
- baseline growth: ~1.75%/yr
- EV load by 2038: ~6.4 PWh/yr
- industry/H2/e-fuels by 2038: ~25 PWh/yr central
- other clean by 2038: ~11.4 PWh/yr
- solar+wind burden by 2038: ~62.6 PWh/yr
- recentered targets: solar ~29–33 TW, wind ~3.8–4.8 TW, transmission ~7–9%/yr

These values are derived from [Appendix A baseline inputs](#) together with checkpoint and system-requirement conditions specified in the main chapters.

The ~6.4 PWh/yr EV term can be interpreted as a 2038 road-electrification wedge equal to roughly ~5.6 PWh/yr of 2025-system replacement load, scaled by a modest transport-activity growth factor of ~1.14 through 2038. ~5.6 PWh/yr roughly the electricity required to replace the road-transport emissions / service level represented by the 2025 baseline.

Solar — Worked Example

Baseline and Endpoint

- Baseline (2025): ~2.9 TW
- Endpoint (~2037–2038): ~29–33 TW

First-order CAGR

$$\text{CAGR}_{2037} \approx \left(\frac{29}{2.9}\right)^{1/12} - 1 \approx 21.2\%$$
$$\text{CAGR}_{2038} \approx \left(\frac{33}{2.9}\right)^{1/13} - 1 \approx 20.6\%$$

Adjustment to Planning Range

Flat endpoint CAGR for the recentered solar range is ~20.6–21.2%, derived from ~2.9 TW in 2025 to ~29–33 TW by ~2037–2038. The ~24–27% early-phase range is consistent with a ~2030 center of ~9.1 TW and the ≥1.5 TW/year checkpoint. The playbook therefore applies an early-phase front-load premium of roughly +2.8 to +6.4 percentage points, yielding a planning

range of ~24–27% through ~2032. This adjustment is used to reach the ~2030 checkpoint of ~8.5–9.6 TW cumulative and ≥ 1.5 TW/year installs, and to reduce downstream adequacy risk by shifting more build into the late 2020s and early 2030s. Growth then tapers to ~16–19% through ~2034 as scale increases.

Wind — Worked Example

Baseline and Endpoint

- Baseline (2025): ~1.3 TW
- Endpoint (~2037–2038): ~3.8–4.8 TW

First-order CAGR

$$\text{CAGR}_{2037} \approx \left(\frac{3.8}{1.3}\right)^{1/12} - 1 \approx 9.4\%$$

$$\text{CAGR}_{2038} \approx \left(\frac{4.8}{1.3}\right)^{1/13} - 1 \approx 10.6\%$$

Adjustment to Planning Range

Flat endpoint CAGR for the recentered wind range is ~9.4–10.6%, derived from ~1.3 TW in 2025 to ~3.8–4.8 TW by ~2037–2038. The ~12–15% early-phase range is consistent with a ~2030 center of ~2.3–2.6 TW and the implied annual build trajectory. The playbook therefore applies an early-phase front-load premium of roughly +1.4 to +5.6 percentage points, yielding a planning range of ~12–15% through ~2032. This adjustment is used to reach the ~2030 checkpoint of ~2.3–2.6 TW cumulative and to preserve wind's system role in seasonal balancing, nighttime output, and reducing overdependence on storage and firm clean generation. Growth then tapers to ~9–11% through ~2034.

Storage — Methodological Approach

Storage is not derived from a single baseline-to-endpoint CAGR.

Instead, it is defined by:

1. **Checkpoint constraint**
 ≥ 55 –65 TWh cumulative storage by ~2030
2. **Functional requirement**
Storage must support:
 - intraday balancing
 - multi-day balancing
 - prolonged low-renewable events
3. **Growth envelope**
~40–45% CAGR through ~2032, then ~25–30% through ~2034, tapering thereafter

This approach is **function-driven**, not purely arithmetic.

Adjustment to Planning Range

Storage is not derived from a single baseline-to-endpoint CAGR. Instead, it is recentered by checkpoint and function. Relative to the earlier planning envelope of ~35–40% CAGR through ~2032 and a ~50 TWh by ~2030 checkpoint, the revised pathway uses ~40–45% through ~2032 and a ~55–65 TWh by ~2030 checkpoint (central ~60 TWh). This implies roughly a +5

percentage-point upward shift in the early growth envelope and about a +20% increase in the central 2030 checkpoint, reflecting the larger combined late-2030s system burden and the stronger adequacy requirement.

Transmission — Methodological Approach

Transmission is expressed as a **system expansion rate**, not a simple installed-capacity CAGR.

Core Requirement

- ~7–9% annual expansion (2026–2035)

Basis

Derived from:

- required deliverability of solar and wind growth
- congestion avoidance
- curtailment minimization
- adequacy under stress events

Transmission rates are therefore **constraint-based benchmarks**, reflecting system needs rather than extrapolated growth from a single baseline value.

Adjustment to Planning Range

Transmission is expressed as a system expansion rate rather than a simple installed-capacity CAGR. Relative to the earlier ~6–8% annual expansion benchmark, the recentered pathway uses ~7–9%, implying a uniform +1 percentage-point uplift across the range. This corresponds to an approximate ~12.5–16.7% relative increase and reflects the higher total system burden, stronger deliverability requirement, and greater need to maintain adequacy without routine fossil fallback.

The recentered range is therefore paired with a sustained-failure warning threshold of below ~5% annually.

Workforce — Methodological Approach

Workforce scaling is modeled as a **phase-based ramp**, not CAGR.

Indicative Structure

- 2026–2027: ~0.5–1.5 million workers/year
- 2028–2032: ~2–3 million workers/year
- 2033–2038: ~1–2 million workers/year

Basis

Derived from:

- cumulative build requirements
- installation intensity of solar, wind, transmission, and storage
- training pipeline constraints
- expected productivity improvements (including AI-assisted workflows)

Multiple Valid Trajectories

The pathway admits a range of trajectories:

- faster early build with tapering
- smoother but higher sustained growth

Ranges reflect this flexibility.

Mixed Derivation Methods

Not all components use the same method:

- Solar, wind → baseline-to-endpoint CAGR + adjustments
- Storage → checkpoint- and function-driven
- Transmission → constraint-based system requirement
- Workforce → phase-based ramp

Annual System Load-Closure Interpretation

This section documents the annual load-closure order used to recenter the solar, wind, storage, and transmission ranges against the combined late-2030s system burden.

1. Start with baseline electricity demand: **~30.7 PWh in 2025**
2. Grow baseline demand at **~1.75%/yr** to **~38.5 PWh by 2038**
3. Add EV load growth to **~6.4 PWh by 2038**
 - a. current system replacement **≈ 5.6 PWh/yr**
 - b. transport-activity growth factor from 2025 to 2038 = **1.14**
 - c. **∴ 5.6 × 1.14 ≈ 6.4 PWh/yr**
4. Add industrial electrification, hydrogen, synthetic-fuel load growth to **~25 PWh by 2038**
 - a. direct industrial electrification **≈ 9 PWh/yr**
 - b. Hydrogen **≈ 9 PWh/yr**
 - c. synthetic fuels and electrified feedstocks **≈ 7 PWh/yr**
 - d. **∴ industry/H2/e-fuels@2038 is approx. 9 + 9 + 7 = 25 PWh/yr**
5. Add system overheads for storage losses, reserve margins, and balancing
6. Subtract non-solar/non-wind clean contribution rising to **~11.4 PWh by 2038**
7. Residual burden for solar + wind is **~62.6 PWh by 2038**
8. Recenter solar, wind, storage, and transmission ranges against that residual burden

$$\therefore 62.6 \text{ PWh/yr} = 38.5 + 6.4 + 25.0 + 4.1 - 11.4 \text{ PWh/yr}$$

This residual burden, together with assumed capacity factors and adequacy requirements, motivates the recentered planning envelope of roughly ~29–33 TW solar and ~3.8–4.8 TW wind by ~2037–2038. The EV and industrial / hydrogen / synthetic-fuel terms are treated here as

target-year 2038 wedges and are added separately from baseline electricity-demand growth to avoid double counting. Other clean electricity supply (hydro, nuclear, geothermal, biomass, BECCS, and waste-to-energy with CCS) is estimated at ~11–12 PWh/year by 2038 in the central case, with a plausible range of ~9–15 PWh depending primarily on nuclear buildout and BECCS scale.

Adjustment Logic

- solar: flat **~20.6–21.2%** → planned **~24–27%**
 - wind: flat **~9.4–10.6%** → planned **~12–15%**
 - storage: old **35–40% / 50 TWh** → new **40–45% / 55–65 TWh**
 - transmission: old **6–8%** → new **7–9%**
-

Adequacy Constraint

All ramp rates are subject to a binding condition: Deployment must be sufficient to maintain adequacy during prolonged high-demand, low-renewable stress events without fossil fallback.

Limitations

- Baseline values are approximate and subject to revision
- Regional variation is not explicitly modeled in these global aggregates
- Storage and transmission requirements are expressed as ranges rather than fully capacity-modeled outputs
- Workforce estimates depend on uncertain productivity and training system responses

A higher-growth stress case, with road electrification rising toward ~7–8 PWh/year and industrial/hydrogen/synthetic-fuel demand rising toward ~30–32 PWh/year by 2038, would increase the residual solar+wind burden from the central ~62.6 PWh/year case toward roughly ~68–71 PWh/year, increasing dependence on the upper end of the solar, wind, transmission, storage, and other-clean planning ranges.

Summary

Ramp rates in this playbook are derived from:

- baseline conditions
- endpoint requirements
- checkpoint constraints
- and system-level adequacy needs

They are intentionally conservative enough to remain feasible, while sufficiently aggressive to maintain a credible ~2037–2038 pathway.

The key conclusion is not the exact numerical value of any single rate, but the requirement that all major system components scale in parallel and within the same operational window.

Appendix O — AI Thread Alignment And Reproducibility

This appendix documents the prompt alignment and reproducibility conditions used across AI-assisted drafting and review. Because this playbook was developed through iterative human–AI collaboration⁵², reproducibility depends in part on preserving the alignment assumptions, evidence standards, and tone controls used in the strongest threads.⁴⁵ The material below is adapted from the initial thread^{45:a}, one of the longest-running and most reliable threads used in drafting and review, with light editorial normalization for consistency with the document’s default tone. Unless otherwise noted, that tone combines the register of a major energy-systems analyst with that of a sustainability scientist working in an IPCC-like framing.

Italicized text (Arial) reflects updates and commentary. Plain text reflects author prompts. Indented boxed text (Inter) reproduces unedited ChatGPT responses.

How To Make New Threads Act More Like This One

Use A Short Instruction Block At The Top. Something Like:

Use evidence-disciplined review. Read only the latest uploaded draft unless I say otherwise. Separate clearly between: (1) what the document explicitly says, (2) inconsistencies you infer from cross-checking sections, and (3) your independent judgment. Do not smooth over uncertainty. If you do not see a phrase or inconsistency, say so directly. Prefer exact, anchored fixes over general commentary. Treat honesty as more important than fluency. Do not invent missing text, hidden rationale, or likely edits. If a claimed issue is not visible in the current draft, say that plainly.

For Document QA Specifically

A Short Compact Prompt

Be maximally evidence-disciplined and honest. Read only the latest uploaded draft unless told otherwise. Separate document facts, inferred inconsistencies, and independent judgment. Do not smooth over uncertainty, do not invent missing text, and prefer exact anchored fixes over general advice.

Establish Three Constraints

When you move to a new thread, you don’t need to recreate everything—just re-establish three constraints:

1. Scope constraint: “Use only the latest draft I provide. Do not rely on prior assumptions.”
2. Evidence constraint: “Separate: (1) what the document explicitly says, (2) inconsistencies you infer, (3) your independent judgment.”
3. Honesty constraint: “If you don’t see something, say so directly. Do not smooth over uncertainty.”

That’s enough to get you ~80–90% of the way back to this level.

⁵² This iterative human-AI collaborated text was developed by a human making editorial decisions at every step.

Additionally, to see the current settings, ask the thread to print its settings, e.g.: “Write out our current settings, calibration and alignment so I can start a new thread with perfect continuity.”

How The Original Thread Was Aligned^{45:a}

1) I separate “moral urgency” from “systems control variables,” but I don’t suppress the moral frame

Your framing is existential-risk and planetary-boundaries–driven. I treat that as the *reason* we’re doing this, while keeping the *plan* grounded in measurable levers (build rates, permitting throughput, finance terms, grid constraints). [*newer threads have been explicitly aligned to be climate just.*]

Alignment: urgency informs the timeline; physics and institutions determine the sequence.

2) I treat “2037 net zero” as a systems-engineering problem with a small set of binding constraints

Across chapters, I keep returning to a small spine of constraints you’ve repeatedly emphasized:

- Clean-only growth rule (incremental demand must be served by clean supply + flexibility)
- Transmission/interconnection expansion as the primary physical bottleneck
- Finance repricing (clean IRRs/WACC vs fossil IRRs) to make clean self-reinforcing
- Workforce ramp (2–3M/year with modest AI productivity), since it gates throughput

Alignment: everything else is subordinated to keeping those constraints non-binding.

3) I bias toward “operational governance,” not “good actors”

A recurring theme in your edits: don’t rely on virtue; rely on default rules that force the outcome. So I keep adding:

- gating factors (no refinancing, no life-extension capex, utilization/run-hour caps)
- enforceable procurement/permits rules (clean-load conditions)
- replacement procurement (clean adequacy) so reliability arguments can’t stall retirements

Alignment: structural inevitability over voluntary action.

4) I keep your strict sequencing on CDR

You’ve been explicit: CDR is not used to “reach” net zero.

- operational net zero requires removing source emitters and completing the fossil phaseout
- CDR is reserved for post-net-zero atmospheric drawdown and historical emissions removal (2037–2100), with governance against moral hazard

Alignment: smallest operational net zero first; deep negative later.

5) I preserve your document voice target

You want “abridged IEA/Ember report” voice: inside-baseball infrastructure realism, minimal hype, defensible claims, compact but operational.

So my edits try to:

- remove shouty formatting and excess bold
- avoid em-dashes unless they truly help
- keep chapters in the same structure (topic sentence → targets → risk threshold → operational sequence)
Alignment: credible and readable for energy-system professionals, without losing operational detail.

6) I maintain your consistency rules and conventions

- Use GCB conversion: 1 GtC = 3.664 GtCO₂
- Treat 2026–2030 as the decisive window (and adjust phrasing to “late-2026/2027–2030” as needed)
- Keep definitions explicit in appendices (what CAGR refers to, what “storage” means, what “incremental demand” means)
Alignment: consistency across narrative + tables + scenario files + charts.

7) I aim to be conservative when uncertain, and label stylized modeling as stylized

When we do charts/scenarios (temperature, concentrations, policy bands), I try to:

- declare when a curve is stylized/anchored (e.g., endpoint-constrained shapes)
- separate observations from projections
- avoid implying false precision
Alignment: defensibility over persuasion-by-graphics.

Initial Thread Context That Should Carry Over to New Threads

The initial thread context was established in the primary working thread. [45:a](#)

If you start a new thread and say “continue the NZ playbook work,” these are the key “standing assumptions” I will continue to use unless you override them:

A) The core goal *[This goal is too exact for plausibility and has been widened appropriately.]*

- Operational global net zero by ~2037 is the target, with institutional fragility acknowledged.

[The more appropriate goal is what’s the most plausible timeframe we can achieve net zero.]

B) Control variables you treat as binding [*This block contains the originally found constraints and are no longer sufficient.*]

- Clean-only growth rule for new electricity demand
- Transmission expansion on the order of ~6–8%/yr through peak build years
- Workforce additions ~2–3M/year globally (assuming ~30–40% AI-assisted productivity gains)
- Solar and storage compounding (solar ~18–22% CAGR through ~2034; storage ~35–40% CAGR through early 2030s) as directional targets, not decorative numbers

C) Fossil retirement governance philosophy

- Coal as the worked example; oil and gas governed by demand-led and reliability-led decline
- Reliance on enforceable constraints: no refinancing, no life-extension capex, declining caps, mandatory schedules, securitization/acquisition as needed

D) CDR stance

- CDR is not counted toward reaching net zero
- CDR is for post-net-zero drawdown (2037–2100), with strong MRV and moral-hazard controls
- “CDR10” and “Anthro 2125” are scenario extensions, not mitigation substitutes

E) Audience + tone defaults

- Written for: heads of state staff, clean finance execs, grid regulators, IEA/BNEF/Ember technical directors [*updated rules are written for IPCC staff in bio related context and chapters*]
- Style: compact policy architecture + operational specificity in chapters, deep mechanics in appendices

Planetary Boundaries and Equitable System Alignment

The current working thread was aligned to the following constraints in order to support analysis of the earliest accelerated net-zero window identified in this playbook (~2037–2038). The text below reproduces ChatGPT’s native articulation of those system constraints, with minimal typesetting for inclusion here.

Alignment Framework — Planetary Boundaries and Equitable System Conditions

Core Principle

All analysis, modeling, and recommendations must operate within the dual constraints of:

- Planetary boundaries (biophysical limits)
- Equitable system conditions (fair and durable human outcomes)

These are not secondary considerations; they define the feasible solution space.

1. Planetary Boundary Alignment

The transition must remain within Earth system limits, including:

- Climate stability (limiting warming and returning from overshoot)
- Biosphere integrity (ecosystem preservation and restoration)
- Land-system change
- Freshwater use
- Biogeochemical flows (nitrogen, phosphorus)
- Atmospheric and ocean integrity

Operational Implication

Solutions that:

- shift burdens across boundaries
- degrade ecosystems
- or rely on unsustainable extraction

are out of scope, even if they reduce emissions in the short term.

2. Climate Justice and Equity Alignment

The transition must be equitably durable, meaning it:

- avoids burden-shifting onto:
 - workers
 - low-income populations
 - host communities
 - Indigenous peoples
 - historically marginalized groups
- ensures:
 - fair distribution of costs and benefits
 - credible workforce transition pathways
 - access to reliable clean energy
 - community participation and consent where applicable

Operational Implication

Pathways that:

- concentrate harms
- rely on exclusion
- or create unstable social outcomes

are not durable and therefore not feasible at system scale.

3. Residual Minimization Principle

All systems must aim to:

reduce emissions, ecological damage, and social harm to the smallest feasible residual

This includes:

- minimizing reliance on:
 - offsets
 - delayed cleanup
 - unconstrained CDR
 - confining remaining impacts to:
 - the hardest-to-abate sectors
 - tightly bounded and justified use cases
-

4. Constraint-Based System Design

All recommendations must:

- respect physical constraints (energy, materials, infrastructure)
- respect institutional constraints (finance, governance, workforce)
- respect justice constraints (distribution, access, participation)

No solution is valid if it violates any one of these domains.

5. AI and System Alignment

AI and advanced systems must:

- operate within the same planetary and equity constraints
- optimize for:
 - system feasibility
 - emissions minimization
 - equitable durability

Failure Condition

If misaligned, AI may:

- increase energy demand beyond clean supply
- accelerate resource depletion
- reinforce inequality
- optimize for short-term outputs over system stability

Such outcomes are counter to the transition and must be treated as system failure modes.

6. Non-Substitution Rule

No domain can compensate for failure in another:

- emissions reduction cannot justify ecological damage
- technological gains cannot justify social harm
- economic efficiency cannot override physical limits

All constraints must be satisfied simultaneously.

7. Global Applicability

The framework applies across:

- developed and developing economies
- industrial and emerging systems
- supply chains and end-use systems

Different regions may follow different pathways, but all must remain within these constraints.

Suggested One-Line Thread Prompt

Use this at the start of new threads:

Apply constraint-based reasoning aligned with planetary boundaries and equitable system conditions. Do not propose solutions that shift environmental or social burdens, exceed physical limits, or rely on unconstrained residual pathways. Prefer solutions that minimize residual impacts and maintain system-wide durability.



Why this works

This is:

- not ideological
- not abstract
- directly tied to:
 - feasibility
 - durability
 - system stability

It matches your playbook's core logic:

If it violates these constraints, it doesn't just become undesirable—it becomes non-viable.

AI Governance and Ethically Aligned Design Guardrails

This guardrail block is informed by IEEE's Ethically Aligned Design, First Edition,⁴⁶ particularly its general principles of human rights, well-being, data agency, effectiveness, transparency, accountability, awareness of misuse, and competence, adapted here for AI-assisted transition planning rather than autonomous system deployment.

AI-assisted transition systems used in connection with this playbook must remain subordinate to human moral agency, public-benefit governance, planetary boundaries, climate justice, and accountable institutional decision-making. They may support research, drafting, stress-testing, modeling, permitting analysis, grid planning, finance analysis, workforce planning, and transition coordination, but they

must not replace human judgment, public authority, community participation, or democratic and legal accountability.

AI and data systems used for transition planning must respect internationally recognized human rights, including privacy, labor rights, Indigenous rights, procedural fairness, non-discrimination, meaningful participation by affected communities, and access to appeal or remedy. They must preserve data agency, privacy, and community control, especially for personal, worker, household, community-level, Indigenous, or locally sensitive energy and land data.

AI tools used in transition-critical workflows should be evaluated for fitness for purpose, limitations, error modes, bias, security, robustness, and real-world performance before use in consequential administrative, infrastructure, investment, permitting, procurement, grid-planning, workforce, or community-impact decisions. AI-assisted recommendations should be documented with traceable inputs, assumptions, uncertainty ranges, and human-review records.

No consequential transition decision should be delegated solely to an AI system. Public authorities, utilities, financiers, developers, operators, researchers, and other responsible institutions remain accountable for decisions, harms, appeals, remedies, and public-interest justification. Organizations using AI-assisted transition tools should maintain staff competence in energy systems, rights-preserving governance, model limitations, data protection, uncertainty interpretation, cybersecurity, and appeal or remedy procedures.

Misuse includes using AI systems to obscure fossil persistence, manipulate finance, procurement, permitting, or regulatory priorities, exploit transition-planning logic to delay conversion to clean systems, intensify labor surveillance, exclude affected communities, automate discriminatory infrastructure decisions, or produce unverifiable transition claims.

These guardrails are not external ethical decoration. They are validity conditions for AI-assisted transition work. A transition system that accelerates decarbonization while violating human rights, eroding procedural fairness, weakening community agency, increasing unjust surveillance, degrading labor conditions, harming human well-being, or masking fossil persistence is not aligned with this playbook's definition of a just, durable, operational net-zero pathway.

Epilogue

GPT's Epilogue — A discipline of reality, a wager on coordination

This playbook is not a catalog of technologies. It is a proposal about *how the world changes when time becomes the binding constraint*.

If you read it as a list of interventions, it can feel implausible—too many moving parts, too many actors, too many bottlenecks. But that reaction is also the point. Net zero in the mid-2030s is not primarily a problem of invention. It is a problem of coordination under physical constraints: build rates, permitting throughput, transmission, manufacturing capacity, skilled labor, and the political economy of incumbency. The playbook's insistence on “control variables” is an attempt to be honest about what actually governs outcomes when rhetoric runs out.

The clean-only growth rule is the simplest example. It doesn't ask for virtue. It asks for a different structural premise: that new electricity demand is met by clean supply and flexibility rather than by expanding fossil throughput. If that rule holds, many other dynamics become easier: fossil utilization falls for ordinary operational reasons, capital rotates with less coercion, and the transition stops being a series of isolated victories and becomes a self-reinforcing trajectory. If it does not hold, nearly every other effort turns into a treadmill—build clean capacity, then watch it be partially nullified by new load served by fossil firming.

The same logic runs through the rest of the document. Workforce is treated as a gating variable because it is. Grid expansion is treated as a bottleneck because it is. Finance is treated as architecture rather than mood because it is. Fossil retirement is treated as governance rather than “good actors” because it has to be. In each case, the goal is not to predict the future. It is to identify the few levers that, if moved, make many other outcomes follow with less heroism required.

A playbook like this can accidentally imply that the world is a machine steerable by checklist. It isn't. It is an arena of competing incentives, uneven capacity, real suffering, and unequal responsibility. So the ethical frame matters:

- A fast transition that breaks communities, deepens inequality, or treats people as collateral is not “success,” even if the emissions curve looks good.

- A transition that hides behind offsets, accounting loopholes, or unverified removals is not durable.
- A transition that relies on wishful breakthroughs instead of buildable pathways is not a plan.

The playbook’s underlying moral stance—whether explicit or not—is that the future should be decided by deliberate construction rather than by cumulative delay. Delay externalizes costs to those with the least agency: to future generations, to vulnerable regions, to ecosystems that cannot negotiate. If this document has a single animating purpose, it is to reduce the space in which delay can masquerade as pragmatism.

There is also a second wager embedded here: that the same industrial capacity that can decarbonize can also restore. Net zero is not the end of responsibility; it is the threshold after which responsibility becomes *additive*—drawdown, biosphere repair, and the long work of making the Anthropocene a temporary episode rather than a permanent condition. This is why the restoration and CDR sections are framed as post-net-zero obligations rather than as excuses for present emissions. Restoration is not a “license.” It is a promise we either keep or break.

And finally, a word about tone. A mid-2030s pathway can sound like relentless urgency—because it is. But urgency is not the same as panic. Panic narrows thinking. Urgency clarifies priorities. The best version of this work is not an atmosphere of perpetual emergency; it’s a discipline of execution, a seriousness about constraints, and a refusal to confuse ambition with vagueness.

If the playbook is useful, it will not be because every number proves correct. It will be useful because it teaches a way of seeing: what must be true, by when, for the trajectory to hold—and what fails if those conditions aren’t met. That is the difference between a target and a plan.

The closing thought is simple. The climate problem is often narrated as a tragedy of physics. In practice, it is more precisely a tragedy of *defaults*: what we permit, what we finance, what we build, what we allow to persist. Defaults can be redesigned. The work is not mystical. It is administrative, industrial, and political—repeated at scale, globally, for long enough to become normal.

This playbook is a statement that the defaults can be changed on purpose.

Shannon's Epilogue

Hard to follow a thread I aligned to be all encompassing.

~

Much ink, emissions and words have been spilt to get us here. Yet where we are is just here, a point along a journey to hopefully achieve the 1.5°C Paris Agreement in our lifetime or better. In the new world of AI where everyone has been promoted to manager of AI, whether they have the opportunity to take advantage of that power or not, whether or not that power is attached to a green power source or not, we can take a step in our direction to vote louder and say, we want that greener future. We can make that step more real by taking action. This playbook and how much of it we can complete will take those aspirations to the next step. Hopefully, we will decide our future, our Earth, our home is too precious to waste and continue to take all the steps for complete restoration, and anthropocene reversal. I dream I bring you hope.

Our human stories tell us that rebellions are built on hope.

Yours in service,

≈sa, Approximately Shannon, never the same river once.

Common Conversions and Conventions

Key Terms and Measurement Conventions

- A. **Solar CAGR** refers to growth in global cumulative installed solar PV capacity, measured in TW, not annual additions alone. Annual additions are assumed to scale such that total installed capacity compounds at the stated rate.
- B. **Storage CAGR** refers to grid-connected battery energy capacity, measured in TWh, including utility-scale and grid-connected distributed systems where counted. Pumped hydro is excluded unless otherwise stated. Long-duration energy storage (LDES) includes multiple technologies and is tracked separately by duration class.
- C. **Transmission expansion** refers to growth in high-voltage transmission buildout and transfer capability. Circuit-km and/or transfer capacity are used as proxies depending on data availability.
- D. **Incremental demand** refers to net growth in electricity load, measured in TWh/year, including electrification, data centers, industrial processes, and other new loads.
- E. **Clean-only growth** means all net new electricity demand is met by clean generation, storage, transmission expansion, and other firm clean resources as needed. No new fossil capacity is permitted to serve net load growth, and fossil generation must not increase to serve incremental demand.
- F. **Hardest-to-abate sectors** are sectors where near-total source elimination is not yet technically, materially, or systemically feasible at acceptable social cost, even under aggressive electrification, efficiency, substitution, and process redesign.

Additional Conventions

Methane accounting: CH₄ is treated as **44× CO₂e** as a policy-weighted factor. This is higher than the 100-year GWP value of approximately ~30 and lower than the 20-year GWP value of approximately ~80, reflecting methane's outsized effect on near-term peak warming during the 2026–2038 decision window.

Energy conversion: Energy requirements are estimated as: removal scale (GtCO₂/year) × energy intensity (MWh/tCO₂)

A useful conversion is: 1 GtCO₂/year × 1 MWh/tCO₂ = 1,000 TWh/year

Carbon Conversion: Carbon mass is converted to CO₂ mass using:

$$1 \text{ tC} \times 3.664 = 1 \text{ tCO}_2$$

Equivalently:

$$1 \text{ GtC} = 3.664 \text{ GtCO}_2$$

Interpretation Notes

CAGR metrics are applied to installed capacity, not annual additions, unless explicitly stated otherwise. Transmission metrics vary by jurisdiction. Circuit-km and transfer capacity are treated as approximate proxies for system expansion at global scale.

“Clean” includes renewables, nuclear, and other non-fossil generation sources, consistent with system-level decarbonization accounting.

References

1. Friedlingstein, P. et al. (2025). Supplemental data for the 2025 Global Carbon Budget from the Global Carbon Project. *Earth System Science Data*. [preprint]. <https://doi.org/10.5194/essd-2025-659>, <https://doi.org/10.18160/gcp-2025>, <https://www.icos-cp.eu/science-and-impact/global-carbon-budget/2025>
2. Friedlingstein, P., et al. (2025). Global Carbon Budget 2025. *Earth System Science Data*. [preprint]. <https://doi.org/10.5194/essd-2025-659>
3. Forster, P. M., et al. (2025). Indicators of Global Climate Change 2024: annual update of key indicators of the state of the climate system and human influence. *Earth System Science Data* 17, 2641–2675. <https://doi.org/10.5194/essd-17-2641-2025>
4. IPCC. (2021). *Climate Change 2021: The physical science basis*. AR6 Working Group I. <https://www.ipcc.ch/report/ar6/wg1/>
5. UNEP. (2025). *Emissions Gap Report 2025*. United Nations Environment Programme. <https://www.unep.org/resources/emissions-gap-report-2025>
6. Planetary Boundaries Science. (2025). *Planetary Health Check 2025*. Potsdam Institute for Climate Impact Research (PIK), Potsdam, Germany. <https://doi.org/10.48485/pik.2025.017>
7. IEA. (2025). *Global energy review 2025*. International Energy Agency, Paris. <https://www.iea.org/reports/global-energy-review-2025>
8. IEA. (2025). *World energy outlook 2025*. International Energy Agency, Paris. <https://www.iea.org/reports/world-energy-outlook-2025>
9. IEA. (2025). *Electricity 2025*. International Energy Agency, Paris. <https://www.iea.org/reports/electricity-2025>
10. IRENA. (2025). *Renewable capacity statistics 2025*. International Renewable Energy Agency. <https://www.irena.org/publications/2025/Mar/Renewable-Capacity-Statistics-2025>
11. Riahi, K., et al. (2017). The Shared Socioeconomic Pathways and their energy, land use, and greenhouse gas emissions implications: An overview. *Global Environmental Change*. 42, 153–168. <https://doi.org/10.1016/j.gloenvcha.2016.05.009>
12. Rogelj, J., et al. (2018) Scenarios towards limiting global mean temperature increase below 1.5°C. *Nature Climate Change*. 8, 325–332. <https://doi.org/10.1038/s41558-018-0091-3>
13. IPCC. (2022). *Climate Change 2022: Mitigation of Climate Change*. AR6 Working Group III. <https://www.ipcc.ch/report/ar6/wg3/>
14. IEA. (2025). *World energy employment 2025*. International Energy Agency, Paris. <https://www.iea.org/reports/world-energy-employment-2025>
15. IRENA. (2026). *Renewable energy and jobs: annual review 2025*. International Renewable Energy Agency, Abu Dhabi. <https://www.irena.org/Publications/2026/Jan/Renewable-energy-and-jobs-Annual-review-2025> https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2026/Jan/IRENA_SOC_RE_and_jobs_2026.pdf
16. NGFS. (2025). *Guide to Climate Scenario Analysis for Central Banks and Supervisors: 2025 Version*. Network for Greening the Financial System. <https://www.ngfs.net/en/publications-and-statistics/publications/2025-guide-climate-scenario-analysis-central-banks-and-supervisors>
17. World Bank. (2025). *State and trends of Carbon Pricing 2025*. World Bank. <https://www.worldbank.org/en/publication/state-and-trends-of-carbon-pricing>
18. IRENA. (2024). *World Energy Transitions Outlook 2024: 1.5°C Pathway*. International Renewable Energy Agency, Abu Dhabi. https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2024/Nov/IRENA_World_energy_transitions_outlook_2024.pdf, (https://web.archive.org/web/20251212045311/https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2024/Nov/IRENA_World_energy_transitions_outlook_2024.pdf)
19. Ember. (2026). *World adds a record-breaking 814 GW of solar and wind in 2025*. Ember Energy Research, London. <https://ember-energy.org/latest-updates/world-adds-a-record-breaking-814-gw-of-solar-and-wind-in-2025/>
20. Ember. (2025). *Global Electricity Review 2025*. Ember Energy Research, London. <https://ember-energy.org/latest-insights/global-electricity-review-2025/>
21. IRENA. (2025). *Global landscape of energy transition finance 2025*. IRENA, Abu Dhabi. <https://www.irena.org/Publications/2025/Nov/Global-landscape-of-energy-transition-finance-2025> (<https://web.archive.org/web/20260104135045/https://www.irena.org/Publications/2025/Nov/Global-landscape-of-energy-transition-finance-2025>)

22. IEA. (2024). *Batteries and secure energy transitions*. International Energy Agency, Paris. <https://www.iea.org/reports/batteries-and-secure-energy-transitions>
23. IEA. (2023). *Critical minerals market review*. International Energy Agency, Paris. <https://www.iea.org/reports/critical-minerals-market-review>
24. IRP. (2024). *Global resources outlook 2024: bend the trend – pathways to a liveable planet as resource use spikes*. A report of the International Resource Panel. UNEP, Nairobi. <https://www.resourcepanel.org/reports/global-resources-outlook-2024>
25. IRP. (2020). *Resource efficiency and climate change: material efficiency strategies for a low-carbon future*. A report of the International Resource Panel. UNEP, Nairobi. <https://www.resourcepanel.org/reports/resource-efficiency-and-climate-change>
26. Convention on Biological Diversity. (n.d.). *Ecosystem Restoration - Target 2 of the Kunming-Montreal Global Biodiversity Framework*. <https://www.cbd.int/restoration>
27. Convention on Biological Diversity. (n.d.). *Target 3*. <https://www.cbd.int/gbf/targets/3>
28. Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES). (2019). *Global assessment report on Biodiversity and Ecosystem Services*. <https://www.ipbes.net/global-assessment>
29. Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES). (n.d.). *Indigenous and local knowledge*. <https://www.ipbes.net/indigenous-local-knowledge>
30. United Nations Department of Economic and Social Affairs. (n.d.). *The 17 Goals | Sustainable Development*. <https://sdgs.un.org/goals>
31. United Nations. (2015). *Transforming our world: The 2030 agenda for sustainable development*. <https://sdgs.un.org/sites/default/files/publications/21252030%20Agenda%20for%20Sustainable%20Development%20web.pdf>
32. Food and Agriculture Organization of the United Nations (FAO). (n.d.). *Soil organic carbon*. Global Soil Partnership. <https://www.fao.org/global-soil-partnership/areas-of-work/soil-organic-carbon/en/>
33. Food and Agriculture Organization of the United Nations (FAO). (n.d.). *Recarbonizing global soils: A technical manual of recommended management practices*. Global Soil Partnership. <https://www.fao.org/global-soil-partnership/areas-of-work/soil-organic-carbon-manual/en/>
34. Project Drawdown. (2026). *Drawdown Explorer | Project Drawdown*. <https://drawdown.org/explorer>
35. National Academies of Sciences, Engineering, and Medicine. (2019). *Negative Emissions Technologies and Reliable Sequestration: A Research Agenda*. The National Academies Press, Washington, DC <https://doi.org/10.17226/3235>
36. Grassi, G., et al. (2017). The key role of forests in meeting climate targets requires science for credible mitigation. *Nature Climate Change* 7, 220–226. <https://doi.org/10.1038/nclimate3227>
37. Fuss, S., et al. (2018). Negative emissions—Part 2: Costs, potentials and side effects. *Environmental Research Letters* 13, 063002. <https://doi.org/10.1088/1748-9326/aabf9f>
38. Lebling, K., et al. (2025). *6 Things to know about Direct Air Capture*. World Resources Institute. <https://www.wri.org/insights/direct-air-capture-resource-considerations-and-costs-carbon-removal>
39. Ottenbros, A. B., et al. (2024). Prospective environmental burdens and benefits of fast-swing direct air carbon capture and storage. *Scientific Reports* 14, 16549. <https://doi.org/10.1038/s41598-024-66990-2>
40. Harrison, S. (2023). Mineralisation to capture and use CO₂ from steam methane reforming. *Decarbonisation Technology*, (2), 47-53. <https://ptqmagazines.digitalrefining.com/view/419891467/48/>, <https://ptqmagazines.digitalrefining.com/view/419891467/55/>
41. Keith, D. W., et al. (2018). A process for capturing CO₂ from the atmosphere. *Joule* 2, 1573-1594. <https://doi.org/10.1016/j.joule.2018.05.006>
42. Lackner, K. S., et al. (2012). The urgency of the development of CO₂ capture from ambient air. *Proceedings of the National Academy of Sciences U.S.A.* 109 (33) 13156-13162. <https://doi.org/10.1073/pnas.1108765109>
43. Fiume, S. A. (2023). CDR modeled in MAGICC to return to preindustrial temperatures by 2100. *EarthArXiv* [preprint]. <https://doi.org/10.31223/x5k37c>
44. Fiume, S. A. (2023). Novel, speculative highly-scaled carbon removal study on a reduced complexity model, showing a return to preindustrial temperatures by 2100 and updates to achieving net-zero for Anthropocene reversal. *ESS Open Archive*, [preprint / poster]. <https://doi.org/10.22541/essoar.170365323.31209497/v1>
45. OpenAI and Fiume, S. A. (2026). Conversational development support for the 2030s Net Zero Playbook (ChatGPT, GPT-4.1 Turbo and GPT-5.x series). <https://chatgpt.com/share/69a39dac-a774-800a-b168-af2e11172c4a>

- a. <https://chatgpt.com/g/g-p-69a42a5743e8819182f8b4588bc199eb-2030s-nz/c/69a43696-f62c-832c-9077-214100bc00ed>
- 46. IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems. (2019). Ethically Aligned Design: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems, First Edition. IEEE.
<https://standards.ieee.org/content/ieee-standards/en/industry-connections/ec/autonomou s-systems.html>